p-Adic statistical field theory of fluctuation effects in hierarchical neural networks

W. A. Zúñiga-Galindo*
University of Texas Rio Grande Valley, School of Mathematical & Statistical Sciences, One West University Blvd, Brownsville, TX 78520, United States
(Dated: Draft of October 26, 2026)

The Buice–Cowan field theory [PRE **75**, 051919 (2007)] describes fluctuations and correlations of neural activity through a path integral whose mean-field limit is the Wilson–Cowan equation and whose critical behavior belongs to the directed percolation class. In its continuum limit the neurons are the points of a Euclidean space and activity spreads by ordinary (local) diffusion. This description encodes neither the hierarchical organization of cortical networks nor their small-world property. We formulate a version of the theory in which the neurons are organized in an infinite, self-similar, tree-like structure, described by the p-adic numbers. The master equation, the operator formulation and the path integral carry over unchanged, and finite networks are exact restrictions of the continuum theory, not approximations of it. The main change is in the linear response. In the Euclidean theory every short-range connectivity kernel reduces to the same local operator, the Laplacian. In the p-adic theory the connectivity kernel itself gives rise to a nonlocal p-adic diffusion operator, the generator of a p-adic heat equation, and different kernels give different operators. For couplings that decay by a fixed factor per level of the hierarchy this is the Taibleson–Vladimirov operator, and the critical theory is an ultrametric Reggeon field theory whose upper critical dimension is twice the decay exponent of the coupling. Hence the rate at which the coupling decays up the hierarchy, and not the dimension of space, decides whether mean-field exponents apply. We obtain a Ginzburg-type criterion, loop-corrected hierarchical Wilson–Cowan equations, log-periodic corrections to scaling, power-law correlations, and the counterparts of the Buice–Cowan results on the phase transition and on self-organization. Exact simulations of the spike model on a tree support the predictions. The avalanche-size exponent takes the mean-field value three-halves when the coupling decays slowly, and decreases continuously toward the directed-percolation value when it decays fast. The simulations also show that the hierarchical network is a small-world network when the decay exponent is smaller than the dimension: connections are highly clustered, while path lengths and spreading times grow slowly, roughly as the logarithm of the number of neurons. The Buice–Cowan theory in its local diffusion limit does not have this property. In a Euclidean space it requires long-range coupling as an additional hypothesis, whereas in the p-adic setting long-range coupling is the only possibility.

I. INTRODUCTION

There are two central hypotheses about the functioning of the brain. The first is that the brain has a hierarchical organization [1, 2]. The second, the *critical brain hypothesis*, is that it operates near a phase transition, where a delicate balance between ordered (synchronized) and disordered (random) states occurs; the neuronal avalanches observed by Beggs and Plenz support this view [3]. Since deep neural networks (DNNs) are inspired by the brain, both hypotheses are also relevant for them [4–6].

Sporns and co-workers proposed a precise form of the hierarchy hypothesis: cortical connections are organized in self-similar, fractal patterns across multiple nested scales [7–9]. This proposal provides a specific geometric mechanism, nested and scale-free self-similarity, by which the brain builds its structural and functional hierarchy. It has three aspects. (i) *Nested connectivity*: pathways link neuronal populations across scales, from whole brain regions down to local minicolumns. (ii) *Self-similarity*: the patterns of connections repeat from one scale to the next. (iii) *Small-world architecture*: cortical networks combine a high local clustering of connections with short paths between distant regions [10, 11], which allows efficient communication at a low wiring cost. As we explain below, nested self-similar connectivity is exactly the structure encoded by the p-adic numbers. It was shown in Ref. [12] that the classical Wilson–Cowan model, in which the neurons are organized as the points of $\mathbb{R}^N$ and interact through Gaussian-type kernels, is not compatible with the small-world property. In this paper we examine the same question for the B-C theory.

Buice and Cowan (B-C) [13] gave a systematic statistical field theory for the critical brain hypothesis. Starting from a Markovian "effective spike" model, they constructed a Doi–Peliti path integral [14, 15]. They showed that its zero-loop (mean-field) limit is a Wilson–Cowan equation and that loops give systematic corrections controlled by a Ginzburg-type criterion. They also argued that the critical point lies in the directed percolation (DP) universality class [16–19]. In B-C the neurons live in $\mathbb{R}^N$ with $N = 2$ or 3. The hierarchy therefore enters, at best, through the choice of kernel. A central problem is to develop new versions of B-C statistical field theories compatible with hypotheses about the hierarchical organization of the brain. This is the objective of this work; however, here, we take only a first step toward it.

* wilson.zunigagalindo@utrgv.edu

Another motivation is the correspondence between neural networks (NNs) and statistical field theories (SFTs). This correspondence is a promising approach for understanding the dynamics of DNNs [20–24]. Since DNNs are inspired by the structure and function of the biological brain, it is natural to conjecture that SFTs of B-C type can play a central role in understanding how large DNNs work. It is relevant to note that B-C developed such a correspondence for cortical networks in the limit of infinitely many neurons, combined with a continuum limit in which the neuron label becomes a position in space [13, 25]. The correspondence between DNNs and lattice SFTs, with a large but finite number of neurons, has been studied intensively [26, 27]. However, standard principles of statistical mechanics suggest that understanding the dynamics of DNNs requires understanding their thermodynamic limit. Based on that, the author has proposed that the continuous SFTs developed by B-C should be taken as the basis for a correspondence between DNNs and SFTs; see Ref. [28] and the references therein.

The present work supports this proposal. In the field-theoretic approaches to DNNs, the natural large parameter is the width n of the layers, or the number of neurons. At fixed depth, the infinite-width limit is a Gaussian theory, and the non-Gaussian corrections come in powers of $1/n$ [26, 27]. The p-adic B-C theory differs in two ways. First, width and depth cannot be separated: a hierarchical network of depth l in dimension N has p^{Nl} neurons, and its thermodynamic limit adds layers to the hierarchy at the same time as it adds neurons, so the infinite-width limit involves infinitely many layers. Second, the limiting theory is not Gaussian: its cubic couplings produce fluctuations that are not small near criticality below the upper critical dimension (Sec. V). The cutoffs of the theory, the depth of the tree and the size of the network, are part of the architecture, and a Ginzburg number plays the role of $1/n$. If the infinite-width limit is understood as a network with one layer and infinitely many neurons, the p-adic SFTs are of a completely different nature from the lattice SFTs of Ref. [26]. We discuss this comparison in Sec. VI A.

Ultrametricity and p-adic numbers in physics. A distance is *ultrametric* if it obeys the strong triangle inequality, $d(x,z) \leq \max\{d(x,y), d(y,z)\}$. A non-empty set endowed with an ultrametric is called an ultrametric space. These spaces constitute the natural mathematical framework for modeling hierarchical systems [29, 30]. In physics they first appeared in the theory of disordered systems. Parisi and collaborators showed that the pure states of the spin-glass phase of mean-field models, described by replica symmetry breaking, are organized in an ultrametric hierarchy [31]. Frauenfelder and co-workers proposed that the conformational substates of proteins are organized in a similar way, so that protein relaxation resembles a random walk on a tree-like landscape of local energy minima [32]. The p-adic numbers are the prototypical ultrametric spaces on which one can do analysis: they carry a measure, a Fourier transform, and pseudodifferential operators [33–37]. This made it possible to formulate replica symmetry breaking p-adically [38], and to describe relaxation in hierarchical energy landscapes by p-adic diffusion equations [39]. Since then, p-adic models have been used in statistical mechanics, quantum field theory and string theory, and the theory of complex systems [30, 33]. p-Adic quantum field theory began in the 1980s under the influence of Vladimirov and Volovich [33], and it has produced an extensive literature [30, 33]. Most of this work concerns formal models, whose connection with measurable phenomena is indirect. To the author's knowledge, the p-adic B-C theory is one of the first p-adic statistical field theories that is derived, without free modeling choices, from the microscopic dynamics of a concrete system, here the master equation of a spiking network, and whose predictions can be compared with simulations and, in principle, with experiments.

p-Adic statistical field theories and neural networks. In a series of papers, the author and collaborators have developed a correspondence between NNs and SFTs in which the fields are defined on a p-adic space instead of a Euclidean one [23, 24, 28, 40, 41]. In this framework, a field is a real-valued function on the boundary of an infinite tree-like structure that provides the architecture of a deep network. The correspondence was first established for deep belief networks [23]: whereas a restricted Boltzmann machine corresponds to a spin glass, a deep, hierarchical architecture corresponds to an ultrametric spin glass, which is described naturally with p-adic numbers. Restricting the energy functional of a p-adic SFT to a finite-dimensional space of test functions gives a discretization that is exact, and that yields a new class of networks, the p-adic convolutional deep Boltzmann machines [40, 41]. Because of the weight sharing imposed by the symmetries of the tree, these networks have $O(\mathcal{N})$ parameters for $\mathcal{N}$ neurons, instead of the $O(\mathcal{N}^2)$ of a fully connected machine. The same ideas were applied to dynamical models: p-adic cellular neural networks, which are the continuum limits of large hierarchical versions of the Chua–Yang networks [42]; hierarchical Wilson–Cowan models [12]; and continuous random networks, for which a dynamic mean-field theory and a criticality condition were obtained [24, 28]. The exact discretization just mentioned is the mechanism used in Sec. III A of the present paper.

A natural way to make a network intrinsically hierarchical is to index neurons by p-adic numbers [33–36]. Working in dimension one ($N = 1$) for simplicity, the unit ball $\mathbb{Z}_p$ is the boundary of a rooted tree in which each vertex has p children. The ultrametric distance $|x-y|_p$ records the depth of the last common ancestor of x and y.

We use the following terminology throughout. The *levels* of the tree play the role of the *layers* of the hierarchy, and we use the words *level* and *layer* synonymously. A vertex at level k, together with all the neurons that de-

scend from it, is a ball of radius p^{-k}. We refer to it simply as a *ball*, or as a *structure of level* k: the ball is built from the p balls of level $k+1$ below it, just as higher structures in the brain are built from lower ones. The *hierarchical distance* between two neurons is $|x-y|_p$: if it equals p^{-k}, the smallest structure containing both neurons is of level k. In previous work [23, 24, 28] it was observed that the B-C theory admits such a formulation. One simply interprets the spatial integrals in Eqs. (12)–(13) of Ref. [13] as integrals over $\mathbb{Q}_p^N$ with respect to the Haar measure. The purpose of this paper is to work out what changes and what does not.

Results and comparison with the B-C theory. Our starting point is the observation that the stochastic model of B-C, its operator formulation and the resulting discrete action do not refer to any geometry. The geometry enters only in the continuum limit, where B-C place the neurons at the points of $\mathbb{R}^N$. We show that if the neurons are instead organized in a tree-like structure, the same limit places them at the points of $\mathbb{Q}_p^N$, and the sums over neurons become integrals with respect to the Haar measure. In contrast with the Euclidean case, the finite network is not an approximation of the continuum theory: the discrete action is exactly the restriction of the continuum action to a finite-dimensional space of functions, a feature shared by other p-adic statistical field theories [23, 24, 28, 40]. Mean-field theory is again a Wilson–Cowan equation, now hierarchical, and the loop expansion of B-C carries over formally unchanged.

The first substantial difference appears in the linear response. In the B-C theory, the expansion of the Fourier transform of the connectivity kernel at small momenta produces a local diffusion term, the Laplacian, the same for every short-range kernel. No such expansion exists over $\mathbb{Q}_p^N$. This does not mean that there is no p-adic diffusion. The theory of p-adic heat equations is vast, and there are many more such equations than in the Euclidean case [30, 34, 36]. What happens is that each connectivity kernel gives rise, in a natural way, to its own nonlocal diffusion operator, a W-operator, which is the generator of a p-adic heat equation. In the Euclidean theory the kernel is reduced to a universal local operator; in the p-adic theory the kernel itself determines the p-adic diffusion operator, a W-operator [36, Chap. 2]. For kernels with compact support, the support of the kernel determines which neurons are truly connected. If the connected neurons form a ball, only its total activity becomes marginal at the critical point, and there are no scale-free correlations across the hierarchy. We also consider *regional networks*, in which the neurons occupy a disjoint union of finitely many (at least two) balls, as in the ultrametric transition networks of Ref. [43]. Each ball represents a region of the cortex in which the spike model holds with its own constant parameters, and the regions are coupled by constant interregional connections. When all regions share the same decay rate and firing-rate function, this is the B-C model with a block-structured connectivity. Region-dependent parameters go beyond the B-C model, whose master equation has a single firing-rate function and a single decay rate for all neurons. The criticality of a regional network is decided by a finite matrix, and its critical behavior is of mean-field type (Lemma 1, Sec. IV A). On a bounded network, the linear dynamics is a Markov process exactly when the distance to criticality is nonnegative, and at the critical point this process becomes conservative: branching exactly compensates decay (Sec. IV C). For isotropic kernels that decay by a fixed factor per level of the hierarchy, $w(\|x\|_p) \propto \|x\|_p^{-N-\sigma}$ at large distances with $\sigma > 0$, the operator is the Taibleson–Vladimirov operator of order σ, which we show to be the unique isotropic and homogeneous choice. The exponent σ is a parameter of the connectivity, not of the dynamics: each time one climbs one level of the hierarchy, the coupling between two neurons is multiplied by $p^{-(N+\sigma)}$, and the total coupling that a neuron receives from the neurons at that level is multiplied by $p^{-\sigma}$. Thus σ measures how fast the coupling decays with hierarchical distance, and it can be read off from the connection weights. It is a new physical parameter with no counterpart in the B-C theory. Throughout the paper σ denotes this decay exponent.

The second difference concerns criticality. B-C argue that the critical point belongs to the DP class, with upper critical dimension 4, and they invoke the high connectivity of cortex to justify mean-field exponents in dimension 2 or 3. In the p-adic theory the effective action at criticality is an ultrametric Reggeon field theory whose upper critical dimension is $N_c = 2\sigma$. Whether mean-field exponents apply is therefore decided by the decay exponent of the hierarchical coupling and not by the embedding dimension. Unlike Euclidean DP with long-range spreading [44], there is no competing local (Laplacian) term, so every $\sigma > 0$ defines a distinct class. The Ginzburg-type criterion of B-C becomes a statement about the level of the hierarchy that fluctuations reach before they relax, and the discrete scale invariance of the hierarchical neuron space, which is invariant only under the dilations $x \mapsto p^{-1}x$, produces log-periodic corrections to scaling, which a Euclidean theory cannot produce. In the mean-field regime the avalanche-size exponent is $3/2$, as in the B-C theory; for $N < 2\sigma$ we predict deviations that depend on σ, and we confirm them by direct simulation of the spike model on a tree (Sec. VIII A).

The third difference concerns the small-world property. For a network with weighted coupling we use two tests: the clustering and the path length of a sparse network whose connections are drawn with probabilities proportional to the coupling, and the time that activity needs to reach half of the neurons in the spike model. With both tests, the hierarchical network is a small-world network when $\sigma < N$: the clustering is high, and path lengths and spreading times grow slowly, roughly as the logarithm of the number of neurons (Sec. IX). For $\sigma > N$ it is not. The B-C theory with the kernels of the classical model, Gaussian or exponential, which for practical purposes have compact support, is not small-world: the

spreading time grows in proportion to the linear size of the network. This agrees with Ref. [12], where the incompatibility is traced to the fact that $\mathbb{R}^N$ has no nontrivial compact subgroups, while in $\mathbb{Q}_p^N$ every ball is one. A finite Euclidean network can be made small-world, by our tests, only by leaving the classical model: by imposing a coupling with a power-law tail on a ring. In $\mathbb{R}^N$ long-range coupling is thus an additional hypothesis, imposed on a theory whose generic limit is local diffusion. In $\mathbb{Q}_p^N$ all diffusion operators are nonlocal, long-range coupling is the only possibility, and it comes with the nested structure of the hierarchy. The threshold $\sigma = N$ differs from the threshold $\sigma = N/2$ for mean-field avalanches: the small-world property is necessary for the exponent $3/2$ in our simulations, but it is not sufficient.

Finally, we give the p-adic counterparts of the phase transition and of the self-organization mechanisms discussed in Ref. [25]. The linear spectrum resolves exactly layer by layer. The zero-loop propagator is computed explicitly and has power-law tails in hierarchical distance, so that correlations away from criticality are not exponentially truncated in space, in contrast with the Euclidean prediction. Power spectra of local averages scale as $\omega^{-(2-N/\sigma)}$ below a crossover frequency set by the size of the averaging region. Standard self-organizing mechanisms, such as synaptic depression, tune the distance to criticality but leave σ unchanged. Throughout, we distinguish results that are exact at zero-loop order from heuristic arguments and from conjectures. In particular, the analysis below the upper critical dimension, which requires the renormalization group, is not carried out here, which is why we regard this work as a first step.

Table I compares the Euclidean B-C theory with the p-adic theory of this paper, item by item: the neuron space and finite networks, the mean-field equation and the spreading operator, the propagator at large distance, the upper critical dimension and the mean-field exponents, scale invariance, correlations and power spectra, the spreading of activity and the small-world property, the avalanche-size exponent, and what self-organization tunes.

Organization of the paper. The main physical results of this work were discussed in Ref. [45]. The present paper is the companion paper to that publication: it contains the complete construction, the proofs, and further results. Section II collects the facts of p-adic analysis that we use. Section III constructs the p-adic effective spike model: the continuum limit of a hierarchical network, the action, mean-field theory, and the free and interacting parts of the action with the corresponding propagators. Section IV studies the linear response and the W-operators selected by the connectivity. Section V derives the effective theory at criticality, its power counting and its mean-field exponents. Section VI gives the Ginzburg-type criterion. Section VII contains the loop-corrected hierarchical Wilson–Cowan equations, and Sec. VIII the avalanche exponents and their numerical simulation. Section IX studies the small-world property. Section X treats the phase transition, correlations, power spectra, rhythms and self-organized criticality. Section XI discusses open problems.

Reading guide. The paper is addressed to physicists, neuroscientists and researchers in artificial intelligence. Readers not familiar with field-theoretic methods for stochastic processes will find in Appendix A a self-contained introduction to the concepts used below: the master equation, the Doi–Peliti formulation, the action, Feynman diagrams, and universality. The appendix follows a simple branching process that contains all the ingredients of the effective spike model, and it ends with a glossary. Appendix B introduces directed percolation and Reggeon field theory, the universality class of the critical point. Appendix D collects the basic facts of p-adic analysis, and Appendix F the p-adic integrals used in the text.

Mathematical format. In some parts of the paper we use the mathematical format of propositions, lemmas, and proofs. This format gives precise control over several new calculations involving p-adic heat equations and the asymptotics of the new p-adic integrals required in perturbative calculations. It does not mean that we present a rigorous mathematical theory of the B-C SFTs; that remains a very difficult open problem.

II. p-ADIC PRELIMINARIES

We collect the facts needed below; see Refs. [33–36]. Definitions and further details (the p-adic norm, balls, the Haar measure, test functions, the Fourier transform and distributions) are given in Appendix D. Throughout, p denotes a fixed prime number. Any nonzero p-adic number $x \in \mathbb{Q}_p \smallsetminus \{0\}$ can be represented as a series of the form $x = p^{\mathrm{ord}(x)} \sum_{j\geq 0} x_j p^j$, with $x_j \in \{0, \dots, p-1\}$ and $x_0 \neq 0$; then $|x|_p = p^{-\mathrm{ord}(x)}$. The p-adic norm $|\cdot|_p$ is ultrametric: $|x+y|_p \leq \max\{|x|_p, |y|_p\}$. This inequality is stronger than the standard triangle inequality, and it has strong implications for the geometry of $\mathbb{Q}_p$. For instance, two balls are either disjoint or one contains the other, which in turn implies that the unit ball $\mathbb{Z}_p = \{|x|_p \leq 1\}$ is an infinite rooted tree: the p-adic distance between two points is determined by the level of their last common ancestor (Fig. 1). Throughout, N denotes the dimension of the neuron space (d in Ref. [13]); we avoid d so as not to write $d^d x$. For $x \in \mathbb{Q}_p^N$ set $\|x\|_p = \max_i |x_i|_p$. Let $d^N x$ denote the Haar measure normalized by $\int_{\mathbb{Z}_p^N} d^N x = 1$; then $d^N(ax) = |a|_p^N \, d^N x$. The sphere $S_k^N = \{\|x\|_p = p^k\}$ has volume $(1 - p^{-N}) p^{kN}$. Hence, for radial $F(x) = F(\|x\|_p)$,

$$\int_{\mathbb{Q}_p^N} F(\|x\|_p)\, d^N x = (1 - p^{-N}) \sum_{k\in\mathbb{Z}} p^{kN} F(p^k). \qquad (1)$$

We warn the reader that the integration theory cannot be reduced to formula (1). Let $\chi_p(y) = e^{2\pi i \{y\}_p}$ be the

TABLE I. Comparison of the Euclidean B-C theory with the p-adic theory of this paper. Entries of the p-adic column are zero-loop (mean-field) results unless stated otherwise. N is the dimension of the neuron space, σ the decay exponent of the hierarchical coupling [Eq. (53)], and μ the distance to criticality.

	Euclidean B-C theory	p-Adic theory
Neuron space	$\mathbb{R}^N$	a subset of $\mathbb{Q}_p^N$, with a hierarchical geometry
Finite network	lattice; an approximation	finite rooted trees, G_l^N; an exact restriction (Prop. 1)
Mean-field equation	Wilson–Cowan	hierarchical Wilson–Cowan [Eq. (20)]
Spreading operator	local diffusion, $-D\nabla^2$	nonlocal p-adic diffusion (W-operator) set by the kernel (Sec. IV A)
Propagator at large distance	Gaussian	power law, $\propto w(\|x\|_p)$ [Eq. (33)]
Upper critical dimension	4	2σ (power counting)
Mean-field exponents	$\nu_\perp = \frac{1}{2}$, $z = 2$	$\nu_\perp = 1/\sigma$, $z = \sigma$
Scale invariance	continuous	discrete ($x \to px$); log-periodic corrections
Correlations for $\mu > 0$	exponentially truncated	power law [Eq. (99)]
Spectrum of a local average	ω^{-2}	$\omega^{-(2-N/\sigma)}$ for $\omega \ll \omega_k$, with ω_k set by the electrode size [Eq. (100)]
Spreading of activity ($\mu < 0$)	constant velocity	constant layers per unit time for $\sigma < N$ (Fig. 6)
Small-world property	no (diffusion limit)	yes for $\sigma < N$ (Sec. IX)
Avalanche-size exponent	3/2 for $N \geq 4$	3/2 for $N \geq 2\sigma$; smaller otherwise (Fig. 5)
Self-organization tunes	μ	μ; σ is unchanged

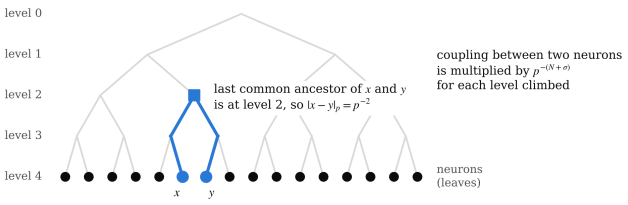


FIG. 1. Neurons organized in a tree-like structure ($p = 2$, depth 4: levels $0, \dots, 4$). The p-adic distance between two neurons is determined by the level of their last common ancestor. With a kernel $w(\|x\|_p) \propto \|x\|_p^{-N-\sigma}$, the exponent σ measures how fast coupling decays with hierarchical distance.

standard additive character (the p-adic analogue of the complex exponential function). For an integrable function $g : \mathbb{Q}_p^N \to \mathbb{C}$, its Fourier transform is defined as $\hat{g}(\xi) = \int \chi_p(\xi \cdot x) g(x)\, d^N x$. The basic identity is

$$\int_{\|x\|_p \leq p^k} \chi_p(\xi \cdot x)\, d^N x = p^{kN}\, \Omega\big(p^k \|\xi\|_p\big)\,, \tag{2}$$

where Ω is the indicator of $[0, 1]$. The Fourier transform of a radial function is radial.

The Taibleson–Vladimirov operator of order $\sigma > 0$ is

$$\boldsymbol{D}^\sigma \phi(x) = \frac{1 - p^\sigma}{1 - p^{-\sigma-N}} \int_{\mathbb{Q}_p^N} \frac{\phi(x - y) - \phi(x)}{\|y\|_p^{\sigma+N}}\, d^N y, \tag{3}$$

with symbol $\|\xi\|_p^\sigma$. It generates a Markov process on $\mathbb{Q}_p^N$ (ultrametric diffusion [34, 37]). Radial convolution operators are diagonalized by the p-adic wavelets [35, 46]

$$\psi_{\gamma n j}(x) = p^{-N\gamma/2}\, \chi_p\big(p^{-1} j \cdot (p^\gamma x - n)\big)\, \Omega\big(\|p^\gamma x - n\|_p\big), \tag{4}$$

with $\gamma \in \mathbb{Z}$, $n \in (\mathbb{Q}_p/\mathbb{Z}_p)^N$ and $j \in \{0, 1, \dots, p-1\}^N \setminus \{(0, \dots, 0)\}$. The wavelet $\psi_{\gamma n j}$ is supported on the ball of radius p^γ centred at $p^{-\gamma} n$, and the wavelets form an orthonormal basis of $L^2(\mathbb{Q}_p^N)$. For radial $w \in L^1$,

$$w \star \psi_{\gamma n j} = \hat{w}\big(p^{1-\gamma}\big)\, \psi_{\gamma n j}, \tag{5}$$

so the eigenvalue depends only on the level of the ball that supports the wavelet, not on its position or on j; see also Ref. [47, Section 4.1].

III. THE p-ADIC EFFECTIVE SPIKE MODEL

A. Neuron geometry: from the discrete master equation to the continuum

This subsection explains why the p-adic theory is a genuine version of the B-C theory and not a modification of it. The main point is that the geometry of the neurons is not fixed by the model. It enters only when the continuum limit is taken.

The master equation has no geometry. The effective spike model of B-C [Sec. II, Eq. (1) of Ref. [13]] is defined for an arbitrary finite set Λ of neurons. Let n_i be the number of effective spikes of neuron $i \in \Lambda$, and w_{ij} the weight with which neuron j innervates neuron i. The probability $P(\mathbf{n}, t)$ of the configuration $\mathbf{n} = (n_i)_{i \in \Lambda}$ obeys

$$\begin{aligned}\partial_t P(\mathbf{n}, t) = \sum_{i \in \Lambda} \Big\{ &\alpha\big[(n_i + 1) P(\mathbf{n}_i^+, t) - n_i P(\mathbf{n}, t)\big] \\ &+ f\big(s_i(\mathbf{n}_i^-)\big) P(\mathbf{n}_i^-, t) - f\big(s_i(\mathbf{n})\big) P(\mathbf{n}, t) \Big\}, \end{aligned} \tag{6}$$

where $s_i(\mathbf{n}) = \sum_j w_{ij} n_j + I$, and $\mathbf{n}_i^\pm$ differs from $\mathbf{n}$ only in the ith entry, which is $n_i \pm 1$, and f is the firing-

rate function. Eq. (6) differs from Eq. (1) of Ref. [13], where the gain term is $f(s_i(\mathbf{n}))P(\mathbf{n}_i^-,t)$, and from the corresponding Hamiltonian, Eq. (A11) there, in which f stands to the left of $(1-\Phi_i^\dagger)$. In Eq. (6) the firing rate is evaluated in the state before the jump, $s_i(\mathbf{n}_i^-) = s_i(\mathbf{n}) - w_{ii}$, which ensures conservation of total probability for any matrix (w_{ij}). The two forms coincide when $w_{ii}=0$; in the continuum limit their difference vanishes (see the discussion after Eq. (12)). Nothing in Eq. (6) refers to a geometry. The set Λ and the matrix (w_{ij}) are arbitrary.

The discrete action has no geometry. The Doi–Peliti construction [Appendices A and B of Ref. [13]] uses one pair of creation and annihilation operators per neuron, $[\Phi_i, \Phi_j^\dagger] = \delta_{ij}$. It is local in the neuron index and applies to any Λ. After the Doi shift and the coherent-state representation, it gives the discrete action (see also Sec. 5.2 of Ref. [23])

$$S_\Lambda[\varphi,\tilde\varphi] = \int_0^\infty dt \sum_{i\in\Lambda} \Big\{ \tilde\varphi_i\, \partial_t \varphi_i + \alpha \tilde\varphi_i \varphi_i - \tilde\varphi_i\, f\Big(\sum_{j\in\Lambda} w_{ij}\big(\tilde\varphi_j\varphi_j + \varphi_j\big) + I\Big)\Big\} - \sum_{i\in\Lambda} \bar n_i\, \tilde\varphi_i(0), \tag{7}$$

where f is the firing rate of the neurons and $\bar n_i \geq 0$ specifies the initial state. The interaction term is the coherent-state form of the Hamiltonian $\sum_i (1-\Phi_i^\dagger)\, f_{\mathrm{NO}}\big(\sum_j w_{ij}\Phi_j^\dagger\Phi_j + I\big)$, with $(1-\Phi_i^\dagger)$ to the left of f (normal ordered as in Ref. [13]), i.e. of Eq. (6), followed by the Doi shift $\Phi_i^\dagger \to 1+\tilde\varphi_i$. Thus Eq. (7) agrees with the action of B-C [Eq. (13) of Ref. [13], in discrete form], although the master equations differ. Normal ordering the Hamiltonian (A11) of Ref. [13], with f to the left of $(1-\Phi_i^\dagger)$, would produce additional terms proportional to w_{ii}, which do not appear in Eq. (13) there. The action of Ref. [13] is therefore the action of the probability-conserving master equation (6). As in Ref. [13], the network starts in an uncorrelated state in which the number of effective spikes of each neuron is Poisson distributed with mean $\bar n_i$:

$$P(\mathbf{n},0) = \prod_{i\in\Lambda} e^{-\bar n_i}\, \frac{\bar n_i^{\,n_i}}{n_i!}, \qquad \bar n_i = \langle n_i(0)\rangle. \tag{8}$$

In the operator representation this state is $\exp\big(\sum_i \bar n_i(\Phi_i^\dagger - 1)\big)|0\rangle$. After the Doi shift it produces the initial-state term $-\sum_i \bar n_i\tilde\varphi_i(0)$ in the action. Other (correlated) initial states give other initial-state terms [13]. The geometry is still absent.

The continuum limit chooses a geometry. To pass to a field theory, one must (i) embed the index set in a space X with a measure dx, (ii) let Λ fill X, and (iii) replace the sums in Eq. (7) by integrals over X. B-C take Λ to be a square lattice $a\mathbb{Z}^N$ with spacing a and $w_{ij} = w(|x_i - x_j|)$. The sums become Riemann sums, $\sum_i a^N F(x_i) \to \int_{\mathbb{R}^N} F(x)\, d^N x$ as $a\to 0$, and the neurons become the points of $\mathbb{R}^N$. This is one particular choice. B-C themselves note that N should be 2 or 3, "or possibly in between to allow for a fractal lattice", and that N reflects how many neighbors a neuron has [13]. The Euclidean geometry is an assumption made at the continuum step, and it is precisely the assumption that suppresses the hierarchical organization of the network.

Hierarchically organized neurons. Instead, take the neurons to be organized hierarchically. For $l \geq 1$ let G_l be the set of numbers of the form

$$i = i_0 + i_1 p + \cdots + i_{l-1} p^{l-1}, \tag{9}$$

where the i_k are p-adic digits, i.e. elements of $\{0,1,\dots,p-1\}$, and let G_l^N be the Cartesian product of N copies of G_l, so that the cardinality of G_l^N is p^{Nl} [23]. The elements of G_l^N are representatives of the quotient $\mathbb{Z}_p^N/p^l\mathbb{Z}_p^N$, and

$$\mathbb{Z}_p^N = \bigsqcup_{i\in G_l^N} \big(i + p^l\mathbb{Z}_p^N\big). \tag{10}$$

Geometrically, G_l is a rooted tree with $l+1$ layers in which each vertex has p children: the truncations i mod p^k, $k=0,\dots,l$, are its vertices, and the elements of G_l are its leaves (layer l). The layer $k=0$ is the root of the tree; see Fig. 1. The set G_l^N can be visualized as a forest of N identical rooted trees; a neuron is an N-tuple of leaves, one from each tree. Each neuron $i\in G_l^N$ is identified with the ball $B_{-l}^N(i) = i + p^l\mathbb{Z}_p^N$ of radius p^{-l} (we use the notation $B_r^N(a)$ of Appendix D for the ball of radius p^r with center a). For $N=1$, two neurons i,j share their first k digits exactly when $|i-j|_p = p^{-k}$, so the ultrametric distance measures how far up the tree one must climb to reach their common ancestor (Fig. 2). For $N>1$, $\|i-j\|_p = p^{-k}$ means that k is the number of leading digits shared in every coordinate, since $\|x\|_p = p^{-\mathrm{ord}(x)}$ with $\mathrm{ord}(x) = \min_i \mathrm{ord}(x_i)$ (Appendix D). B-C's homogeneity hypothesis, that w_{ij} depends only on the distance between i and j [13], becomes

$$w_{ij} = w(\|i-j\|_p) \qquad (i\neq j), \tag{11}$$

for a radial kernel $w\in L^1$, and it is invariant under the translations of the finite group G_l^N. For the self-coupling of the smallest structure we set

$$w_{ij} = p^{Nl} \int_{B_{-l}^N(j)} w(\|x_i - y\|_p)\, d^N y, \tag{12}$$

which agrees with Eq. (11) for $i\neq j$ (by ultrametricity $\|x_i - y\|_p = \|i-j\|_p$ for $y\in B_{-l}^N(j)$). For $i=j$ it defines w_{ii} as the average coupling inside $B_{-l}^N(i)$. Excluding self-coupling, as in Ref. [23], changes the theory by a term of order $\int_{\|y\|_p\leq p^{-l}} w\, d^N y$, which vanishes as $l\to\infty$.

The continuum limit of the hierarchical network. Let $\mathcal{D}^l \subset \mathcal{D}(\mathbb{Z}_p^N)$ be the space of functions constant on the

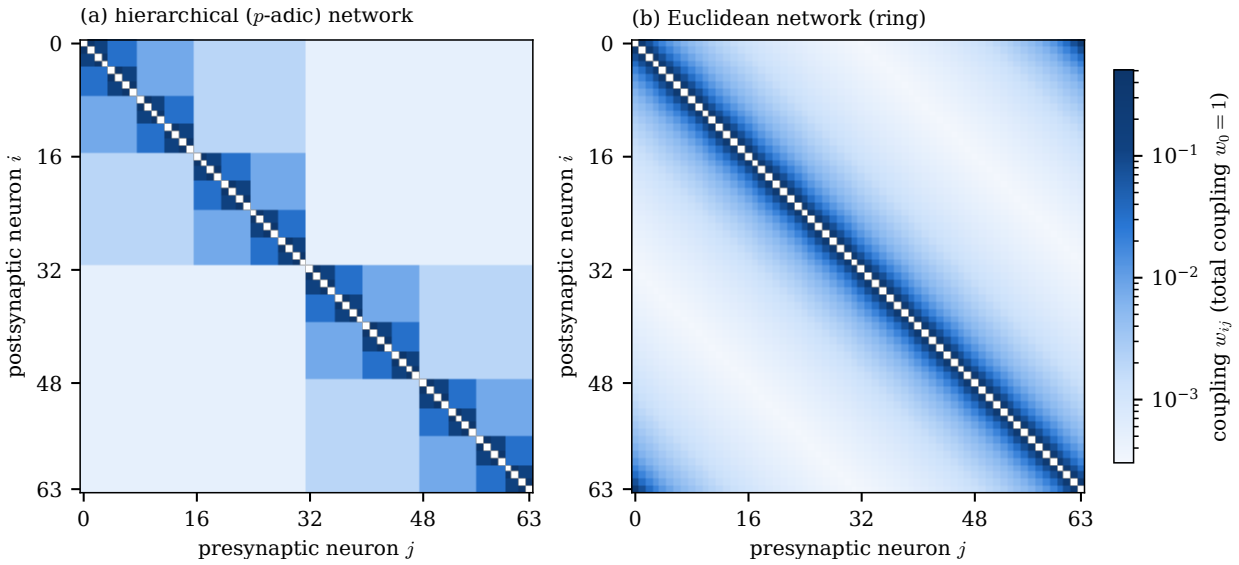


FIG. 2. Connection matrices w_{ij} of two networks of 64 neurons, with $\sigma = 1$ (logarithmic color scale, common to both panels; the diagonal is excluded). In both networks the proportionality constant is fixed by the condition that the total coupling received by a neuron is $w_0 = \sum_{j\neq i} w_{ij} = 1$. (a) The hierarchical network G_l with $p = 2$, $l = 6$, $N = 1$ and $w_{ij} = C_p\,|i-j|_p^{-(1+\sigma)}$, where $C_p = 1/2016$. The neurons are ordered along the boundary of the tree, so that every ball is a block of consecutive indices; the nested blocks are the balls of levels $5, 4, \ldots, 1$. (b) A Euclidean network: a ring with $w_{ij} = C_E\,|i-j|^{-(1+\sigma)}$, where $|i-j|$ is the distance along the ring and $C_E \simeq 0.310$. The hierarchical matrix is constant on blocks, which is the property used in Proposition 1.

balls $B_{-l}^N(i)$, $i \in G_l^N$, supported in the N-dimensional unit ball $\mathbb{Z}_p^N$. Then $\dim \mathcal{D}^l = p^{Nl}$, and

$$\mathcal{D}^l \hookrightarrow \mathcal{D}^{l+1} \hookrightarrow \mathcal{D}(\mathbb{Z}_p^N) = \bigcup_l \mathcal{D}^l. \tag{13}$$

Here, $\mathcal{D}(\mathbb{Z}_p^N)$ denotes the space of locally constant functions with support in the N-dimensional unit ball. Each element of G_l^N represents the smallest structure of the network, a population whose size is proportional to its volume p^{-Nl}. This is the same interpretation as B-C's effective spikes, which count active neurons in a small region. We therefore relate the discrete variables to densities by

$$\begin{aligned} \varphi_i &= p^{-Nl}\varphi(x_i), \qquad & \tilde\varphi_i &= \tilde\varphi(x_i), \\ \bar n_i &= p^{-Nl}\bar n(x_i), & & \end{aligned} \tag{14}$$

with $\varphi, \tilde\varphi, \bar n \in \mathcal{D}^l$ and $x_i \in B_{-l}^N(i)$. This is the analogue of the factors a^N in the Euclidean limit. The firing-rate function must be scaled in the same way. In the master equation (6) and in the action (7) of the network G_l^N we take, for all neurons, the firing-rate function

$$f_{(l)} = p^{-Nl} f, \tag{15}$$

where f is the firing-rate function of the continuum theory: the rate at which effective spikes are produced in the ball $B_{-l}^N(i)$ is the rate per unit volume, f, times the volume p^{-Nl} of the ball. The function $f_{(l)}$ depends on the depth l of the network but not on the neuron. At fixed l the factor p^{-Nl} can be absorbed into f; it matters only for $l \to \infty$.

Proposition 1 (Exact discretization)**.** *Let w be radial and integrable, and let S be the action* (18) *below with $\mathbb{Q}_p^N$ replaced by $\mathbb{Z}_p^N$. For fields $\varphi(\cdot,t), \tilde\varphi(\cdot,t), \bar n \in \mathcal{D}^l$ and the identifications* (12)*,* (14) *and* (15)*,*

$$S_{G_l^N}[\varphi_i, \tilde\varphi_i] = S[\varphi, \tilde\varphi]\big|_{\mathcal{D}^l}. \tag{16}$$

Proof. For $g \in \mathcal{D}^l$, $\int_{\mathbb{Z}_p^N} g\, d^N x = p^{-Nl} \sum_{i\in G_l^N} g(x_i)$ exactly. This turns the terms $\tilde\varphi_i \partial_t \varphi_i$, $\alpha\tilde\varphi_i\varphi_i$ and $\bar n_i \tilde\varphi_i(0)$ into the corresponding integrals. The product $\tilde\varphi\varphi$ lies in $\mathcal{D}^l$, and translation invariance of the Haar measure gives $w \star \mathcal{D}^l \subset \mathcal{D}^l$ with $(w\star g)(x_i) = \sum_j p^{-Nl} w_{ij}\, g(x_j)$. Hence the argument of f in Eq. (7) equals $\big(w \star (\tilde\varphi\varphi + \varphi) + I\big)(x_i)$, and $\sum_i \tilde\varphi_i\, f_{(l)}(\cdot) = p^{-Nl} \sum_i \tilde\varphi(x_i)\, f(\cdot) = \int \tilde\varphi\, f(\cdot)\, d^N x$. □

Proposition 1 is not specific to the present model. The fact that the discrete theory is an exact restriction of the continuum theory to a finite-dimensional space of test functions is a common feature of p-adic statistical field theories; see Refs. [23, 24, 28, 40]. We include the statement and its proof here because the rest of the paper depends on it: it is the core of the construction. For a hierarchically organized network, the passage to the continuum replaces the sums over G_l^N by integrals with respect to the Haar measure of $\mathbb{Z}_p^N$. The neurons become the points of $\mathbb{Z}_p^N$ (for $N = 1$, the leaves of an infinite

rooted tree). To reach $\mathbb{Q}_p^N$, use $p^{-l}\mathbb{Z}_p^N/p^l\mathbb{Z}_p^N$ in place of $\mathbb{Z}_p^N/p^l\mathbb{Z}_p^N$. For $l \geq 1$,

$$p^{-l}\mathbb{Z}_p^N = \bigsqcup_{i\in\widetilde{G}_l^N} \big(i + p^l\mathbb{Z}_p^N\big), \tag{17}$$

where $\widetilde{G}_l$ is the set of numbers of the form $i = i_{-l}p^{-l} + \cdots + i_0 + i_1 p + \cdots + i_{l-1}p^{l-1}$, with p-adic digits i_k, and $\widetilde{G}_l^N$ is the Cartesian product of N copies of $\widetilde{G}_l$, of cardinality p^{2Nl}. Multiplying $\widetilde{G}_l$ by p^l one obtains the finite rooted tree G_{2l}. The finite network is therefore of the same kind as before, with depth $2l$, and the balls $p^{-l}\mathbb{Z}_p^N$ exhaust $\mathbb{Q}_p^N$ as $l \to \infty$. Note that the points of $\mathbb{Q}_p^N$ are not the leaves of a rooted tree: since the balls grow without bound, there is no root. The points of $\mathbb{Q}_p^N$ are organized in a tree-like structure, which is the union of the rooted trees of the balls $p^{-l}\mathbb{Z}_p^N$.

In the Euclidean setting, the thermodynamic limit (infinite volume at fixed spacing) and the continuum limit (spacing tending to zero at fixed volume) are logically independent, and a lattice of spacing a approximates the continuum only as $a \to 0$. In Ref. [13] the two limits are taken together and not distinguished. In the p-adic setting they are a single limit. On $\mathbb{Z}_p^N$, the network G_l^N has p^{Nl} neurons and spacing p^{-l}, so $l \to \infty$ sends the number of neurons to infinity and the spacing to zero simultaneously, at fixed volume. The exhaustion by the balls $p^{-l}\mathbb{Z}_p^N$ makes the volume grow and the spacing shrink with the same parameter l. Moreover, by Proposition 1, every step of this limit is exact.

In general, the neuron space is a subset of $\mathbb{Q}_p^N$, determined by the support of w. Its geometry can be very general, but it is always hierarchical. In general one may consider a network topology consisting of a finite number of clusters, each of them a finite rooted tree, with connections between the clusters. In the simulations of Secs. VIII A and IX we use a single finite rooted tree, which is acceptable and convenient.

A hierarchical network is not a tree. A tree is a particular architecture: a graph whose edges are the connections. Hierarchical architectures are tree-like structures, topologies made of trees, such as a single rooted tree ($\mathbb{Z}_p^N$), a union of rooted trees with no root ($\mathbb{Q}_p^N$), or a finite number of connected clusters. In them the trees organize the neurons; they are not the connections. The neurons are the leaves, the internal vertices are modules (balls), and the couplings (11) join every pair of neurons, with a strength fixed by the level of their nearest common ancestor [Fig. 2(a)]. In particular, the model is not directed percolation on a Cayley tree (Bethe lattice). There, the units sit on all the vertices, activity spreads along the edges, and the number of sites within graph distance d grows exponentially in d. That lattice is effectively infinite-dimensional, and DP on it is mean-field. Here the ball of radius p^k has Haar volume p^{kN}, so the neuron space has finite dimension N. This is why the upper critical dimension $N_c = 2\sigma$ of Sec. V is finite, and why fluctuation effects appear for $N < 2\sigma$.

There is also a structural difference from the Euclidean case. On $a\mathbb{Z}^N$ the lattice action agrees with the continuum action only as $a \to 0$, because Riemann sums are approximations and convolution does not preserve piecewise-constant functions. On G_l^N the identity (16) holds *for every* l. The discrete theory is the continuum theory restricted to the finite-dimensional space $\mathcal{D}^l$.

B. The continuum action

By the results of Sec. III A, in the limit $l \to \infty$, the discrete action $S_{G_l^N}[\varphi_i, \tilde{\varphi}_i]$ tends to the continuous action of the p-adic effective spike model

$$\begin{aligned} S[\varphi,\tilde{\varphi}] = \int_{\mathbb{Q}_p^N} d^N x \int_0^\infty dt \Big\{ &\tilde{\varphi}\,\partial_t\varphi + \alpha\tilde{\varphi}\varphi \\ &- \tilde{\varphi}\, f\big(w \star (\tilde{\varphi}\varphi + \varphi) + I\big)\Big\} \\ - \int_{\mathbb{Q}_p^N} d^N x\, \bar{n}(x)\, \tilde{\varphi}(x,0), \end{aligned} \tag{18}$$

with $(w \star g)(x) = \int_{\mathbb{Q}_p^N} w(\|x - y\|_p)\, g(y)\, d^N y$. Here $\bar{n}(x) \geq 0$ is the *initial density of effective spikes*. The initial state is uncorrelated, and the number of effective spikes in a small ball B is Poisson distributed with mean $\int_B \bar{n}(x)\, d^N x$. It is the continuum limit of the discrete means $\bar{n}_i$ of Eq. (8), via $\bar{n}_i = p^{-Nl}\bar{n}(x_i)$ [Eq. (14)]. In particular $\langle\varphi(x,0)\rangle = \bar{n}(x)$, so $\bar{n}$ is the initial condition of the mean field (Sec. III C). The generating functional, Feynman rules, effective action and loop expansion of Secs. III–IV of Ref. [13] are formally identical.

C. Mean-field theory: the Wilson–Cowan equation

As in Ref. [13], mean-field theory is the zero-loop approximation of the theory, i.e. the lowest order of the steepest-descent evaluation of the path integral. In field theory this order is usually called *tree level*, since it corresponds to Feynman diagrams without closed loops; we avoid this name, because in this paper "tree" always refers to the hierarchical organization of the neurons. At this order the moments are the solutions of the classical equations of motion $\delta S/\delta\tilde{\varphi} = 0$, $\delta S/\delta\varphi = 0$, and all loop corrections are neglected. We write $a(x,t) = \langle\varphi(x,t)\rangle$ and $\tilde{a}(x,t) = \langle\tilde{\varphi}(x,t)\rangle$ for the zero-loop means.

The initial-state term of Eq. (18) is not integrated over time. To treat it on the same footing as the other terms, we write it as a source localized at $t = 0$:

$$\begin{aligned} &- \int_{\mathbb{Q}_p^N} d^N x\, \bar{n}(x)\, \tilde{\varphi}(x,0) \\ &\quad = - \int_{\mathbb{Q}_p^N} d^N x \int_0^\infty dt\; \bar{n}(x)\, \delta(t)\, \tilde{\varphi}(x,t). \end{aligned} \tag{19}$$

Here the time integral runs over $[0,\infty)$, as in all the actions of this paper, and $\delta(t)$ is understood as a unit mass

at the endpoint $t = 0$, i.e. $\int_0^\infty \delta(t)\, g(t)\, dt = g(0)$. The equation $\delta S/\delta\varphi = 0$ is linear and homogeneous in $\tilde\varphi$. With the final condition inherited from the Doi shift, its solution is $\tilde a = 0$, reflecting conservation of probability [cf. Eq. (5) of Ref. [13]]. Setting $\tilde\varphi = \tilde a = 0$ in $\delta S/\delta\tilde\varphi(x,t) = 0$ kills every term in which the derivative does not act on the explicit prefactor $\tilde\varphi$ in front of f. The result is

$$\partial_t a(x,t) = -\alpha\, a(x,t) + f\big((w \star a)(x,t) + I\big) + \bar n(x)\, \delta(t), \tag{20}$$

with $a(x,t) = 0$ for $t < 0$. Equation (20) is the Wilson–Cowan equation of the effective spike model, written with the initial state as a source. In the present setting the convolution is taken over $\mathbb{Q}_p^N$ with respect to the Haar measure, and we call it the *hierarchical Wilson–Cowan equation*; deterministic equations of this type were studied in Ref. [12].

The term $\bar n\, \delta(t)$. The source $\bar n(x)\delta(t)$ encodes the initial state. Since $a = 0$ for $t < 0$, integrating Eq. (20) over a small interval around $t = 0$ gives that a jumps from 0 to $\bar n(x)$ at $t = 0$, and for $t > 0$ the source vanishes. Therefore, for $t \geq 0$, Eq. (20) is equivalent to the Cauchy problem

$$\partial_t a = -\alpha a + f(w \star a + I) \quad (t > 0), \qquad a(x,0) = \bar n(x), \tag{21}$$

which is the form given in Eq. (44) of Ref. [13], with the initial condition $a(x,0) = \bar n(x)$, and the form used in the rest of the paper. The condition $a = 0$ for $t < 0$ requires $f(I) = 0$. From now on, we assume that $f(0) = 0$ and $I = 0$. In Ref. [13] the external input I is kept general and f is expanded about I. The condition $f(I) = 0$ is not stated there; it is only implicit. Section IV C of Ref. [13] considers "perturbations around a homogeneous equilibrium solution" and speaks of "the absorbing state", and Sec. V describes the transition "from the absorbing (nonfluctuating) fully quiescent state to a spontaneously active state". A quiescent absorbing state exists only if $f(I) = 0$.

Mean field as an approximation of the exact equation for the mean. Equivalently, taking the expectation of the operator form of the master equation gives the exact equation $\partial_t\langle\varphi\rangle = -\alpha\langle\varphi\rangle + \langle f(w \star \varphi + I)\rangle$ for $t > 0$ [Eqs. (111)–(112) of Ref. [25]]. This is not a closed equation for the mean $\langle\varphi\rangle$: since f is nonlinear, the term $\langle f(w \star \varphi + I)\rangle$ depends on all the higher moments of φ, whose equations involve still higher moments, and so on (Appendix A 2). Mean-field theory is the approximation

$$\langle f(w \star \varphi + I)\rangle \approx f(w \star \langle\varphi\rangle + I), \tag{22}$$

in which the average of f is replaced by f of the average. It amounts to assuming that all moments factorize into products of the mean, i.e. that fluctuations are neglected, and it turns the exact equation into a closed equation for the mean, which is Eq. (21) [Eq. (113) of Ref. [25]]. The loop expansion of Secs. III D and VII replaces this uncontrolled approximation by a systematic expansion: the one-loop correction adds to the right-hand side of Eq. (21) a term coupling a to the two-point correlation [Eq. (82)].

Discrete version. On the finite network G_l^N, with the identifications (14) and (15), Eq. (20) becomes the system $\dot a_i = -\alpha a_i + f\big(p^{-Nl}\sum_j w_{ij}a_j + I\big) + p^{Nl}\bar n_i\, \delta(t)$, $i \in G_l^N$, for the densities $a_i = p^{Nl}\langle n_i\rangle$. The space $\mathcal{D}^l$ is invariant under the convolution with w and under the pointwise action of f, so this system is exactly the restriction of Eq. (20) to $\mathcal{D}^l$: if $\bar n \in \mathcal{D}^l$, then $a(\cdot,t) \in \mathcal{D}^l$ for all t. Existence, uniqueness and positivity of the solutions for integrable kernels were established in Ref. [12]. For general initial data the discrete solutions approximate the continuous one on every finite time interval, with an error controlled by the error in the initial data. The rigorous construction of path integrals for actions involving W-operators, as limits of finite-dimensional ones, was carried out in Ref. [48].

D. Free and interacting actions

Reference [13] splits the action into a free part S_{free} and an interacting part S_i [their Eqs. (14)–(15)] but gives the two parts only in words. Since the propagators used below depend on this choice, we state it explicitly. There are two distinct splits: one about the vacuum $\varphi = \tilde\varphi = 0$ (used for the perturbative expansion of Sec. III of Ref. [13]), and one about the mean field (used for the loop expansion of Sec. IV). They coincide only when the mean field vanishes, $a \equiv 0$, i.e. when the network is at the absorbing state $\varphi = 0$, which requires $f(I) = 0$ [Eq. (39)].

The firing-rate function. The firing-rate function $f : \mathbb{R} \to \mathbb{R}$ is a function with a convergent Taylor expansion on the whole real line, nonnegative for $s \geq 0$, and satisfying $f(0) = 0$, $f'(0) > 0$ and $f''(0) < 0$. Nonnegativity is required only for $s \geq 0$: since $w \geq 0$ and $I = 0$, the input of a neuron is never negative (see Remark 1). A prototype of such a function is

$$f(s) = f_{\max}\big(1 - e^{-s/s_0}\big), \qquad f_{\max}, s_0 > 0. \tag{23}$$

This function is real analytic on the whole real line, so the Taylor expansion (25) below converges for every argument and the vertex expansion is well defined. It also provides the smoothness that B-C require for the renormalization argument [Sec. V of Ref. [13]]. Its derivatives at the origin are $f^{(n)}(0) = (-1)^{n+1} f_{\max}/s_0^n$. In particular,

$$f(0) = 0, \quad f'(0) = \frac{f_{\max}}{s_0} > 0, \quad f''(0) = -\frac{f_{\max}}{s_0^2} < 0. \tag{24}$$

These are exactly the conditions used in the critical analysis: the condition $f(0) = 0$ gives an absorbing state [Eq. (39)]; $f'(0) > 0$ gives branching and propagation; and $f''(0) < 0$ (saturation) gives the relative minus sign between the cubic vertices of Eq. (59), which B-C identify

as necessary for the DP fixed point. With these values, the distance to criticality is $\mu = \alpha - f'(0)\,\hat{w}(0) = \alpha - f_{\max}\hat{w}(0)/s_0$, and B-C's rescaling of the fields [Eq. (58) of Ref. [13]] involves the factor

$$\Big(\frac{2f'(0)}{\hat{w}(0)\,|f''(0)|}\Big)^{1/2} = \Big(\frac{2s_0}{\hat{w}(0)}\Big)^{1/2}.$$

Remark 1 (Firing-rate function versus activation function). In neural-network and rate models, an *activation function* (a sigmoid, tanh, ReLU, ...) maps a unit's (or node's) input to its output (or state), determined by the inputs connected to the node and their weights. That output can take negative values; e.g., tanh represents deviations from a baseline rate in the symmetric range $(-1, 1)$. The function f in the effective spike model differs in three ways.

1. *It is a transition rate, not a state.* In the master equation (6), $f(s_i)$ is the probability per unit time that neuron i emits another effective spike. It has units of inverse time and must be nonnegative wherever it is evaluated. The state of the neuron is the spike count n_i, not f.

2. *It becomes an activation function only in mean field.* In the Wilson–Cowan equation (21), obtained at zero-loop order, f plays the role of a deterministic activation function acting on the mean input $w\star a + I$. Beyond zero-loop order, f enters through all its derivatives, as the vertices $V_{n,m}$ of Eq. (29), and it fixes the statistics of the fluctuations, not only the mean response.

3. *Criticality constrains it at a point.* The requirements $f(I) = 0$, $f'(I) > 0$ and $f''(I) < 0$ concern the behavior of f at the operating point, not its global shape. The hyperbolic tangent fails them. Nonnegativity forces $I = 0$, where $\tanh''(0) = 0$. The cubic aggregation vertex then vanishes at zero-loop order and the leading saturating term is quartic ($\tanh'''(0) = -2$), which is the setting of tricritical directed percolation rather than DP.

Nonnegativity and analyticity together have a consequence. A real analytic function with $f(0) = 0$ and $f'(0) > 0$ is negative for small $s < 0$, so the inputs must satisfy $s \geq 0$. We therefore take $w \geq 0$ and $I = 0$. As B-C note, inhibition cannot be distinguished from the decay rate α in the effective spike model [13], so net inhibition is represented through α. Models with negative weights require a nonnegative extension of f to $s < 0$ (e.g. $f = 0$ there), which is not analytic at $s = 0$.

Expansion about the vacuum. Expand the firing-rate function about $I = 0$:

$$f\big(w\star(\tilde{\varphi}\varphi+\varphi)\big) = \sum_{n\geq 1}\frac{f^{(n)}(0)}{n!}\Big(w\star(\tilde{\varphi}\varphi)+w\star\varphi\Big)^n. \quad (25)$$

The sum starts at $n = 1$ because $f(0) = 0$. The *free action* is the part of Eq. (18) bilinear in φ and $\tilde{\varphi}$:

$$\begin{aligned} S_{\text{free}}[\varphi,\tilde{\varphi}] &= \int_{\mathbb{Q}_p^N} d^N x \int_0^\infty dt\; \tilde{\varphi}\,(\partial_t + \alpha)\,\varphi \\ &\quad - f'(0)\int_{\mathbb{Q}_p^N} d^N x \int_0^\infty dt\; \tilde{\varphi}\,(w\star\varphi) \\ &\equiv \int_{\mathbb{Q}_p^N} d^N x \int_0^\infty dt\; \tilde{\varphi}\,L\,\varphi, \end{aligned} \quad (26)$$

with

$$L = \partial_t + \alpha - f'(0)\,w\star. \quad (27)$$

The derivation of Eq. (26) *uses* $f(0) = 0$. The zeroth-order term of the expansion (25), $f(0)\int\tilde{\varphi}$, vanishes [Eq. (24)], and $f'(0) = f_{\max}/s_0$ for the prototype (23). The condition $f(0) = 0$ is also what makes the vacuum a solution: the classical equations of motion are $\delta S/\delta\varphi = 0$ and $\delta S/\delta\tilde{\varphi} = 0$ (Sec. III C); for $\tilde{\varphi} = 0$ they reduce to Eq. (20), and in the absence of initial spikes ($\bar{n} = 0$) the vacuum $\varphi = \tilde{\varphi} = 0$ solves them if and only if $f(0) = 0$. Then the action splits as $S = S_{\text{free}} + S_i$, with S_{free} the purely bilinear form (26), and S_i contains no term linear in $\tilde{\varphi}$ other than the initial state. If $f(0) \neq 0$, the action contains in addition the source $-f(0)\int\tilde{\varphi}$. The vacuum is then not stationary, and Eq. (26) is no longer the free action for fluctuations about a solution. This source cannot be absorbed into a normalization constant: after the Doi shift every term of Eq. (18) contains at least one factor of $\tilde{\varphi}$, so $S[\varphi, 0] = 0$ and $Z[0, 0] = 1$ (conservation of probability), and a term linear in $\tilde{\varphi}$ shifts the mean activity. In that case one must expand about the mean field instead (see below).

The *interacting action* is the remainder, $S_i = S - S_{\text{free}}$. Using the binomial theorem in Eq. (25),

$$S_i[\varphi,\tilde{\varphi}] = \sum_{n\geq 1}{\sum_{m=0}^{n}}' V_{n,m}[\varphi,\tilde{\varphi}] - \int_{\mathbb{Q}_p^N} d^N x\,\bar{n}(x)\,\tilde{\varphi}(x,0), \quad (28)$$

$$\begin{aligned} V_{n,m} &= -\frac{f^{(n)}(0)}{n!}\binom{n}{m} \\ &\quad\times\int_{\mathbb{Q}_p^N} d^N x\int_0^\infty dt\;\tilde{\varphi}\,\big(w\star(\tilde{\varphi}\varphi)\big)^m\big(w\star\varphi\big)^{n-m}. \end{aligned} \quad (29)$$

The prime on the sum means that the term $(n, m) = (1, 0)$ is omitted, since it is already contained in S_{free}. The vertex $V_{n,m}$ has n incoming legs (factors of φ) and $m + 1$ outgoing legs (factors of $\tilde{\varphi}$). This is the vertex structure described in Ref. [13], where "the $n = 1$, $m = 0$ term has already been incorporated into the propagator." The only term of Eq. (28) that is a source rather than an interaction is the initial-state term $-\int\bar{n}\,\tilde{\varphi}(\cdot, 0)$, which has no time integral and only outgoing legs (the bold dots in the figures of Ref. [13]). The term $n = 0$

of the expansion, $-f(0)\int d^N x \int_0^\infty dt\,\tilde\varphi$, would be a second source describing spontaneous firing (see above); it is absent because $f(0)=0$.

Fourier conventions on $\mathbb{Q}_p^N$. We use $\hat g(\xi)=\int_{\mathbb{Q}_p^N}\chi_p(\xi\cdot x)\,g(x)\,d^N x$, with the Haar measure normalized as in Sec. II. This measure is self-dual, so the inverse is $g(x)=\int_{\mathbb{Q}_p^N}\chi_p(-\xi\cdot x)\,\hat g(\xi)\,d^N\xi$ and the convolution theorem reads $\widehat{w\star g}=\hat w\,\hat g$. The Dirac distribution $\delta(x)$ on $\mathbb{Q}_p^N$ has $\hat\delta=1$. For a real radial kernel w, $\hat w$ is real and radial, because w is invariant under $x\mapsto -x$.

Zero-loop propagator of $S_{\rm free}$. The propagator $\Delta(x,t)=\langle\varphi(x,t)\tilde\varphi(0,0)\rangle_{S_{\rm free}}$ is the Green's function of L: $L\Delta=\delta(x)\,\delta(t)$ [Eq. (15) of Ref. [13]]. In Fourier variables $\partial_t\hat\Delta=-\lambda_0(\xi)\hat\Delta+\delta(t)$. Hence

$$\hat\Delta(\xi,t)=\theta(t)\,e^{-\lambda_0(\xi)t},\qquad \lambda_0(\xi)=\alpha-f'(0)\,\hat w(\xi),\tag{30}$$

where θ denotes the Heaviside step function, $\theta(t)=1$ for $t>0$ and $\theta(t)=0$ for $t<0$, with the convention $\theta(0)=0$. This convention is the one of Ref. [13]. It follows from normal ordering of the operators in the Hamiltonian, i.e. it is the Itô prescription, and it implies that the propagator does not connect a field to itself at equal times.

A note on the "response field". The name of $\tilde\varphi$ can be misread. The field $\tilde\varphi$ is not the response of the network: it marks the point where a stimulus is applied, and φ the point where the response is measured. Injecting extra spikes at rate $h(x',t')$ adds the term $\int h\,\tilde\varphi$ to $-S$, so h enters as a source of $\tilde\varphi$, and $\delta\langle\varphi(x,t)\rangle/\delta h(x',t')=\langle\varphi(x,t)\tilde\varphi(x',t')\rangle=\Delta(x-x',t-t')$, which vanishes unless $t>t'$. In particular, $\langle\varphi(x',t')\tilde\varphi(x,t)\rangle=0$ for $t>t'$, since an average in which $\tilde\varphi$ carries the latest time vanishes.

The propagator in position space. Since w is integrable, the Riemann–Lebesgue theorem implies that $\hat w(\xi)\to 0$ as $\|\xi\|_p\to\infty$, so $\lambda_0(\xi)\to\alpha$. Hence, for each fixed $t>0$, $e^{-\lambda_0(\xi)t}\to e^{-\alpha t}\neq 0$ as $\|\xi\|_p\to\infty$: as a function of ξ, $e^{-\lambda_0(\xi)t}$ tends to a nonzero constant and is therefore not integrable over $\mathbb{Q}_p^N$, and the inverse Fourier transform must be taken in the sense of distributions:

$$\Delta(x,t)=\theta(t)\,\mathcal{F}^{-1}_{\xi\to x}\big(e^{-\lambda_0(\xi)t}\big),\tag{31}$$

where $\mathcal{F}$ is the Fourier transform on the space $\mathcal{D}'(\mathbb{Q}_p^N)$ of distributions on $\mathbb{Q}_p^N$. Writing $e^{-\lambda_0(\xi)t}=e^{-\alpha t}\big[1+\big(e^{f'(0)t\,\hat w(\xi)}-1\big)\big]$ and using $\mathcal{F}^{-1}(\hat w^{\,n})=w^{\star n}$ gives

$$\Delta(x,t)=\theta(t)\,e^{-\alpha t}\big[\delta(x)+K_t(x)\big],\qquad K_t=\sum_{n\ge1}\frac{\big(f'(0)\,t\big)^n}{n!}\,w^{\star n}.\tag{32}$$

The series converges in $L^1(\mathbb{Q}_p^N)$, since $\|w^{\star n}\|_1\le\|w\|_1^n$, so K_t is an integrable function. The two terms of $\Delta(x,t)$ in Eq. (32) have a direct interpretation. The δ term is the injected spike decaying in place at rate α. K_t sums the chains of n synaptic transmissions. Because $e^{f'(0)t\hat w}-1$ is radial, K_t can be computed explicitly as a sum over spheres (Proposition 4 of Appendix F). At large hierarchical distance, the propagator is proportional to the kernel w: it decays with the distance exactly as the coupling between neurons does,

$$\Delta(x,t)\simeq\theta(t)\,f'(0)\,t\,e^{-\lambda_0(0)t}\,w(\|x\|_p),\qquad \|x\|_p\to\infty,\tag{33}$$

for nonnegative, radially nonincreasing w (Proposition 4). The p-adic numbers appear in Eqs. (30)–(33) only through the fact that the convolution and the Fourier transform are taken over $\mathbb{Q}_p^N$. The decomposition (32) holds in the Euclidean theory as well; what is specific to $\mathbb{Q}_p^N$ is the explicit radial evaluation and the exact tail (33). Figure 3 compares the two propagators. The comparison already suggests why the B-C theory in its diffusion limit is not compatible with the small-world property: its propagator is cut off at the diffusion length, so that a spike has no influence on distant neurons at short times, whereas the p-adic propagator reaches all hierarchical distances at every time. A power-law tail is necessary for the small-world property but not sufficient; the question is examined in Sec. IX.

Ultraviolet behavior. The δ term in Eq. (32) controls the ultraviolet behavior. A product of propagators at coinciding points, such as $\Delta(x-y,\tau)^2$ in a bubble diagram, contains δ^2 and is not defined. Correspondingly, the momentum integral $\int e^{-\lambda_0(q)\tau}e^{-\lambda_0(\eta-q)\tau}\,d^N q$ diverges, because its integrand tends to $e^{-2\alpha\tau}$ as $\|q\|_p\to\infty$. Such diagrams are finite only when the vertices supply factors of $\hat w$, as in the loop terms of B-C, where every vertex acts through $w\star$, or with an ultraviolet cutoff. The finite network G_l^N provides the cutoff naturally: on $\mathcal{D}^l$ the distribution δ is replaced by $p^{Nl}\,\Omega(p^l\|x\|_p)$, and momenta are restricted to $\|\xi\|_p\le p^l$. This makes precise B-C's remark that the integrals are "well behaved in the ultraviolet because of the weight function, if not by a neural lattice cutoff" [13].

Expansion about the mean field. For the loop expansion, Ref. [13] shifts the fields by their means, $\varphi\to a+\varphi$, $\tilde\varphi\to\tilde a+\tilde\varphi$, with $a(x,t)=\langle\varphi(x,t)\rangle$ and $\tilde a=0$. After the shift, φ and $\tilde\varphi$ denote the fluctuations about the mean. The action then takes the form $S=S_0[a]+S_F[a;\varphi,\tilde\varphi]$ [Eqs. (33)–(35) of Ref. [13]], where S_0 is the action evaluated at the mean and S_F, the action of the fluctuations, contains all the terms that depend on φ and $\tilde\varphi$. Let

$$s(x,t)=(w\star a)(x,t)+I.\tag{34}$$

The argument of f becomes $s+w\star\varphi+w\star(a\tilde\varphi)+w\star(\tilde\varphi\varphi)$. We expand f in a Taylor series about s (not about I), multiply by $-\tilde\varphi$, and collect the terms of S_F according to their degree in the fluctuation fields φ and $\tilde\varphi$:

1. The term linear in $\tilde\varphi$ collects the drift and the initial-state term, Eq. (19): $\int d^N x\int_0^\infty dt\;\tilde\varphi\big[\partial_t a+\alpha a-f(s)-\bar n\,\delta(t)\big]$. It vanishes at zero-loop order, because a then solves Eq. (20). In this way the initial-state term is absorbed into the equation that

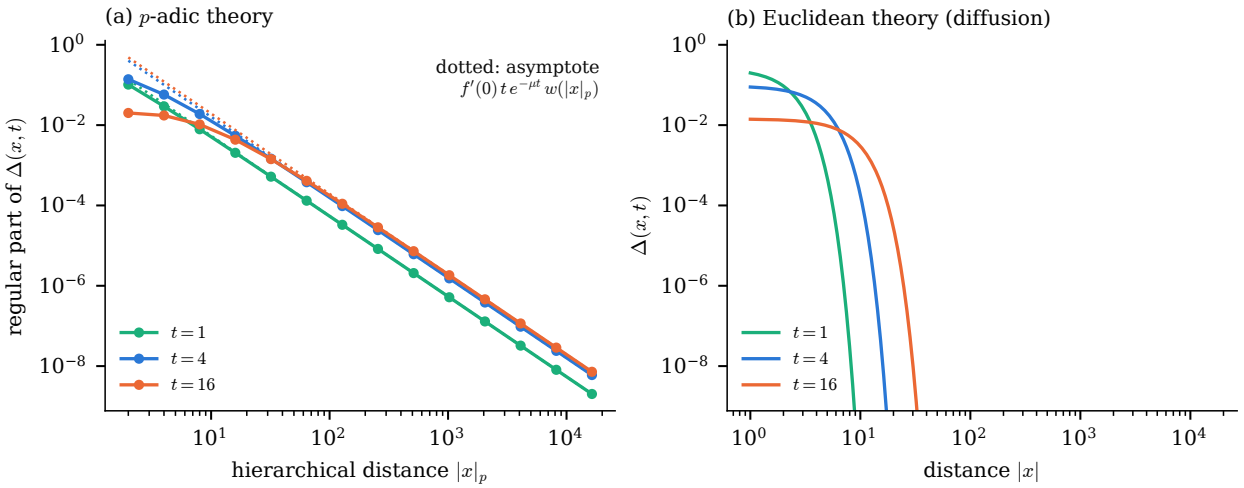


FIG. 3. Zero-loop propagator at three times. (a) Regular part $e^{-\alpha t}K_t(x)$ of the p-adic propagator (32), computed from the explicit formula (F5) of Appendix F, for $p=2$, $N=1$, a kernel that is constant for $|x|_p \le 1$ and decays as $|x|_p^{-(1+\sigma)}$ with $\sigma=1$ beyond, $\alpha=1$ and $f'(0)w_0=0.9$ (so $\mu=0.1$). Dotted lines: the asymptote (33). The computation is for the continuum theory on $\mathbb{Q}_p$, not for a finite tree: the series of Eq. (F5) are summed over 300 levels, which is numerically exact. The neuron space $\mathbb{Q}_p$ is the disjoint union of infinitely many unit balls; it is not compact, and it is not a regional network in the sense of Lemma 1, which requires finitely many regions. Restricting it to the ball of radius 2^{14} centered at the origin, $\{|x|_2 \le 2^{14}\} = 2^{-14}\mathbb{Z}_2$, which is compact and is the disjoint union of the 2^{14} unit balls $i+\mathbb{Z}_2$, $i \in 2^{-14}\mathbb{Z}_2/\mathbb{Z}_2$, with the same α and $f'(0)$, changes the plotted values by less than 2×10^{-5} and only cuts off the tail at the size of the network. The points are the values at $|x|_p = 2^k$, $k=1,\dots,14$, i.e. at the 14 levels of the hierarchy above the unit ball $\mathbb{Z}_p$, on which the kernel is constant. (b) The Euclidean propagator of the B-C theory on the real line, $e^{-\mu t}(4\pi D t)^{-1/2}e^{-x^2/4Dt}$, with the same μ and $D=1$, for $1 \le |x| \le 2^{14}$. In both panels the unit of distance is the short-distance scale of the model (the radius of the unit ball in (a), the length $\sqrt{D/\alpha}$ in (b)), and a ball of radius p^k has volume p^k in $\mathbb{Q}_p$ (radii are powers of p), while an interval of radius r has length $2r$ in $\mathbb{R}$, so the two horizontal scales measure comparable numbers of neurons. The p-adic propagator has a power-law tail at every time; the Euclidean one is cut off at the diffusion length $\sqrt{4Dt}$.

determines a: it cancels against the other terms of the bracket and does not remain in the shifted action. After the shift, $\bar{n}$ no longer appears in the action as a source; it enters only through a, i.e. through $s = w\star a + I$.

2. The terms of second degree, which we denote by $S_F^{(2)}$, are

$$S_F^{(2)} = \int d^N x \int_0^\infty dt\, \Big\{ \tilde\varphi\,(\partial_t+\alpha)\varphi - f'(s)\,\tilde\varphi\,(w\star\varphi)\Big\} - \int d^N x \int_0^\infty dt\; f'(s)\,\tilde\varphi\, w\star(a\tilde\varphi). \tag{35}$$

The first line defines the response propagator of the shifted theory, $\Delta_a(x,t;x',t') = \langle\varphi(x,t)\tilde\varphi(x',t')\rangle$ [Eq. (47) of Ref. [13]]:

$$\partial_t\Delta_a(x,t;x',t') = -\alpha\Delta_a(x,t;x',t') + \delta(x-x')\,\delta(t-t') + f'\big(s(x,t)\big)\,(w\star\Delta_a)(x,t;x',t'), \tag{36}$$

where $w\star$ acts on the first spatial argument. When a is not homogeneous, i.e. when $a(x,t)$ depends on the position x, the coefficient $f'(s(x,t))$ depends on x and Δ_a is not translation invariant: it is not a function of $x-x'$ alone, which is why it is written as a function of both points. Equation (36) contains no term $\bar{n}\,\delta(t)$, for two reasons. First, the initial-state term has already been absorbed into the equation for the mean [item 1]. Second, Δ_a is a *response function*: it is the solution of the linearization of Eq. (20) about a, driven by a unit impulse $\delta(x-x')\delta(t-t')$ applied at (x',t'). The initial state is part of the background a about which one linearizes, not of the perturbation. The inhomogeneous term $\bar{n}\,\delta(t)$ of Eq. (20) drops out of the linearization because it does not depend on a. The second line of Eq. (35) has two factors of $\tilde\varphi$ and no φ. It is not part of the propagator. Following Ref. [13], we treat it as the $(2,0)$ vertex

$$-f'(s(x))\,\tilde\varphi(x)\int_{\mathbb{Q}_p^N} w(\|x-y\|_p)\,a(y)\,\tilde\varphi(y)\,d^N y, \tag{37}$$

which is Eq. (54) of B-C; it generates the zero-loop two-point correlation [their Eq. (55)] and the source term in Eq. (83) below.

3. Cubic and higher terms, such as $-f'(s)\,\tilde\varphi\, w\star(\tilde\varphi\varphi)$ and $-\frac{1}{2}f''(s)\,\tilde\varphi\,(w\star\varphi)^2$, are the vertices of the loop expansion. The latter produces the one-loop term in Eq. (82).

Homogeneous states and the absorbing state. For a homogeneous stationary mean field $a(x,t)=a^*$ we have $s=\hat w(0)a^*+I$, where a^* solves $\alpha a^*=f(\hat w(0)a^*+I)$. Eq. (36) then gives

$$\hat\Delta_{a^*}(\xi,t)=\theta(t)\,e^{-\lambda_{a^*}(\xi)t},\qquad \lambda_{a^*}(\xi)=\alpha-f'\big(\hat w(0)a^*+I\big)\,\hat w(\xi). \tag{38}$$

In general $\lambda_{a^*}\neq\lambda_0$: the propagator of the shifted theory is not the propagator (30) of the free action $S_{\rm free}$ of Eq. (26), which was obtained by expanding about the vacuum. The two propagators coincide when $a^*=0$. However, $a^*=0$ is a stationary state only if

$$f(I)=0, \tag{39}$$

in which case there is no constant source and $\varphi=0$ is an absorbing state. We recall that an absorbing state is a state that the dynamics can reach but cannot leave: if no neuron has an effective spike and $f(I)=0$, no spike is ever created, and the network remains silent forever. If $f(I)>0$, the constant source $-f(I)\int d^Nx\int_0^\infty dt\,\tilde\varphi$ drives spontaneous activity, there is no absorbing state, and the DP-type analysis of Secs. V–VIII does not apply directly. Condition (39) is implicit in the critical analysis of Ref. [13], which treats "the absorbing (nonfluctuating) fully quiescent state". We impose it explicitly whenever we discuss criticality.

IV. LINEAR RESPONSE: p-ADIC DIFFUSION

We linearize about a homogeneous stationary state a^*. By Sec. III D, the relevant propagator is that of the shifted theory, Eq. (38). For the critical analysis below we take the absorbing state $a^*=0$, which requires $f(I)=0$ [Eq. (39)]; for $I=0$ this is the condition $f(0)=0$. Then

$$\hat\Delta(\xi,t)=\theta(t)\,e^{-\lambda(\xi)t},\qquad \lambda(\xi)=\lambda_0(\xi)=\alpha-f'(0)\,\hat w(\xi), \tag{40}$$

and Eq. (40) is also the zero-loop propagator (30) of the free action $S_{\rm free}$.

In real space the same result is obtained by separation of variables, in any dimension N. The linearized equation is

$$\partial_t u(x,t)+\alpha\,u(x,t)-f'(0)\,(w\star u)(x,t)=0, \tag{41}$$

which is $Lu=0$ with the operator L of Eq. (27). We look for solutions of the form $u(x,t)=\psi_{\gamma nj}(x)\,T(t)$, where $\psi_{\gamma nj}$ is the wavelet defined in Eq. (4). By Eq. (5),

$$\partial_t T(t)+\alpha\,T(t)-f'(0)\,\hat w\big(p^{1-\gamma}\big)\,T(t)=0, \tag{42}$$

so that $T(t)=T(0)\,e^{-\lambda(p^{1-\gamma})t}$, where $\lambda(p^{1-\gamma})$ denotes the value of $\lambda(\xi)$ for $\|\xi\|_p=p^{1-\gamma}$. Since the wavelets form an orthonormal basis of $L^2(\mathbb{Q}_p^N)$, the solution of the Cauchy problem for Eq. (41) with initial datum $u(x,0)=u_0(x)\in L^2(\mathbb{Q}_p^N)$ is

$$u(x,t)=\sum_{\gamma,n,j}c_{\gamma nj}\,e^{-\lambda(p^{1-\gamma})t}\,\psi_{\gamma nj}(x),\qquad t\ge0, \tag{43}$$

with $c_{\gamma nj}=\int_{\mathbb{Q}_p^N}u_0(x)\,\overline{\psi_{\gamma nj}(x)}\,d^Nx$. The linear response thus decomposes into independent *hierarchical modes*, one for each ball, with relaxation rate set by the level of the ball.

A. The operator selected by the connectivity

Recall that $\mu=\alpha-f'(0)\,\hat w(0)$ is the bare distance to criticality (B-C denote this bare value by μ_0 and reserve μ for the renormalized parameter), and introduce the operator

$$\boldsymbol{W}\phi(x)=f'(0)\int_{\mathbb{Q}_p^N}\big(\phi(x)-\phi(x-y)\big)\,w(\|y\|_p)\,d^Ny \tag{44}$$

for $\phi\in\mathcal{D}(\mathbb{Q}_p^N)$. This operator is pseudodifferential with radial symbol $A_w(\xi)=A_w(\|\xi\|_p)$:

$$\boldsymbol{W}\phi(x)=\mathcal{F}^{-1}_{\xi\to x}\big(A_w(\|\xi\|_p)\,\mathcal{F}_{x\to\xi}\phi\big), \tag{45}$$

where

$$A_w(\xi)=f'(0)\big(\hat w(0)-\hat w(\xi)\big)=f'(0)\int_{\mathbb{Q}_p^N}w(\|y\|_p)\,\big(1-\chi_p(\xi\cdot y)\big)\,d^Ny. \tag{46}$$

With these definitions, the relaxation rate $\lambda(\xi)=\lambda_0(\xi)$ of Eqs. (40) and (30) can be written exactly, for every $\xi\in\mathbb{Q}_p^N$, as

$$\lambda(\xi)=\mu+A_w(\xi). \tag{47}$$

Operators of this form, and the corresponding heat equations $\partial_t u+\boldsymbol{W}u=0$, are the W-operators studied in Refs. [36, 43]. Their symbols are nonnegative and radial, and the heat semigroups are Markovian. The free part of the action is therefore $\tilde\varphi(\partial_t+\mu+\boldsymbol{W})\varphi$ *with no approximation*. The linear dynamics of the p-adic B-C theory is the heat equation determined by the synaptic kernel w, not by an independent choice of "Laplacian".

The operator $\boldsymbol{W}$ of Eq. (44) is also defined over $\mathbb{R}^N$, replacing $\|y\|_p$ by the Euclidean norm $|y|$, and its symbol is given by Eq. (46) with $\chi_p(\xi\cdot y)$ replaced by $e^{iq\cdot y}$. Then $\lambda(q)=\mu+A_w(q)$ holds exactly, and at $\mu=0$ the linearized equation

$$\partial_t u(x,t)+\boldsymbol{W}u(x,t)=0,\qquad x\in\mathbb{R}^N,\ t\ge0, \tag{48}$$

is a nonlocal diffusion equation. B-C did not consider this equation; instead, they expand $\hat w(q)=\hat w(0)-\frac12 w_2q^2+\dots$. If w has a finite second moment $w_2=\int x_1^2\,w(|x|)\,d^Nx$, then $\hat w(q)=\hat w(0)-\frac12 w_2|q|^2+o(|q|^2)$,

and the symbol of $\boldsymbol{W}$ is $A_w(q) = D|q|^2 + o(|q|^2)$, with $D = \frac{1}{2}f'(0)\,w_2$. Since $|q|^2$ is the symbol of $-\nabla^2$, $\boldsymbol{W}$ and $-D\nabla^2$ have the same symbol to leading order as $q \to 0$; equivalently, expanding $\phi(x-y)$ in Eq. (44) to second order gives $\boldsymbol{W}\phi = -D\nabla^2\phi + \dots$ for ϕ that varies slowly on the range of w. In this sense the long-wavelength limit of $\boldsymbol{W}$ is a diffusion operator, with diffusion constant proportional to the second moment of w [B-C's diffusion length is $l_d^2 = D/\mu$, Eq. (52) of Ref. [13]]. Analyticity of $\hat{w}$ at the origin, which for $w \geq 0$ holds only if w decays exponentially, is not needed. B-C assume short-range connectivity and use the moments $w_2, w_4, \dots$ [Eqs. (50)–(52) of Ref. [13]]. Kernels with power-law tails, $w(x) \sim |x|^{-N-\sigma}$, never have an analytic Fourier transform: $\hat{w}(0) - \hat{w}(q)$ contains a term proportional to $|q|^\sigma$ ($|q|^\sigma \log|q|$ for even integer σ). For $\sigma > 2$ the second moment is finite, the term q^2 dominates, $|q|^\sigma$ is an irrelevant correction, and the critical behavior is that of DP. For $\sigma < 2$ the second moment is infinite, $\hat{w}(0) - \hat{w}(q) \propto |q|^\sigma$, the nonlocal term is relevant, and the critical behavior is that of DP with long-range spreading [44, 49]. The dividing line between local and nonlocal diffusion is thus $\sigma = 2$, the finiteness of the second moment, not analyticity. There is no such mechanism over $\mathbb{Q}_p^N$, because a nonconstant locally constant function has no Taylor expansion. There are infinitely many inequivalent heat equations on $\mathbb{Q}_p^N$, and each connectivity kernel selects one. The infrared behavior therefore depends on how $A_w(\xi)$ behaves as $\|\xi\|_p \to 0$. We distinguish the cases below.

Networks on a single ball. Suppose that w is supported in the ball $B_r^N = p^{-r}\mathbb{Z}_p^N$, with r the smallest possible integer. The support of w determines the neurons that are really connected: $w(\|x-y\|_p) = 0$ whenever x and y lie in distinct balls of radius p^r, so the neurons outside a ball of radius p^r do not interact with those inside it. We therefore take the network to be the ball B_r^N, and discard the neurons outside it. A function on B_r^N is identified with a function on $\mathbb{Q}_p^N$ by extending it by zero outside B_r^N, so that $L^\rho(B_r^N) \subset L^\rho(\mathbb{Q}_p^N)$ for $\rho = 1, 2$. This is possible because of the topology of $\mathbb{Q}_p^N$: B_r^N is open and compact, so the extension by zero preserves local constancy, and B_r^N is an additive subgroup, so $B_r^N + \operatorname{supp} w \subset B_r^N$ and the dynamics never leaves B_r^N. In this way networks with different supports can be treated within the same space $\mathbb{Q}_p^N$. The network on a ball is the case $K = B_r^N$ of Sec. IV C. In Fourier variables, $\hat{w}(\xi) = \hat{w}(0)$ for $\|\xi\|_p \leq p^{-r}$, since $\chi_p(\xi \cdot x) = 1$ on the support of w, so $A_w(\xi) = 0$ and $\lambda(\xi) = \mu$ on a whole ball of momenta. For a function ϕ supported in B_r^N, $\hat{\phi}$ is constant on this ball and equal to $\int \phi$. Hence the only mode of the network with $A_w = 0$ is its total activity, which is conserved at $\mu = 0$ (Sec. IV C). All the other modes, the wavelets supported in B_r^N, are damped, since A_w is bounded below by a positive constant on $\|\xi\|_p > p^{-r}$. There is no diverging correlation length, and the critical behavior is that of a single well-mixed population (mean field), which becomes sharp only in the limit of infinitely many neurons in the ball. A connected hierarchical network extending over several balls requires connections between them: constant couplings between regions (Lemma 1 below), or tails that extend over all levels of the hierarchy (Lemma 2).

Regional networks. A more realistic model couples the balls. Suppose that the neurons occupy a disjoint union of finitely many balls, at least two,

$$K = \bigsqcup_{a \in A} B_{l_a}^N(a), \qquad B_{l_a}^N(a) = a + p^{-l_a}\mathbb{Z}_p^N, \tag{49}$$

with A a finite index set of cardinality $|A| \geq 2$ as in Ref. [43]. Then K is compact and open. The case of infinitely many balls is different. For instance, $\mathbb{Q}_p^N$ is the disjoint union of the unit balls $a + \mathbb{Z}_p^N$, with a running over $\mathbb{Q}_p^N/\mathbb{Z}_p^N$; this neuron space is not compact, the matrix M below becomes infinite, and Lemma 1 does not apply. This is the continuum theory, treated separately. We abbreviate $B_a = B_{l_a}^N(a)$, the ball of radius p^{l_a} with center a. Each ball B_a represents a *region* of the cortex in which the spike model holds with constant parameters: a decay rate α_a, a firing-rate function f_a with $f_a(0) = 0$, $f_a'(0) > 0$ and $f_a''(0) < 0$, and a radial, integrable and continuous connectivity $w_a(\|x-y\|_p) \geq 0$ between neurons $x, y \in B_a$. Neurons in different regions are coupled with constant strength: $w(x,y) = c_{ab} \geq 0$ for $x \in B_a$, $y \in B_b$, $a \neq b$. The linearized dynamics is

$$\partial_t u(x,t) = -\alpha_a\, u(x,t) + f_a'(0) \int_K w(x,y)\, u(y,t)\, d^N y \tag{50}$$

for $x \in B_a$, where $w(x,y) = w_a(\|x-y\|_p)$ if $x, y \in B_a$. Transitions between regions are branching events whose offspring land in another region: the elementary events are those of Eq. (A1), with a block-structured connectivity K_{ij}. For $\alpha_a = \alpha$ and $f_a = f$, the model is the B-C model with block-structured connectivity. Region-dependent parameters extend the B-C master equation (6); the derivation of the Doi–Peliti action goes through unchanged with indexed rates. A translation-invariant radial kernel restricted to K is a particular regional network (Appendix E). Set $\omega_a = \int_{B_{l_a}^N} w_a(\|z\|_p)\, d^N z$, let $\hat{w}_a$ be the Fourier transform of $w_a 1_{B_{l_a}^N}$, and define the matrix

$$M_{aa} = -\alpha_a + f_a'(0)\,\omega_a, \qquad M_{ab} = f_a'(0)\,\mathrm{vol}(B_a)\,c_{ab}, \tag{51}$$

for $a \neq b$.

Lemma 1 (Regional networks)**.**

1. *Every wavelet $\psi_{\gamma n j}$ supported in B_a is an eigenfunction of the right-hand side of Eq.* (50)*, with eigenvalue $-\alpha_a + f_a'(0)\,\hat{w}_a(p^{1-\gamma}) \leq M_{aa}$.*

2. *The regional activities $U_a(t) = \int_{B_a} u(y,t)\, d^N y$ satisfy $dU_a/dt = \sum_b M_{ab} U_b$.*

3. *The leading growth rate of Eq.* (50) *is the spectral abscissa* $s(M)$*, i.e. the largest real part of the eigenvalues of* M*. The network is critical if and only if* $s(M) = 0$*.*

Proof. See Appendices E 1–E 3. □

With finitely many regions, the critical point $s(M) = 0$ is that of a finite network of well-mixed regions. The modes inside each region have growth rates at most $M_{aa} \le s(M)$, generically strictly smaller, so at criticality they remain damped. There is no diverging correlation length inside a region, and the critical behavior is of mean-field (branching-process) type (Fig. 4). When $c_{ab} = 0$, the regions are independent networks of the kind discussed above. Scale-free critical behavior across levels requires coupling across infinitely many levels of the hierarchy, as for the power-law tails of the next lemma.

Lemma 2 (Hierarchical power-law tails)**.** *Suppose* w *is radial and integrable, and* $w(x) = C_w \|x\|_p^{-N-\sigma}$ *for* $\|x\|_p > p^r$*, with* $C_w > 0$ *and* $\sigma > 0$*. Then, for all* $\|\xi\|_p \le p^{-r}$*,*

$$\hat{w}(0) - \hat{w}(\xi) = c_\sigma \, \|\xi\|_p^\sigma, \qquad c_\sigma = C_w \, \frac{1 - p^{-\sigma-N}}{p^\sigma - 1}. \quad (52)$$

Equivalently, $A_w(\xi) = \kappa_W \|\xi\|_p^\sigma$ *with* $\kappa_W = f'(0)\, c_\sigma$*, i.e.* $\boldsymbol{W}$ *and* $\kappa_W \boldsymbol{D}^\sigma$ *have the same symbol on the ball* $\|\xi\|_p \le p^{-r}$*.*

Proof. See Appendix E 5. □

The hypothesis of Lemma 2 has a direct hierarchical meaning. On $\mathbb{Q}_p^N$, $\|x\|_p$ takes the values p^j, and a power law in $\|x\|_p$ is the same as *geometric decay per level of the tree*:

$$\frac{w(p^{j+1})}{w(p^j)} = p^{-(N+\sigma)} \qquad (j \ge r). \quad (53)$$

The coupling between two neurons is attenuated by the same factor each time their common ancestor lies one layer closer to the root. The next lemma shows that this self-similarity, together with isotropy, is what singles out the Taibleson–Vladimirov operator.

In the next lemma we use the following notation. $\boldsymbol{A}$ denotes a pseudodifferential operator of the form

$$\boldsymbol{A}\phi = \mathcal{F}^{-1}_{\xi \to x}\big(\hat{A}(\xi)\, \mathcal{F}_{x \to \xi}\phi\big), \qquad \phi \in \mathcal{D}(\mathbb{Q}_p^N),$$

whose symbol $\hat{A} : \mathbb{Q}_p^N \smallsetminus \{0\} \to \mathbb{C}$ is a continuous function; such an operator commutes with translations. $\boldsymbol{D}^\sigma$ denotes the Taibleson–Vladimirov operator of order $\sigma > 0$, Eq. (3), i.e. the pseudodifferential operator with symbol $\|\xi\|_p^\sigma$. $GL_N(\mathbb{Z}_p)$ denotes the group of $N \times N$ matrices k with entries in $\mathbb{Z}_p$ and $|\det k|_p = 1$; it acts on $\mathbb{Q}_p^N$ by $\xi \mapsto k\xi$, and $\|k\xi\|_p = \|\xi\|_p$ for every $k \in GL_N(\mathbb{Z}_p)$. Indeed, since the entries of k are in $\mathbb{Z}_p$, the strong triangle inequality gives $\|k\xi\|_p \le \|\xi\|_p$; since $|\det k|_p = 1$, the entries of $k^{-1} = (\det k)^{-1} \operatorname{adj}(k)$ are also in $\mathbb{Z}_p$, so $\|\xi\|_p = \|k^{-1} k\xi\|_p \le \|k\xi\|_p$.

Lemma 3 (Uniqueness of $\boldsymbol{D}^\sigma$)**.** *Let* $\boldsymbol{A}$ *be a pseudodifferential operator with symbol* $\hat{A}(\xi)$*, as above. Assume:*

1. Isotropy: $\hat{A}(k\xi) = \hat{A}(\xi)$ *for all* $k \in GL_N(\mathbb{Z}_p)$*. For the kernel operators* (44) *this is the statement that* w *depends only on* $\|y\|_p$*.*

2. Homogeneity of degree σ: $\hat{A}(a\xi) = |a|_p^\sigma \hat{A}(\xi)$ *for all* $a \in \mathbb{Q}_p^\times$*. Equivalently,* $\boldsymbol{A}[\phi(a\,\cdot)] = |a|_p^\sigma (\boldsymbol{A}\phi)(a\,\cdot)$*.*

Then $\hat{A}(\xi) = \hat{A}(1)\, \|\xi\|_p^\sigma$*, where* 1 *denotes any vector of norm one, i.e.* $\boldsymbol{A} = \hat{A}(1)\, \boldsymbol{D}^\sigma$*.*

Proof. The group $GL_N(\mathbb{Z}_p)$ acts transitively on each sphere $\{\|\xi\|_p = p^k\}$, so by (1), $\hat{A}(\xi) = g(\|\xi\|_p)$ for a function g on $p^{\mathbb{Z}}$. Taking $a = p^{-k}$ in (2) gives $g(p^k) = p^{k\sigma} g(1)$. □

B. Isotropy, self-similarity and universality

The two hypotheses of Lemma 3 are idealizations of features of cortical hierarchies, and each mirrors a familiar Euclidean symmetry.

Isotropy: no preferred branch. Each $g \in GL_N(\mathbb{Z}_p)$ maps every ball onto a ball of the same radius and preserves inclusions, so it acts as an automorphism of the tree: it fixes the balls centered at the origin and permutes, consistently at every level, the branches below each vertex. It also preserves the hierarchical distance. Invariance under this group means that the coupling between two neurons depends only on their hierarchical distance, i.e. on the level of their last common ancestor, and not on the ball (structure) of the hierarchy to which they belong. Physically, all the balls of a given level are equivalent: the pattern of connections inside a ball of radius p^k, and between that ball and the rest of the network, is the same for every ball of radius p^k, so no subtree is wired differently from its siblings. For $N \ge 2$, isotropy also means that the N "directions" of branching are equivalent; the anisotropic case is discussed in Remark 2. For $N = 1$, condition (1) is automatic ($GL_1(\mathbb{Z}_p) = \mathbb{Z}_p^\times$), and homogeneity alone forces $\boldsymbol{D}^\sigma$. Isotropy is the hierarchical counterpart of rotational invariance in $\mathbb{R}^N$, which makes a Euclidean kernel depend only on $|x - y|$.

Homogeneity: self-similarity across levels. Multiplication by $a \in \mathbb{Q}_p^\times$ is a change of hierarchical level. For example, $x \mapsto p^{-1}x$ maps the ball B_k^N of radius p^k onto the ball B_{k+1}^N, i.e. it moves one level up the tree. Homogeneity of degree σ says that the dynamics looks the same at every level, up to a rescaling of the rates: the modes supported on balls of radius p^k (the wavelets $\psi_{\gamma nj}$ with $\gamma = k$, for which $\|\xi\|_p = p^{1-k}$) relax at a rate $p^{-k\sigma}$ times that of the modes supported on balls of radius 1. Indeed, $\psi_{\gamma nj}$ is an eigenfunction of $\boldsymbol{D}^\sigma$ with eigenvalue $\|\xi\|_p^\sigma = p^{(1-\gamma)\sigma}$, as in Eq. (5), and in the solution (43) of the Cauchy problem it decays as $e^{-\lambda(p^{1-\gamma})t}$,

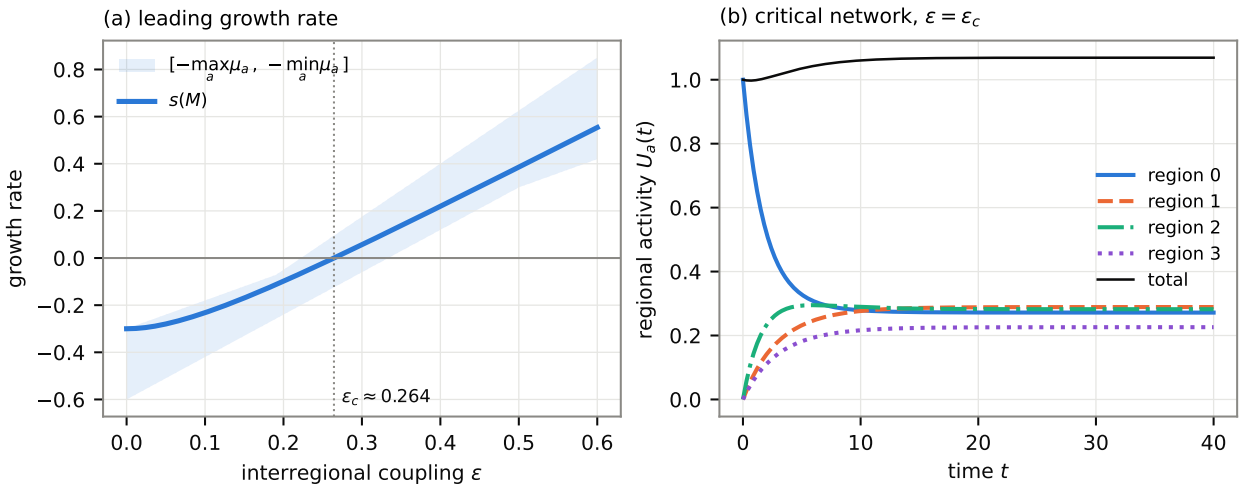


FIG. 4. A regional network (Lemma 1) with four regions, $K = \bigsqcup_{a=0}^{3} B^1_{-2}(a)$, $B^1_{-2}(a) = a + 4\mathbb{Z}_2$ ($p = 2$, $N = 1$). Inside each region the kernel is constant, so the wavelet modes of region a relax at rate α_a. The parameters are $\alpha_a = (1, 1.2, 0.8, 1.5)$, $f'_a(0) = (1, 1.5, 0.8, 1.2)$ and $f'_a(0)\,\omega_a = (0.6, 0.7, 0.5, 0.9)$, and the interregional couplings are $c_{ab} = \varepsilon\, d_{ab}^{-2}$, with $d_{ab} = |a - b|_2 \in \{1/2, 1\}$. (a) The leading growth rate $s(M)$ as a function of ε, and the band $[-\max_a \mu_a, -\min_a \mu_a]$ of Appendix E. The network is critical at $\varepsilon_c \approx 0.264$, where $\mu_a \approx (0.003, -0.095, -0.017, 0.124)$: the μ_a are not equal, so the critical dynamics is not conservative. (b) Regional activities $U_a(t)$ at ε_c, with all activity initially in region 0. They converge to the Perron eigenvector of M and then stay constant. The total activity also tends to a constant, 1.07, different from its initial value 1.

with $\lambda(p^{1-\gamma}) = \mu + \kappa_W p^{(1-\gamma)\sigma}$ for $A_w = \kappa_W \|\xi\|_p^\sigma$. Physically, the connectivity is self-similar: climbing one level of the hierarchy always attenuates the coupling by the same factor $p^{-(N+\sigma)}$ [Eq. (53)]. This is aspect (ii) of the hypothesis of Sporns and co-workers recalled in the Introduction: the patterns of connections repeat from one scale to the next. Equivalently, the hierarchy has no intrinsic scale apart from the microscopic cutoff.

What Lemma 3 says. Together, "no preferred branch" and "no preferred level" leave a single operator up to a constant, $\kappa_W \boldsymbol{D}^\sigma$, whose only parameter is the per-level attenuation σ. In $\mathbb{R}^N$, isotropy and homogeneity alone also leave a one-parameter family, the fractional Laplacians $(-\nabla^2)^{\sigma/2}$, whose symbols are $|q|^\sigma$; locality, or a finite second moment of the kernel, then fixes $\sigma = 2$, i.e. the Laplacian [50, 51]. Over $\mathbb{Q}_p^N$ there is no locality condition to impose, and σ remains a physical parameter of the connectivity.

Why it matters for criticality. A real connectome is not exactly self-similar, and homogeneity is an idealization. It is relevant because at a critical point only the large-scale behavior matters: by power counting, the deviations of an asymptotically homogeneous kernel from the homogeneous one are irrelevant (see the classes below). For this reason $\kappa_W \boldsymbol{D}^\sigma$ is not the literal connectivity, but the representative of a class of kernels, as we now explain.

Universality classes of W-operators. What matters for critical behavior is the infrared behavior of A_w. Using the power counting of Sec. V (momentum dimension 1 for $\|\xi\|_p$), we distinguish three classes.

1. *Asymptotically homogeneous kernels:* $w(p^j) = C_w p^{-j(N+\sigma)}\big(1 + O(p^{-j\delta})\big)$ as $j \to \infty$, with $\delta > 0$. This condition imposes an important restriction on the symbol of the operator. By Eq. (46), $A_w(\xi) = f'(0) \int_{\mathbb{Q}_p^N} w(\|y\|_p)\big(1 - \chi_p(\xi \cdot y)\big)\, d^N y$. Let $\|\xi\|_p = p^m$, and split the integral over the spheres S_j^N, on each of which w is constant. The factor $1 - \chi_p(\xi \cdot y)$ vanishes for $\|y\|_p \le p^{-m}$, since then $|\xi \cdot y|_p \le 1$; by Eq. (2), $\int_{S_j^N} \chi_p(\xi \cdot y)\, d^N y$ equals $-p^{-mN}$ for $j = 1 - m$ and 0 for $j > 1 - m$. Together with the volumes of the spheres [Eq. (1)], this gives the exact formula

$$A_w(\xi) = f'(0)\Big[(1 - p^{-N}) \sum_{j \ge 1-m} p^{jN} w(p^j) + p^{-mN} w(p^{1-m})\Big], \qquad \|\xi\|_p = p^m. \tag{54}$$

The symbol is thus a sum over the levels $j \ge 1 - m$, i.e. over the hierarchical distances larger than $\|\xi\|_p^{-1}$. Inserting the asymptotic form of w, the sum is geometric, and as $\|\xi\|_p \to 0$ (i.e. $m \to -\infty$),

$$A_w(\xi) = \kappa_W \|\xi\|_p^\sigma \big(1 + O(\|\xi\|_p^\delta)\big), \qquad \|\xi\|_p \to 0, \tag{55}$$

with $\kappa_W = f'(0)\, c_\sigma$ as in Lemma 2: the leading terms reproduce c_σ, and the corrections $O(p^{-j\delta})$, summed over $j \ge 1 - m$, are of relative order $p^{(m-1)\delta} \propto \|\xi\|_p^\delta$. The correction has degree $\sigma + \delta > \sigma$ and is irrelevant by the power counting of Sec. V. All such kernels are expected to share the critical behavior of $\kappa_W \boldsymbol{D}^\sigma$. In this sense $\boldsymbol{D}^\sigma$ represents a *universality class labelled by* σ, as ∇^2 represents all

short-range kernels in the Euclidean theory. The difference is that σ is not fixed by analyticity. It is a physical parameter of the connectivity, the per-level attenuation in Eq. (53).

2. *Slowly varying corrections:* $w(p^j) = C_w p^{-j(N+\sigma)}\ell(j)$ with ℓ slowly varying (e.g., a power of j). Then $A_w(\xi) \sim \kappa_W \|\xi\|_p^\sigma \ell(-\log_p \|\xi\|_p)$, and we expect the exponents of the $\boldsymbol{D}^\sigma$ class with logarithmic corrections.

3. *Faster than any power:* if $w(p^j)p^{j(N+\sigma)} \to 0$ for every $\sigma > 0$, there is no scale-invariant infrared limit. Compact support is the extreme case (see the discussion before Lemma 1).

The statements above are exact at zero-loop order. That the corrections in class 1 remain irrelevant beyond zero-loop order is expected from power counting but not proved here.

Remark 2 (Anisotropic hierarchies). For $N \geq 2$, B-C's homogeneity assumption requires only translation invariance, so the kernel need not be radial. Dropping isotropy but keeping homogeneity allows other operators. An example is the class of pseudodifferential operators with symbol $|\mathfrak{f}(\xi)|_p^\beta$, where $\mathfrak{f}$ is an elliptic homogeneous polynomial of degree m [36]. Ellipticity gives $C_1\|\xi\|_p^m \leq |\mathfrak{f}(\xi)|_p \leq C_2\|\xi\|_p^m$, so these operators have the same power counting as $\boldsymbol{D}^\sigma$ with $\sigma = m\beta$, and $N_c = 2\sigma$ is unchanged. They model hierarchies in which the N "directions" of branching are not equivalent. Whether they fall in the same universality class as $\boldsymbol{D}^\sigma$ is an open question.

Summary. The long-distance linear dynamics is the heat equation of the W-operator selected by the connectivity. For isotropic, self-similar connectivity (and, we expect, for every asymptotically homogeneous kernel) this is ultrametric diffusion generated by $\kappa_W \boldsymbol{D}^\sigma$ [34, 36, 39]. Its Green function is the transition density of a Markov process on the tree, and it decays as a power of $\|x\|_p$ rather than as a Gaussian. The exponent σ is, in principle, measurable from connectomic data as a function of hierarchical distance.

C. Master equation and criticality on networks supported on a single ball

The condition $\mu \geq 0$ has a probabilistic meaning: it is the condition under which the linear dynamics on a network supported on a single ball, or on a regional network, is a Markov process. Here, we discuss the simplest case, in which the neurons occupy a ball $K = B_m^N$ containing the support of w. The general case, in which K is a finite disjoint union of balls, as in Ref. [43], is treated in Appendix E. We assume that w is continuous.

Since K is an additive subgroup, $K + \operatorname{supp} w \subset K$, so a solution of Eq. (41) with initial datum supported in K stays supported in K for all $t \geq 0$. Moreover, $\int_K w(\|x-y\|_p)\, d^N y = \hat{w}(0)$ for every $x \in K$, and Eq. (41) restricted to K takes the form of a master equation,

$$\partial_t u(x,t) = \int_K \big\{ j(x\,|\,y)\, u(y,t) - j(y\,|\,x)\, u(x,t) \big\}\, d^N y \tag{56}$$

for $x \in K$, $t \geq 0$, with transition density rates

$$j(x\,|\,y) = f'(0)\, w(\|x-y\|_p),$$
$$j(y\,|\,x) = \frac{\alpha}{\hat{w}(0)}\, w(\|x-y\|_p).$$

The condition $j(x\,|\,y) \leq j(y\,|\,x)$ for all $x, y \in K$ (Hypothesis 1 of Ref. [43]) is equivalent to $f'(0)\hat{w}(0) \leq \alpha$, i.e. $\mu \geq 0$. Under this condition, by Theorem 3.1 of Ref. [43] (stated there for $N = 1$ for simplicity; the results extend to arbitrary dimension N), Eq. (56) generates a Feller semigroup on the space of continuous functions defined on K, $C(K, \mathbb{R})$, and the solution of the Cauchy problem is $u(x,t) = \int_K u_0(y)\, p_t(x, dy)$, where p_t is the transition function of a Markov process with state space K. The condition is also necessary: since $\boldsymbol{W} 1_K = 0$ on K, the solution with $u_0 = 1_K$ is $e^{-\mu t} 1_K$, so $p_t(x, K) = e^{-\mu t}$, which exceeds 1 when $\mu < 0$. Three cases follow.

- $\mu = 0$: the process is conservative. The dynamics is the heat equation $\partial_t u + \boldsymbol{W} u = 0$, a random walk on the hierarchy with no loss of activity: the rate $f'(0)\hat{w}(0)$ at which active neurons trigger new activity equals the rate α at which activity decays (branching ratio $m = f'(0)\hat{w}(0)/\alpha = 1$), so the total activity $\int_K u(x,t)\, d^N x$ is conserved.

- $\mu > 0$: the walk is killed at rate μ, i.e. it is sent to a cemetery state outside K, which is an absorbing state [43]; in the present context it corresponds to the quiescent state.

- $\mu < 0$: there is no Markov interpretation. u is the mean density of a supercritical branching random walk and grows like $e^{|\mu| t}$; in this regime the linearization ceases to be valid, and the nonlinear terms of f, which saturate the growth, must be kept already at the level of mean-field theory.

Thus the critical point $\mu = 0$ is exactly the point at which the linear dynamics on a ball K containing the support of w becomes a conservative Markov process. When K is a disjoint union of balls, the killing rate is constant on each ball, and a conservative dynamics is always critical (Appendix E).

Comparison with $\mathbb{R}^N$. Over $\mathbb{R}^N$ the situation is different. A bounded set K is not a subgroup, and a solution with initial datum supported in K is, for every $t > 0$, supported on all of $\mathbb{R}^N$ (for $w \geq 0$ positive near the origin). A bounded region of cortex must therefore be modeled by restricting the integral to K. The resulting master equation has the symmetric rates $f'(0)\, w(|x-y|)$ and the killing rate $\alpha - f'(0) \int_K w(|x-y|)\, d^N y$, which depends on x and grows near the boundary of K. The

linear dynamics is then not translation invariant, and at $\mu = 0$ it is not conservative. Over $\mathbb{Q}_p^N$, a ball has no boundary in this sense, and on a disjoint union of balls the killing rate is constant on each ball.

V. CRITICALITY: ULTRAMETRIC REGGEON FIELD THEORY

Physical picture. We consider the network without external drive, $f(I) = 0$ [Eq. (39)], so that the silent state $a \equiv 0$, where $a = \langle\varphi\rangle$ is the mean activity (Sec. III C), is *absorbing*: once all neurons are silent, they stay silent. The absorbing state is an idealization. Here "silent" means that there are no effective spikes, the activity that propagates through the network, not that there is no electrical activity at all. Real cortex is never silent: even at rest it receives weak input and fires spontaneously. This corresponds to a small source $f(I) > 0$, which turns the absorbing state into a stationary state of low activity, $a^* \simeq f(I)/\mu$ (Sec. X D). Resting cortical activity is described by this weakly driven state near $\mu = 0$, in which spikes produced by the input start avalanches that spread and die out; the absorbing state is its limit as $I \to 0$. A single spike triggers on average $m = f'(0)\hat{w}(0)/\alpha$ further spikes (the branching ratio), and $\mu = \alpha(1 - m)$. For $\mu > 0$ ($m < 1$) every avalanche of activity dies out, and the network returns to the silent state. For $\mu < 0$ ($m > 1$) activity can sustain itself. Its growth is stopped by the *saturation* of f: the firing rate is bounded, $f \le f_{\max}$, so the response grows less than proportionally to the input, and $f''(0) < 0$ [Eq. (24)]. The network then settles in an active state of density $\rho > 0$. The value $\mu = 0$ separates the two phases. This is a nonequilibrium phase transition into an absorbing state. When the neurons occupy Euclidean space $\mathbb{R}^N$, as in B-C, it belongs to the universality class of directed percolation (Appendix B). Here the neurons occupy $\mathbb{Q}_p^N$; the transition is of the same type, but its critical behavior is governed by the hierarchical connectivity, through the operator $\boldsymbol{W}$. At $\mu = 0$ avalanches of all sizes occur, and their statistics follow power laws. Near $\mu = 0$ the activity is small and slowly varying, while its correlation length and time diverge. Mean-field theory (the Wilson–Cowan equation) neglects the fluctuations of this sparse activity, and the question is whether they change the scaling laws. To answer it we expand the action about $\varphi = \tilde{\varphi} = 0$, which, since $a \equiv 0$ and $\tilde{a} = 0$, is the expansion about the mean field (Sec. III D), in powers of the fields, and keep only the terms that matter at large scales. The quadratic terms, $\tilde{\varphi}(\partial_t + \mu + \boldsymbol{W})\varphi$, describe the linear dynamics of Sec. IV: propagation of activity through the hierarchy and its decay. The cubic terms describe the two elementary processes that the linear dynamics misses. *Branching*, $\tilde{\varphi}^2\varphi$: the spikes triggered by one active neuron are correlated, which is the source of fluctuations; their strength is proportional to the activity itself (multiplicative noise). *Aggregation*, or saturation, $\tilde{\varphi}\varphi^2$: two active inputs to the same neuron produce less than twice the response, which limits the growth for $\mu < 0$ and fixes ρ. Quartic and higher terms are irrelevant near the upper critical dimension (power counting below). The resulting action, Eq. (59), is Reggeon field theory with the Laplacian replaced by the W-operator. It describes both phases: the absorbing phase, in which $\varphi = 0$ is stable, and the active phase, in which its classical equation has the stable solution $\rho \propto |\mu|$. The rest of this section rescales the fields so that the two cubic couplings are equal, which exhibits the rapidity-reversal symmetry, and uses power counting to find when the fluctuations generated by branching change the mean-field exponents: this happens for $N < N_c = 2\sigma$.

The effective action. We follow B-C [Sec. V of Ref. [13]]. Among the vertices of the interacting action (Sec. III D), the relevant ones near the critical point are the cubic vertices $-f'(0)\,\tilde{\varphi}\,w \star (\tilde{\varphi}\varphi)$ (branching) and $-\frac{1}{2}f''(0)\,\tilde{\varphi}\,(w \star \varphi)^2$ (aggregation, or saturation). For the slowly varying fields that dominate at criticality, $w\star$ may be replaced by $\hat{w}(0)$ in these vertices; the difference contains the operator $\boldsymbol{W}$ and is irrelevant. This gives $-f'(0)\hat{w}(0)\,\tilde{\varphi}^2\varphi + \frac{1}{2}|f''(0)|\hat{w}(0)^2\,\tilde{\varphi}\varphi^2$. B-C then rescale the fields [Eq. (58) of Ref. [13]],

$$\tilde{\psi} = c\,\tilde{\varphi}, \qquad \psi = c^{-1}\varphi, \qquad c = \Big(\frac{2f'(0)}{\hat{w}(0)\,|f''(0)|}\Big)^{1/2}. \tag{57}$$

The quadratic part of the action depends only on the product $\tilde{\varphi}\varphi = \tilde{\psi}\psi$ and is unchanged, while the two cubic coefficients become equal in magnitude: $f'(0)\hat{w}(0)/c = \frac{1}{2}|f''(0)|\hat{w}(0)^2 c = g/2$, with

$$g = \big(2f'(0)\,|f''(0)|\,\hat{w}(0)^3\big)^{1/2}. \tag{58}$$

Keeping the relevant interactions, the result for a general connectivity kernel is

$$S_{\rm eff}^{W}[\psi,\tilde{\psi}] = \int d^N x \int_0^\infty dt \,\Big\{\tilde{\psi}\big(\partial_t + \mu + \boldsymbol{W}\big)\psi + \frac{g}{2}\big(\tilde{\psi}\psi^2 - \tilde{\psi}^2\psi\big)\Big\}, \tag{59}$$

with $\boldsymbol{W}$ the W-operator (44) selected by the connectivity, μ the distance to criticality and g the coupling (58). The relative minus sign between the cubic terms comes from $f''(0) < 0$. Eqs. (57) and (58) give the bare values of μ and g; beyond zero-loop order both are modified by fluctuations (renormalized), as in B-C. Their Eq. (59) writes the cubic term as $g(\varphi^2\tilde{\varphi} - \tilde{\varphi}^2\varphi)$, with a coupling g for which they give no explicit expression; their rescaling, Eq. (58) of Ref. [13], gives it the bare value $g/2$, with our g of Eq. (58). We keep the notation $\boldsymbol{W}$ throughout: the kinetic operator is not an independent modeling choice, it is fixed by w. For the scaling analysis we assume from now on that $\boldsymbol{W}$ is in the class of asymptotically homogeneous kernels (Sec. IV B). This is the class in which scale-free critical behavior occurs. Kernels with compact support couple only finitely many levels, and their critical behavior is of mean-field (branching-process) type (Lemma 1); more generally, kernels that

decay faster than any power have no scale-invariant infrared limit (class 3 of Sec. IV B). For an asymptotically homogeneous kernel,

$$A_w(\xi) = \kappa_W \|\xi\|_p^\sigma \big(1 + O(\|\xi\|_p^\delta)\big), \qquad \|\xi\|_p \to 0, \quad (60)$$

with $\kappa_W > 0$, $\sigma > 0$ and $\delta > 0$. The Taibleson–Vladimirov operator $\kappa_W \boldsymbol{D}^\sigma$ is the exactly homogeneous representative of this class (Lemma 3). It corresponds to isotropic connectivity with per-level attenuation $p^{-(N+\sigma)}$. Only the leading term of Eq. (60) enters the power counting below; the correction is irrelevant.

Rapidity reversal. The action (59) is invariant under the transformation

$$\psi(x,t) \to -\tilde\psi(x,-t), \qquad \tilde\psi(x,t) \to -\psi(x,-t), \quad (61)$$

called *rapidity reversal*, which exchanges the density field ψ and the response field $\tilde\psi$ and reverses time. Indeed, after an integration by parts in t the quadratic term is invariant because $\boldsymbol{W}$ is self-adjoint (w is radial), and the two cubic terms are exchanged because their coefficients have equal magnitude, which is what the rescaling (57) achieves. As in B-C [Sec. V of Ref. [13]], this is an emergent symmetry of the critical theory; the initial-state term of the full action breaks it. The symmetry relates two order parameters of the absorbing-state transition. In the active phase ($\mu < 0$),

$$\rho = \lim_{t\to\infty} \langle \psi(x,t)\rangle \propto (-\mu)^\beta, \qquad P_\infty \propto (-\mu)^{\beta'}, \quad (62)$$

where ρ is the stationary density of activity, starting from a fully active network, and P_∞ is the probability that the activity started by a single seed (one spike) never dies out. The exponents β and β' are defined by Eq. (62); they are the exponents β and β' of Eq. (60) of Ref. [13]. B-C state these definitions, the mean-field values $\beta = \beta' = 1$ and the relation $\beta = \beta'$ without derivation: all of this is standard directed percolation material, taken from the literature (Ref. [19], which is Ref. [9] of B-C). We derive the mean-field values in Appendix C and justify $\beta = \beta'$ below. The density is a one-point function of ψ, while the survival probability of a seed is a one-point function of $\tilde\psi$ (the response to a localized initial perturbation). Rapidity reversal maps one onto the other, so $\beta = \beta'$, as in DP [19, 52]. The same argument gives equal exponents for the decay in time of the density and of the survival probability at $\mu = 0$ (Appendix B 4).

Power counting. Assign momentum dimension 1 to $\|\xi\|_p$, and write $[t] = -z$ for the scaling dimension of time, where z is the dynamic exponent. In the quadratic part of Eq. (59), ∂_t and $\boldsymbol{W}$ (whose symbol is $\kappa_W\|\xi\|_p^\sigma$ at small $\|\xi\|_p$) are added, so they have the same dimension; hence, at zero-loop order, $z = \sigma$, the order of the Taibleson–Vladimirov operator. The quadratic part is dimensionless if $[\psi] + [\tilde\psi] = N$, and rapidity reversal gives $[\psi] = [\tilde\psi] = N/2$. Requiring the cubic term $\int d^N x\, dt\, \frac{g}{2}\tilde\psi\psi^2$ to be dimensionless gives $-N - z + 3N/2 + [g] = 0$, where $-N - z$ is the dimension of the measure $d^N x\, dt$; thus

$$[g] = \sigma - \frac{N}{2}. \quad (63)$$

Equation (59) keeps only the cubic vertices. The interacting action of Sec. III D also contains quartic and higher vertices: by Eq. (25), the term $n = 2$ contains $-f''(0)\,\tilde\varphi\,(w \star (\tilde\varphi\varphi))(w\star\varphi)$, of type $\tilde\varphi^2\varphi^2$, and the term $n = 3$ contains $-\frac{1}{6}f'''(0)\,\tilde\varphi\,(w\star\varphi)^3$, of type $\tilde\varphi\varphi^3$. For a quartic coupling u the same count gives $-N - z + 4N/2 + [u] = 0$, i.e. $[u] = \sigma - N$, which is negative near $N = 2\sigma$. These vertices are therefore irrelevant near the upper critical dimension, and dropping them in Eq. (59) is justified there. For N well below 2σ (in particular for $N \le \sigma$, where $[u] \ge 0$), that they remain irrelevant at the non-Gaussian fixed point is an assumption, the same one made for Euclidean directed percolation below $d = 4$, and it is not proved here. The cubic coupling is relevant for $N < 2\sigma$, so

$$N_c = 2\sigma. \quad (64)$$

Equivalently, in terms of the spectral dimension $d_s = 2N/\sigma$ of the p-adic diffusion, Eq. (95) of Sec. X, mean-field theory holds for $d_s \ge 4$, as in the Euclidean theory with local coupling. More generally, consider a term $\int d^N x \int_0^\infty dt\, \tilde\psi^{\,b}\psi^{c-1}\,\boldsymbol{D}^a\psi$, where $\boldsymbol{D}^a$ is the Taibleson–Vladimirov operator (3) of order $a \ge 0$ (with $\boldsymbol{D}^0$ the identity), i.e. the operator with symbol $\|\xi\|_p^a$; here a is real, not necessarily an even integer. The dimension of the integral is $-N - z + a + (b+c)N/2$, and its coupling has the opposite dimension. With $z = \sigma$, the coupling is relevant or marginal iff

$$N \le \frac{2(\sigma - a)}{b + c - 2}. \quad (65)$$

For $\sigma = 2$ this is Eq. (132) of Ref. [25]. For $a = 0$, cubic vertices give $N_c = 2\sigma$ and quartic vertices give $N_c = \sigma$, so quartic and higher couplings are irrelevant near $N_c = 2\sigma$. There is no local gradient term available over $\mathbb{Q}_p^N$. A symbol term $\|\xi\|_p^{\sigma'}$ with $\sigma' > \sigma$, for example from a faster-decaying component of the kernel, is irrelevant. Therefore, unlike Euclidean Lévy-flight DP, where ∇^2 takes over for $\sigma \ge 2$ [44], *every* $\sigma > 0$ defines a distinct theory. In particular the p-adic line ($N = 1$) is below its upper critical dimension whenever $\sigma > 1/2$.

Mean-field exponents. At zero-loop order, the stationary density in the active phase ($\mu < 0$) is $\rho = 2|\mu|/g$, and a mode of momentum ξ relaxes at the rate $\lambda(\xi) = \mu + A_w(\xi) \simeq \mu + \kappa_W\|\xi\|_p^\sigma$. These two facts give

$$\beta = \beta' = 1, \qquad \nu_\parallel = 1, \qquad \nu_\perp = \frac{1}{\sigma}, \qquad z = \sigma. \quad (66)$$

Compare the Euclidean values $(1, 1, 1, \frac{1}{2}, 2)$ used by B-C. The exponents of the total activity (β, β', $\nu_\parallel$) coincide with those of B-C; the spatial ones ($\nu_\perp$, z) carry the decay exponent σ, and coincide with them for $\sigma = 2$. The derivation is given in Appendix C.

Interpretation for cortex. In the Euclidean theory $N_c = 4$, so for $N = 2, 3$ fluctuations dominate sufficiently close to the critical point, and the exponents are not those of mean-field theory. B-C argued indirectly that cortex should nevertheless display mean-field DP exponents: since each neuron receives input from very many others over long distances, the fluctuation correction in their Ginzburg criterion [Eq. (52) of Ref. [13]] carries a small prefactor, so mean-field behavior should be observed over a wide range of μ and fail only in a narrow region around $\mu = 0$. In this argument mean-field theory is a good approximation, not an exact result.

In the p-adic theory the argument becomes the sharp statement $\sigma \le N/2$: for $N \ge 2\sigma$ the mean-field exponents are exact asymptotically, arbitrarily close to the critical point, and for $N < 2\sigma$ they eventually fail, however small the fluctuation region. A simple Ginzburg estimate, derived in Appendix C 8, shows where the inequality comes from. In the active phase $\rho = 2|\mu|/g$, and the noise of DP is multiplicative, with strength of order $g\rho$. Averaged over a correlation volume $\xi_\perp^N$ and a correlation time $\xi_\parallel$, the relative fluctuations are

$$\frac{\langle \delta\rho^2 \rangle}{\rho^2} \simeq \frac{g\,\xi_\parallel}{\rho\,\xi_\perp^N} \simeq g^2\,|\mu|^{N/\sigma - 2}, \tag{67}$$

by Eq. (66) ($\xi_\parallel = 1/|\mu|$, $\xi_\perp = (\kappa_W/|\mu|)^{1/\sigma}$). This ratio vanishes as $\mu \to 0$ exactly when $N > 2\sigma$, the condition found by power counting and by the one-loop integral of Sec. VI.

The inequality has a direct reading in terms of connectivity. Let W_k be the total coupling that a neuron receives from the neurons at hierarchical distance p^k [Eq. (87)]. For the power-law kernel, $W_{k+1}/W_k = p^{-\sigma}$: each level up the hierarchy contributes p^σ times less total input than the level below. Since each vertex of the tree has p^N children, the condition $\sigma \le N/2$ reads

$$\frac{W_{k+1}}{W_k} = p^{-\sigma} \;\ge\; p^{-N/2} = \frac{1}{\sqrt{p^N}}\,:$$

mean-field theory holds when the input from successive levels of the hierarchy falls off no faster than one over the square root of the branching number. If distant levels still contribute strongly (small σ), each neuron averages over a large and growing population, and fluctuations average out. If the coupling is concentrated on nearby levels (large σ), fluctuations are not averaged away, and the exponents depart from mean field; the simulations of Sec. VIII A show this trend in the avalanche-size exponent. Whether hierarchical cortex is mean-field is therefore a measurable question about how fast the synaptic input to a neuron decays with hierarchical distance, not about the dimension of physical space.

Beyond mean field. For $N < 2\sigma$ the cubic coupling is relevant, fluctuations dominate sufficiently close to the critical point, and the exponents differ from Eq. (66). Computing them requires a renormalization-group analysis of Eq. (59), which we do not attempt here. In this respect we follow B-C, who do not compute the exponents either: they identify their effective action with Reggeon field theory and take its critical behavior, directed percolation, from the literature [Sec. V of Ref. [13]]. That step is not available for Eq. (59), whose critical behavior below $N_c = 2\sigma$ is, to our knowledge, not known. The renormalization group over the p-adics is well developed, from Dyson's hierarchical model [53, 54] and the p-adic renormalization group [55] to rigorous constructions with anomalous dimensions [56] and the p-adic $O(n)$ models [57]; these provide the natural framework for this step. At present, the simulations of Sec. VIII A give numerical estimates of the avalanche-size exponent below N_c. Like B-C, we assume that the relative minus sign between the cubic terms of Eq. (59), which reflects the saturation of f, persists beyond zero-loop order.

VI. A p-ADIC GINZBURG CRITERION

Origin of the criterion. The B-C criterion [Eqs. (50)–(53) of Ref. [13]] compares the zero-loop and one-loop contributions to the vertex $\Gamma^{(1,1)}$ (Fig. 8 of Ref. [13]).

The vertex $\Gamma^{(1,1)}$. Let $\Gamma[a, \tilde{a}]$ be the effective action of B-C [Eqs. (36)–(41) of Ref. [13]]: the Legendre transform of $W = \ln Z$ with respect to the sources, regarded as a functional of the means $a = \langle\varphi\rangle$, $\tilde{a} = \langle\tilde{\varphi}\rangle$. Its stationary points are the exact means, and its derivatives are the proper (one-particle-irreducible) vertices. At the quiescent state we set

$$\Gamma^{(1,1)}(x,t;x',t') = \left.\frac{\delta^2\Gamma}{\delta\tilde{a}(x,t)\,\delta a(x',t')}\right|_{a=\tilde{a}=0}. \tag{68}$$

It is the inverse of the full propagator:

$$\int \Gamma^{(1,1)}(x,t;y,s)\,\langle\varphi(y,s)\tilde{\varphi}(x',t')\rangle\,d^N y\,ds = \delta(x-x')\,\delta(t-t').$$

At zero loops $\Gamma = S$, and $\Gamma^{(1,1)}$ is the kernel of the operator $\partial_t + \alpha - f'(0)\,w\star$ of the free action. By translation invariance in space and time, $\Gamma^{(1,1)}$ depends only on $(x - x', t - t')$. With $\hat{F}(\xi,\omega) = \int \chi_p(\xi\cdot x)\,e^{i\omega t}F(x,t)\,d^N x\,dt$ and $\alpha - f'(0)\hat{w}(\xi) = \mu + A_w(\xi)$ [Eq. (46)], one has, to one loop,

$$\begin{aligned}\hat{\Gamma}^{(1,1)}(\xi,\omega) &= -i\omega + \mu + A_w(\xi) - \Sigma(\xi,\omega),\\ \Sigma(\xi,\omega) &= f''(0)f'(0)\\ &\quad\times\int_{\mathbb{Q}_p^N}\frac{\hat{w}(\eta)^2\,\hat{w}(\xi-\eta)\,d^N\eta}{-i\omega + 2\mu + A_w(\eta) + A_w(\xi-\eta)}.\end{aligned} \tag{69}$$

The self-energy Σ is the loop of Fig. 8 of Ref. [13]. It is built from the branching vertex $f'(0)\,\tilde{\varphi}\,w\star(\tilde{\varphi}\varphi)$ and the merging vertex $\frac{1}{2}f''(0)\,\tilde{\varphi}\,(w\star\varphi)^2$, joined by two propagators $\theta(t)e^{-(\mu+A_w)t}$; the time integral gives the denominator, and the two ways of joining the legs cancel the

factor $\frac{1}{2}$. Define

$$\mu_R(\mu) := \hat{\Gamma}^{(1,1)}(\xi=0,\omega=0) = \mu - f''(0)f'(0)\,J(\mu),$$
$$J(\mu) = \int_{\mathbb{Q}_p^N} \frac{\hat{w}(\xi)^3}{2\big(\mu + A_w(\xi)\big)}\,d^N\xi, \tag{70}$$

to one loop. The bare parameter μ is the decay rate of the uniform mode in mean-field theory; μ_R is its decay rate in the full theory, i.e. the inverse of the static response of the network to a uniform stimulus. Equivalently, $\mu_R = 1/\langle S\rangle$, where $\langle S\rangle = \int dt \int d^N x\,\langle\varphi(x,t)\tilde{\varphi}(0,0)\rangle = 1/\hat{\Gamma}^{(1,1)}(0,0)$ is the mean total number of spikes triggered by one injected spike (the mean avalanche size), fluctuations included; in mean-field theory $\langle S\rangle = 1/\mu$. Thus μ_R is the renormalized distance to criticality, and the critical point is where it vanishes. There are no factors $(2\pi)^{-N}$, because the Fourier transform uses the character χ_p and the self-dual Haar measure (Appendix D).

B-C's Eq. (50) compares the two terms of Eq. (70). Its symbol $\gg$ compares magnitudes: for a saturating f, $f''(0) < 0$, and the one-loop term, taken with its sign, has the opposite sign to μ. This is why $|f''(0)|$ appears below, as in B-C's Eq. (52); the sign of $f''(0)$ fixes the direction of the shift of the critical point. Taken literally at zero momentum, B-C's condition $|\mu| \gg |f''(0)f'(0)J(\mu)|$ fails as $\mu \to 0$ whenever $J(0)$ is finite ($N > \sigma$, see below), even for $N > 2\sigma$, where mean-field exponents are exact. It detects the shift of the critical point, not a breakdown of mean-field theory. The criterion must therefore be applied to the distance $\mu - \mu_c$ from the true critical point, as below; B-C do this implicitly by keeping only the infrared singular part of the integral (their Eq. (51)). B-C write the prefactor as $\frac{1}{2}f''f'$, omitting the factor 2 from the two pairings (as they omit the combinatorial factor in their Eq. (25)). We checked the factor on a single neuron with birth rate $f'(0)\,n + \frac{1}{2}f''(0)\,n(n-1)$ and death rate αn. There, the one-loop self-energy $f''(0)f'(0)/(-i\omega + 2\mu)$ predicts the decay rate $\mu - f''(0)f'(0)/\mu + O(f''(0)^2)$ of the mean, and the leading eigenvalue of the master equation agrees; B-C's prefactor would halve the correction.

For $N > \sigma$ the value $J(0)$ is finite. Near $\xi = 0$, by Eq. (60), the integrand behaves as $\hat{w}(0)^3/(2\kappa_W\|\xi\|_p^\sigma)$, and summing over spheres gives $\sum_{k\le 0} p^{k(N-\sigma)}$, which converges if and only if $N > \sigma$. At intermediate $\|\xi\|_p$, $A_w > 0$ by Lemma 4(1) below. At large $\|\xi\|_p$, $A_w(\xi) \to f'(0)\hat{w}(0) > 0$, and $\int|\hat{w}|^3 < \infty$ if, for instance, $w \in L^1 \cap L^2$, since then $|\hat{w}|^3 \le \|\hat{w}\|_\infty|\hat{w}|^2$; alternatively, the cutoff of Sec. VI A makes this region finite.

The critical point is the value of the bare parameter $\mu = \alpha - f'(0)\hat{w}(0)$ at which the infinite network on $\mathbb{Q}_p^N$, with fluctuations included, undergoes the absorbing-state transition. Physically, μ is set by α, $f'(0)$, and w. In mean-field theory, this happens at $\mu = 0$. With fluctuations, to one loop, it happens at $\mu_c = f''(0)f'(0)J(0)$. On a finite network, there is no sharp transition (Sec. VI A), so the statement refers to $\mathbb{Q}_p^N$. Equivalently, μ_c is defined by $\mu_R(\mu_c) = 0$, i.e. $\mu_c = f''(0)f'(0)\,J(\mu_c) \simeq f''(0)f'(0)\,J(0)$; the last step is valid to one loop, since $J(\mu) - J(0) \to 0$ as $\mu \to 0$ for $N > \sigma$. For a saturating f, $f''(0) < 0$ and $\mu_c < 0$, i.e. $\alpha < f'(0)\hat{w}(0)$ at the transition: the network needs more excitation than mean-field theory predicts. Fluctuations lower the mean firing rate, as Jensen's inequality requires for a concave f. The shift is non-universal, like the shift of the critical temperature in equilibrium theory, and says nothing about the approach to the transition. That is controlled by

$$\mu_R \simeq (\mu - \mu_c)\big[1 - f''(0)f'(0)\,J'(\mu)\big].$$

Mean-field theory is reliable when the one-loop term in the bracket is small, $c\,|J'(\mu)| \ll 1$ with $c = |f''(0)|f'(0)$, i.e.

$$\frac{|f''(0)|\,f'(0)}{2}\int_{\mathbb{Q}_p^N}\frac{\hat{w}(\xi)^3}{\big(\mu + A_w(\xi)\big)^2}\,d^N\xi \ll 1. \tag{71}$$

The singular dependence on μ comes from small $\|\xi\|_p$. By Eqs. (46) and (60), $\hat{w}(\xi) = \hat{w}(0) - A_w(\xi)/f'(0)$ and $A_w(\xi) = O(\|\xi\|_p^\sigma)$, so $\hat{w}(\xi)^3 = \hat{w}(0)^3\big(1 + O(\|\xi\|_p^\sigma)\big)$. The correction contributes at most a multiple of $\int A_w/(\mu + A_w)^2\,d^N\xi$, which is smaller than the leading term by a factor $O\big(\mu^{\min(1,\,2-N/\sigma)}\big)$ (times $|\ln\mu|$ if $N = \sigma$). Replacing $\hat{w}(\xi)^3$ by $\hat{w}(0)^3$ therefore does not change the leading singular behavior, and by Eq. (58) the criterion becomes

$$\frac{g^2}{4}\,I_2(\mu) \ll 1, \qquad I_2(\mu) = \int_{\|\xi\|_p \le p^\Lambda}\frac{d^N\xi}{\big(\mu + A_w(\xi)\big)^2}, \tag{72}$$

with $\Lambda \in \mathbb{Z}$, and g as in Eq. (58). For $N \le \sigma$, $J(0)$ diverges and there is no shift to subtract; the direct ratio $c\,J(\mu)/\mu$ in Eq. (70) then behaves as $\mu^{N/\sigma-2}$, the same power as below.

The cutoff. Since w is radial, $A_w(\xi)$ depends only on $\|\xi\|_p$. Write $A_w(p^k)$ for its value on the sphere $S_k^N = \{\|\xi\|_p = p^k\}$. By Lemma 5 of Appendix F,

$$I_2(\mu) = (1 - p^{-N})\sum_{k=-\infty}^{\Lambda}\frac{p^{kN}}{\big(\mu + A_w(p^k)\big)^2}. \tag{73}$$

The ultraviolet limit of Eq. (73) is controlled by the terms involving larger positive powers of p^k (associated with small balls $p^k\mathbb{Z}_p^N$), while the infrared limit is controlled by powers of p^k, with $k \to -\infty$. The cutoff is needed because A_w is bounded: $0 \le A_w(\xi) \le 2f'(0)\hat{w}(0)$, and $A_w(\xi) \to f'(0)\hat{w}(0)$ as $\|\xi\|_p \to \infty$ by the Riemann–Lebesgue lemma. Without a cutoff the sum would diverge at large k. This divergence is created by the replacement $\hat{w}(\xi)^3 \to \hat{w}(0)^3$, since $\hat{w}(\xi) \to 0$; it is not a property of the model. The natural values of Λ are discussed in Sec. VI A.

Lemma 4 (Infrared asymptotics of I_2)**.** *Let $w \ge 0$ be radial and integrable, and $f'(0) > 0$. Assume Eq. (60) in the form $A_w(\xi) = \kappa_W\|\xi\|_p^\sigma\big(1+\varepsilon(\xi)\big)$, with $|\varepsilon(\xi)| \le C\|\xi\|_p^\delta$ for $\|\xi\|_p \le p^{-r}$, for some $r \in \mathbb{Z}$ and $C, \delta > 0$. Fix $\Lambda \in \mathbb{Z}$. Then:*

1. *$A_w(\xi) > 0$ for every $\xi \neq 0$.*

2. *$I_2(0) < \infty$ if and only if $N > 2\sigma$.*

3. *If $N < 2\sigma$, then, as $\mu \to 0^+$,*

$$I_2(\mu) = (1-p^{-N})\,\kappa_W^{-N/\sigma}\,\mu^{N/\sigma-2} \times \Big[\Phi(s) + O\big((\mu/\kappa_W)^{\delta/\sigma}\big) + O\big(\mu^{2-N/\sigma}\big)\Big], \quad (74)$$

where $\mu/\kappa_W = p^{\sigma s}$, $s \in \mathbb{R}$, and

$$\Phi(s) = \sum_{k\in\mathbb{Z}} \frac{p^{(k-s)N}}{\big(1+p^{\sigma(k-s)}\big)^2}. \quad (75)$$

If $N + \delta = 2\sigma$, the first error term carries an extra factor $|\ln\mu|$; if $N + \delta > 2\sigma$, it is absorbed into the second. The constant in the second error term depends on Λ.

The proof is given in Appendix F. Three remarks:

- No hypothesis beyond Eq. (60) is needed. In particular, positivity of A_w away from the origin is a consequence of Eq. (60), not an extra assumption. If only $\varepsilon(\xi) \to 0$ is known, Eq. (74) still holds with an error $o(1)$ inside the bracket.

- The limit $\mu \to 0$ is taken at fixed Λ: the constant in the second error term grows like $p^{\Lambda N}$. The power law (60) is used only in the infrared.

- For the power-law tails of Lemma 2, $\varepsilon \equiv 0$ on $\|\xi\|_p \le p^{-r}$, and the first error term is absent. In the Euclidean case a power-law tail always produces correction terms.

The function Φ is periodic in $s = \sigma^{-1}\log_p(\mu/\kappa_W)$, with period 1. Define the ultrametric diffusion length

$$\ell_W = \Big(\frac{\kappa_W}{\mu}\Big)^{1/\sigma}, \qquad k^* = \log_p \ell_W. \quad (76)$$

Here $\mu = \alpha - f'(0)\hat{w}(0)$ is the bare distance to criticality (Sec. V). Near the shifted critical point it should be read as $\mu - \mu_c$, which is $\simeq \mu_R$ in the region where the criterion holds; the two differ by the shift $\mu_c = O(g^2)$. By Eqs. (72) and (74), the mean-field criterion reads

$$\frac{1-p^{-N}}{4}\,\frac{g^2}{\kappa_W^2}\,\ell_W^{2\sigma-N}\,\Phi(s) \ll 1. \quad (77)$$

Recall that the coupling g of Eq. (59) of Ref. [13], for which B-C give no explicit expression, has the bare value $g/2$, with our g of Eq. (58); this follows from their rescaling, Eq. (58) of Ref. [13] (Sec. V).

Physical meaning of the criterion. The left-hand side of Eq. (77) is the one-loop term in the bracket of $\mu_R \simeq (\mu-\mu_c)[1 - f''(0)f'(0)J'(\mu)]$, i.e. the relative size of the fluctuation correction to the response of the network. Since $\langle S\rangle = 1/\mu_R \simeq (\mu-\mu_c)^{-1}[1 + f''(0)f'(0)J'(\mu)]$, it is also the relative correction to the mean avalanche size: mean-field theory is reliable when fluctuations change $\langle S\rangle$ by a small fraction. Its two factors have a direct reading. The coupling $g^2 \propto f'(0)\,|f''(0)|$ measures the strength of the spike-count noise that survives in the dynamics of the mean: it comes from the branching of activity, $f'(0)$, and from the saturation of the firing rate, $f''(0)$, and it vanishes for a linear f, for which mean-field theory is exact. The factor $\ell_W^{2\sigma-N}$ measures how much of this noise is averaged out. A perturbation of the quiescent state lives a time of order $\ell_W^\sigma/\kappa_W = 1/\mu$ and, during that time, spreads over a ball of radius ℓ_W, which contains a number of neurons proportional to ℓ_W^N; its noise is averaged over that ball and that time. Indeed, since $\ell_W^{2\sigma-N}/\kappa_W^2 = \kappa_W^{-N/\sigma}\mu^{N/\sigma-2}$, the left-hand side of Eq. (77) is, up to the factor $\Phi(s)$, the relative fluctuation of the activity in a correlation volume during a correlation time, Eq. (67) of Sec. V. Mean-field theory is reliable when these fluctuations are small: when each neuron, through the connections it receives, averages over many independent sources of noise before the perturbation relaxes. As $\mu \to 0$, the ball of radius ℓ_W climbs the hierarchy, up to level $k^* = \log_p \ell_W$. If $N > 2\sigma$, the number of neurons it contains grows faster than the noise accumulates, and the criterion improves; this is the hierarchical form of the condition $W_{k+1}/W_k = p^{-\sigma} \ge p^{-N/2}$ of Sec. V. If $N < 2\sigma$, the connectivity decays too fast with hierarchical distance, the effective number of independent inputs does not keep up, and fluctuations dominate sufficiently close to the critical point. On a finite network the ball cannot exceed the network itself, which leads to the Ginzburg number of Sec. VI A.

Two features are new:

1. *Hierarchical reading.* k^* is the level of the tree that a fluctuation reaches before relaxing. Mean-field theory fails when fluctuations propagate high enough in the hierarchy, i.e. when small balls start to act coherently with balls that are far away in the hierarchical distance.

2. *Log-periodicity.* The factor $\Phi(s)$ reflects discrete scale invariance under $x \to px$: observables near criticality carry modulations that are periodic functions of $\ln\mu$ with period $\sigma\ln p$, i.e. invariant under $\mu \mapsto p^\sigma\mu$. The Fourier series of Φ, derived in Appendix F, gives the size of the modulation. With $a = N/\sigma$ and $\omega_n = 2\pi n/(\sigma\ln p)$,

$$\Phi(s) = \frac{1}{\sigma\ln p}\sum_{n\in\mathbb{Z}} \Gamma(a - i\omega_n)\,\Gamma(2-a+i\omega_n)\,e^{-2\pi i n s}, \quad (78)$$

where Γ denotes Euler's Gamma function, not the vertex function $\Gamma^{(1,1)}$ of Eqs. (68)–(69). The mean

value of Φ is $(1-a)\pi/\big(\sigma \ln p \sin(\pi a)\big)$, equal to $1/(\sigma \ln p)$ for $a = 1$. The oscillating part decreases like $|\omega_1| e^{-\pi|\omega_1|}$, with $\omega_1 = 2\pi/(\sigma \ln p)$, so it is exponentially small unless $\sigma \ln p$ is large. For $N = \sigma = 1$, the relative amplitude of the oscillation of Φ is about 5×10^{-11} for $p = 2$, 2×10^{-3} for $p = 7$, 0.07 for $p = 31$, and 0.24 for $p = 101$. The value for $p = 7$ is of the same order as the amplitude found in the avalanche simulations of Sec. VIII. Log-periodicity is thus a qualitative signature of a p-adic (discrete-hierarchical) architecture that a Euclidean theory cannot produce, but it is quantitatively observable only when $\sigma \ln p$ is large.

A. Natural cutoffs

Two scales enter the criterion (77). The ultraviolet cutoff p^Λ controls non-universal quantities: the shift μ_c of the critical point (defined after Eq. (70)) and the amplitude of the $O(\mu^{2-N/\sigma})$ correction in Eq. (74). The infrared scale is the size L (diameter) of the network ($L = p^r$, $r \in \mathbb{Z}$). The universal content of this section, namely $N_c = 2\sigma$, the power $\mu^{N/\sigma-2}$, and the function Φ, does not depend on either, provided the scale ℓ_W at which the criterion is evaluated lies many levels of the hierarchy away from both: $p^{-\Lambda} \ll \ell_W \ll L$. Since the terms of $\Phi(s)$ decay geometrically away from the level $k = -\log_p \ell_W$, the cutoffs then change these results only by relative amounts of order $p^{-(2\sigma-N)(\Lambda+\log_p \ell_W)}$ and $(\ell_W/L)^N$. This scale separation is the real assumption behind any Ginzburg criterion; B-C make it implicitly when they extract the infrared singular behavior of their Eq. (51). In the p-adic model both cutoffs are natural, and both are exact.

The network cutoff. Fields on the finite network G_l^N belong to $\mathcal{D}^l$: they are supported in $\mathbb{Z}_p^N$ and constant on the balls of radius p^{-l}. By Appendix D, their Fourier transforms are supported in $\|\xi\|_p \le p^l$ and constant on the balls of radius 1. Hence $\Lambda = l$. The momenta form the finite set $p^{-l}\mathbb{Z}_p^N/\mathbb{Z}_p^N$, one class per neuron: $\xi = 0$, and $(1-p^{-N})p^{kN}$ classes on each sphere $\|\xi\|_p = p^k$, $1 \le k \le l$. By Proposition 1 this is not an approximation. Suppose the network keeps only the connections inside $\mathbb{Z}_p^N$, i.e., it uses $w_{\rm in} = w\, 1_{\mathbb{Z}_p^N}$, and set $\mu_l = \alpha - f'(0)\hat{w}_{\rm in}(0)$. The characters $\chi_p(\xi\cdot x)$, $\xi \in p^{-l}\mathbb{Z}_p^N$, are then exact eigenfunctions of the linearized dynamics on $\mathbb{Z}_p^N$, with decay rates $\mu_l + A_{w_{\rm in}}(\xi)$, and the finite-network version of Eq. (73) is

$$I_2^{(l)}(\mu_l) = \frac{1}{\mu_l^2} + (1-p^{-N}) \sum_{k=1}^{l} \frac{p^{kN}}{\big(\mu_l + A_{w_{\rm in}}(p^k)\big)^2}. \quad (79)$$

For a power law inside the network, $w_{\rm in}(x) = C_w \|x\|_p^{-N-\sigma}$ for $p^{1-l} \le \|x\|_p \le 1$, Eq. (54) gives, for $1 \le k \le l$,

$$A_{w_{\rm in}}(p^k) = \kappa_W p^{k\sigma} - c_\infty, \qquad c_\infty = f'(0)\, C_w\, \frac{1-p^{-N}}{p^\sigma - 1}, \quad (80)$$

where $c_\infty < \kappa_W p^\sigma$ is the coupling lost to the outside of $\mathbb{Z}_p^N$. Three consequences follow:

- The finite network is critical at $\mu_l = 0$, i.e. $\alpha = f'(0)\hat{w}_{\rm in}(0)$. Only the uniform mode becomes critical: every mode with $\xi \neq 0$ keeps a decay rate bounded below, $\mu_l + A_{w_{\rm in}}(\xi) \ge A_{w_{\rm in}}(p) = \kappa_W p^\sigma - c_\infty > 0$ (a spectral gap), and relaxes in a finite time even at $\mu_l = 0$.

- The scaling form (74) holds only while the relevant momentum $\|\xi\|_p \approx \ell_W^{-1}$ lies between p and p^l, i.e., in the *scaling window* $p^{-l} \ll \ell_W \ll 1$ (equivalently, $\kappa_W \ll \mu_l \ll \kappa_W p^{l\sigma}$): the diffusion length must be large compared with the size of a neuron (a leaf of the tree) and small compared with the size of the network. When ℓ_W becomes comparable to the size of the network, the zero mode $1/\mu_l^2$ dominates. This is finite-size rounding: on a finite network an absorbing-state transition is never sharp. The depth l is the natural finite-size parameter.

- On a network whose neuron space is the ball $B_r^N = p^{-r}\mathbb{Z}_p^N$, of diameter $L = p^r$ (here $r = 0$, $L = 1$), the left-hand side of (77) is largest at $\ell_W \approx L$. Mean-field theory therefore fails somewhere inside the scaling window if and only if the Ginzburg number

$$\mathrm{Gi} = \frac{g^2}{\kappa_W^2}\, L^{2\sigma-N} \quad (81)$$

is of order one or larger. If $\mathrm{Gi} \ll 1$, mean-field theory holds throughout the window, and the approach to $\mu_l = 0$ is governed by the fluctuations of the zero mode, i.e. by the finite size of the network. On $\mathbb{Q}_p^N$ ($L = \infty$) and for $N < 2\sigma$, mean-field theory fails sufficiently close to criticality, whatever the value of g.

The exhaustion $p^{-l}\mathbb{Z}_p^N \uparrow \mathbb{Q}_p^N$ of Sec. III A removes both cutoffs at once: the momenta run over $p^{1-l} \le \|\xi\|_p \le p^l$.

The kernel cutoff. In $\mathbb{Q}_p^N$ the Fourier transform exchanges the scale of local constancy with the radius of the support (Appendix D). A radial w is constant on each sphere, so if it is constant on the ball B_{-m}^N it is constant on all balls of radius p^{-m}, and then $\hat{w}(\xi) = 0$ for $\|\xi\|_p > p^m$. In this case the integrand of the exact criterion vanishes for $\|\xi\|_p > p^m$, the cutoff $\Lambda = m$ is supplied by w itself, and $A_w(\xi) = f'(0)\hat{w}(0)$ exactly for $\|\xi\|_p > p^m$. This makes precise B-C's remark that the integrals are "well behaved in the ultraviolet because of the weight function, if not by a neural lattice cutoff" [13]. A Euclidean kernel cannot do this: a compactly supported $\hat{w}$ forces w to be analytic, hence not compactly

supported. Biologically, p^{-m} is the size of the smallest population that the connectivity distinguishes. The dual statement concerns the support. If w is supported in a ball of radius p^r, then $A_w(\xi) = 0$ for $\|\xi\|_p \le p^{-r}$. In an ultrametric space this is literal: each neuron interacts only inside its own ball of radius p^r, and the network splits into disconnected pieces. All the modes with $\|\xi\|_p \le p^{-r}$ become massless together, $I_2 \sim \mu^{-2}$, and each piece behaves as a finite network, with no scaling regime. Eq. (60) excludes this case (Lemma 4): a power-law criterion requires connections at all hierarchical distances.

Comparison with field theories of artificial neural networks. In the field-theoretic approaches to DNNs, the natural large parameter is the width n of the layers, or the number of neurons [20–22, 26, 27]. At fixed depth, the infinite-width limit is a Gaussian theory, and the non-Gaussian corrections come in powers of $1/n$; in Ref. [26] the effective expansion parameter is the depth-to-width ratio. Other works regularize the perturbative integrals with a cutoff in input space and obtain the Wilsonian running of the couplings by requiring that physical predictions do not depend on it [21, 58]. The p-adic setting differs in two ways. First, width and depth cannot be separated. The network G_l^N has depth l and p^{Nl} neurons, and the thermodynamic limit $l \to \infty$ (Sec. III A) adds layers to the hierarchy at the same time as it adds neurons: the infinite-width limit involves infinitely many layers. Second, the limiting theory is not Gaussian. The continuum action has the cubic couplings of Eq. (59), and for $N < 2\sigma$ the fluctuations they produce are not small near criticality. The cutoffs $\Lambda = l$ and L are part of the architecture, and the Ginzburg number (81) plays the role that $1/n$, or the depth-to-width ratio, plays in the works above. In addition, the fluctuations here come from the stochastic spiking dynamics, not from the random initialization of the weights. If the infinite-width limit is understood as a network with one layer and infinitely many neurons, the p-adic SFTs are of a completely different nature from the lattice SFTs of Ref. [26]. This supports the proposal of Ref. [28] that continuous SFTs of B-C type, rather than lattice SFTs with a large but finite number of neurons, should be the basis of the correspondence between DNNs and SFTs.

VII. LOOP-CORRECTED HIERARCHICAL WILSON–COWAN EQUATIONS

B-C's one-loop equations [Eqs. (48)–(49) and (54)–(57) of Ref. [13]] carry over. Let $s(x,t) = (w \star a)(x,t) + I$ and let $C(x,y,t)$ be the equal-time connected correlation of φ (a factorial cumulant of n). At one loop,

$$\begin{aligned}\partial_t a(x) =& -\alpha a(x) + f(s(x)) \\ &+ \tfrac{1}{2} f''(s(x)) \int_{\mathbb{Q}_p^N}\int_{\mathbb{Q}_p^N} w(\|x-y\|_p)\, w(\|x-z\|_p)\, C(y,z)\, d^N y\, d^N z, \qquad (82)\\ \partial_t C(x,y) =& -2\alpha C(x,y) + f'(s(x))\,(w \star_1 C)(x,y) + f'(s(y))\,(w \star_2 C)(x,y) \\ &+ f'(s(x))\, w(\|x-y\|_p)\, a(y) + f'(s(y))\, w(\|x-y\|_p)\, a(x), \qquad (83)\end{aligned}$$

where $\star_i$ denotes convolution over $\mathbb{Q}_p^N$ in the ith argument, e.g. $(w \star_1 C)(x,y) = \int_{\mathbb{Q}_p^N} w(\|x-u\|_p)\, C(u,y)\, d^N u$. The source term in Eq. (83) comes from the $(2,0)$ vertex (37), and the one-loop term in Eq. (82) from the $-\frac{1}{2} f''(s)\, \tilde{\varphi}\, (w \star \varphi)^2$ vertex of Sec. III D. Equations (82)–(83) coincide with the generalized activity equations (3.23)–(3.24) of Ref. [59], written for a radial kernel on $\mathbb{Q}_p^N$.

Two consequences:

- *Correlation matrices between balls.* Expanding C in the p-adic wavelets $\psi_{\gamma nj}$, $C = \sum C_{(\gamma nj),(\gamma' n' j')} \psi_{\gamma nj} \otimes \psi_{\gamma' n' j'}$, gives correlations between balls at arbitrary levels. These objects can be compared directly with functional-connectivity matrices.
- *Mathematical program.* Well-posedness of Eqs. (82)–(83) on $\mathbb{Z}_p^N$, convergence of level-l truncations, positivity of a for saturating f, and stability of hierarchical patterns. B-C remark that N controls pattern formation; here patterns are indexed by balls, and the question becomes which balls synchronize.

VIII. AVALANCHES

An *avalanche* is the cascade of activity triggered by a single spike in an otherwise quiescent network. Two conditions make the background quiescent: $\bar{n} = 0$, i.e. there are no spikes initially, and $f(0) = 0$, i.e. there is no spontaneous firing, so that the quiescent state is absorbing. The seed spike may make other neurons fire, each of these may trigger further spikes, and so on, until the activity dies out and the network returns to the absorbing state. An avalanche is thus a transient excur-

sion from the quiescent state; it changes the activity but not the parameters α, $f'(0)$ and w, and hence not the distance μ to criticality. Avalanches are a probe, not a mechanism: their statistics, for example the mean size $\langle S\rangle = 1/\mu_R$ and the cutoff of the size distribution, reveal how close the network is to the critical point, but they do not change μ. Only a slow, activity-dependent plasticity acting on the parameters, i.e. a change of the synaptic strengths driven by the activity itself, such as synaptic depression, can bring the network there (self-organized criticality, Sec. X G). The *size* S of the avalanche is the total number of spikes, and its *duration* T is the time until extinction. In the language of directed percolation, an avalanche is a cluster grown from a single seed (Appendices B 1–B 2), and its statistics are controlled by the distance to the DP transition:

- in the absorbing phase ($\mu > 0$), a spike has on average fewer than one descendant; avalanches are finite, and $P(S)$ is cut off exponentially beyond a cutoff size S_c: $P(S) \simeq S^{-\tau_S} G(S/S_c)$, where $G(u)$ tends to a constant for $u \ll 1$ and decays exponentially for $u \gg 1$. For $N \le N_c$, $S_c \sim \mu^{-(\nu_\parallel + N\nu_\perp - \beta)}$ (hyperscaling); in mean-field theory, $N \ge N_c$, $S_c \sim \mu^{-2}$, the value of this formula at $N = N_c = 2\sigma$;
- in the active phase ($\mu < 0$), a finite fraction $P_\infty \sim |\mu|^{\beta'}$ of the avalanches never stops;
- at the critical point, the branching ratio is one, the cutoff size S_c diverges, and the distributions become power laws, $P(S) \sim S^{-\tau_S}$ and $P(T) \sim T^{-\tau_T}$; the mean avalanche size $\langle S\rangle = 1/\mu_R$ (Sec. VI) diverges.

Only at the critical point are avalanches scale-free: in the absorbing phase the cutoff S_c sets a characteristic size, and in the active phase some avalanches never stop. Scale-free avalanches are thus the signature of a network poised at the DP transition, and their exponents are universal: they are fixed by the DP exponents β, β', $\nu_\parallel$ and $\nu_\perp$. This is why B-C interpreted the avalanches observed in cortical slices [3] as evidence of a DP transition, and why τ_S tests the universality class.

Following B-C [Eqs. (60)–(64)], the avalanche-size distribution $P(S) \sim S^{-\tau_S}$ is obtained from the path integral with $\bar{n} = 0$ and a single seed. In terms of the survival exponent $\delta = \beta'/\nu_\parallel$ and the initial-slip exponent ϑ, $\tau_S = 1 + \delta/(1 + \vartheta + \delta)$ [60]. Below N_c, hyperscaling gives $\vartheta = (N\nu_\perp - \beta - \beta')/\nu_\parallel$, and then

$$\tau_S = 1 + \frac{\beta'}{\nu_\parallel + N\nu_\perp - \beta}. \tag{84}$$

- For $N \ge 2\sigma$ (mean field), $\vartheta = 0$ and $\delta = 1$, so $\tau_S = 3/2$ and the duration exponent is 2. The p-adic theory reproduces the Beggs–Plenz exponent [3], as B-C's does. At $N = 2\sigma$ the hyperscaling expression also gives $1 + 1/(1 + 2 - 1) = 3/2$, consistently.
- For $N < 2\sigma$, the exponents acquire $O(\varepsilon)$ corrections that depend on σ, and $\tau_S \neq 3/2$. This is confirmed by the simulations of Sec. VIII A.

A. Numerical simulations

The statement that τ_S departs from 3/2 for $N < 2\sigma$ rests on power counting. We tested it by simulating the effective spike model directly.

Model and method. The network is the tree G_l with $p = 2$, $N = 1$ and depth $l = 40$ (2^{40} neurons, so that the boundary of the network is not reached on the simulated time scales, except marginally for the smallest σ). Two neurons $i \neq j$ have a last common ancestor in the tree; the neurons below it form the smallest ball containing both. If this ball has 2^k neurons ($k = 1, \dots, l$), i.e. if the common ancestor lies k levels above the leaves, then $|i - j|_2 = 2^{k-l}$. The coupling between them is $w \propto 2^{-k(1+\sigma)} \propto |i - j|_2^{-(1+\sigma)}$ [this is Eq. (53) with $N = 1$], with no self-coupling, normalized to total coupling $w_0 = 1$. Each effective spike decays at rate $\alpha = 1$, and neuron i emits a spike at rate $f(s_i)$ with the firing rate function Eq. (23), $s_0 = 0.25$. The control parameter is $b = f'(0)w_0/\alpha$; mean-field theory places the critical point at $b = 1$, i.e. $\mu = 0$. The simulation is exact in continuous time: since $f(s) \le f'(0)s$, each spike attempts to trigger a spike at rate b at a neuron chosen with probability w_{ij}/w_0, and the attempt is accepted with probability $f(s_i)/[f'(0)s_i]$. An avalanche starts from a single spike and ends when no spike is left. Its size S is the total number of spikes.

For each σ the critical value b_c was located by bisection, requiring that the mean number of spikes $\mathcal{N}(t)$ be a pure power law of t up to $t = 10^3$. At b_c we generated 2×10^5 avalanches, followed up to $t = 10^3$. We measured the exponents δ and ϑ of the survival probability $P(t) \propto t^{-\delta}$ and of $\mathcal{N}(t) \propto t^{\vartheta}$ for $30 \le t \le 10^3$, and the exponent τ_S by a maximum-likelihood fit of the size distribution for $10 \le S \le 2000$. As a control we ran the same program on a Euclidean network, a chain with nearest-neighbor coupling, and on the model without saturation (a pure branching process). The exact exponents are known in both cases.

Methods. The simulations follow standard methods for absorbing-state transitions [19]. The continuous-time dynamics is simulated exactly, by an event-driven (Gillespie-type) algorithm [61] combined with thinning [62], as described above, so that the saturation of f is treated without approximation. Each run starts from a single spike, as in the spreading method of Grassberger and de la Torre [63], and the critical point is located by requiring that $\mathcal{N}(t)$ be a pure power law. The exponents δ and ϑ are obtained by least-squares fits on logarithmic scales; τ_S is obtained by maximum likelihood for a power law truncated to the fitting window [64], with bootstrap errors, and independently from the scaling relation $\tau_S = 1 + \delta/(1 + \vartheta + \delta)$ [60]. The

TABLE II. Simulation results at the critical point. b_c: critical value of $b = f'(0)w_0/\alpha$ (mean-field value: 1). δ, ϑ: survival and growth exponents. τ_S (fit): maximum-likelihood fit of the size distribution; statistical errors are below 0.005. τ_S (scaling): from $\tau_S = 1 + \delta/(1 + \vartheta + \delta)$. In parentheses, the exact or known values of $(\delta, \vartheta, \tau_S)$ for the two controls: the critical branching process and DP in one dimension. The two controls show that the fitted τ_S is systematically too large by about 0.03.

	b_c	δ	ϑ	τ_S (fit)	τ_S (scaling)
Pure branching (exact: 1, 0, 3/2)	1.000	1.02	-0.06	1.53	1.52
Tree, $\sigma = 0.25$	1.106	0.99	+0.03	1.52	1.49
Tree, $\sigma = 0.5$	1.371	0.90	-0.01	1.46	1.48
Tree, $\sigma = 0.75$	1.739	0.65	+0.16	1.39	1.36
Tree, $\sigma = 1$	2.169	0.49	+0.24	1.32	1.28
Tree, $\sigma = 1.5$	3.157	0.28	+0.32	1.23	1.17
Euclidean chain (DP: 0.159, 0.314, 1.108)	4.848	0.16	+0.31	1.14	1.11

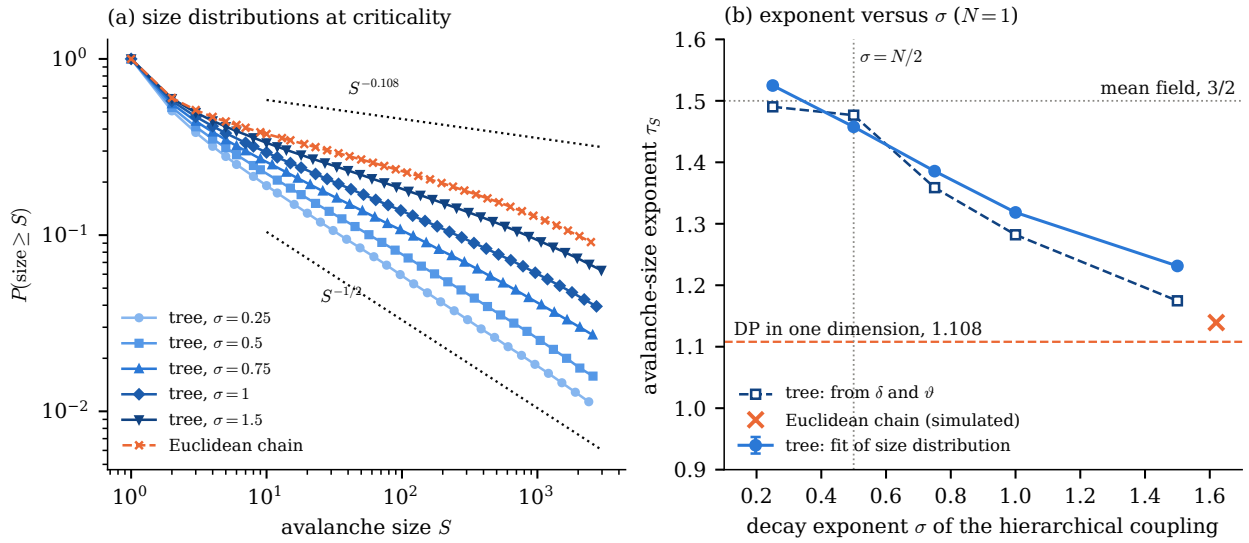


FIG. 5. Avalanches at the critical point. (a) Probability that an avalanche has size at least S, for the hierarchical network with five values of σ and for a Euclidean chain with nearest-neighbor coupling. Dotted lines: the mean-field law $S^{-1/2}$ ($\tau_S = 3/2$) and the law $S^{-0.108}$ of directed percolation in one dimension. (b) Avalanche-size exponent τ_S against σ. Filled circles: fit of the size distribution. Open squares: from the measured δ and ϑ. Cross: the Euclidean chain with nearest-neighbor coupling (simulated, fit of the size distribution); it has no parameter σ and is drawn to the right of the tree data. Dashed horizontal line: the value $\tau_S = 1.108$ of directed percolation in one dimension [60], not a simulation. The vertical line marks $\sigma = N/2$, where $N = N_c$. Parameters as in Table II.

two controls, with known exponents, calibrate the systematic bias of the fit. The simulation code, the data and the scripts that produce the figures are available at `https://wazuniga.github.io/padic-BC/`. We used Claude (Anthropic; model claude-opus-5-5) to help implement the numerical simulation code of this section and of Sec. IX, for some well-known techniques in statistical physics [19, 61–63]. The control was carried out as discussed above.

Results. Table II and Fig. 5 show three things.

1. *The controls are reproduced.* The chain gives $\delta \simeq 0.16$ and $\vartheta \simeq 0.31$, the values of directed percolation in one dimension. The pure branching process gives $\delta \simeq 1$ and $\vartheta \simeq 0$. In both cases the fitted τ_S exceeds the exact value by about 0.03, which we take as the systematic error of the fit in this window.

2. *Mean-field behavior for $\sigma \leq N/2$.* For $\sigma = 0.25$ the exponents are those of the branching process, $\tau_S \simeq 3/2$. At $\sigma = 0.5 = N/2$ they are close to the mean-field values, with small deviations of the size expected from logarithmic corrections at the upper critical dimension.

3. *Non-mean-field behavior for $\sigma > N/2$.* For $\sigma = 0.75$, 1 and 1.5, τ_S decreases steadily, from about 1.36–1.39 to about 1.18–1.23, and ϑ becomes positive. The exponents vary continuously with σ and move toward the short-range values as σ grows.

These results agree with $N_c = 2\sigma$, and they show that

the hierarchical decay exponent σ controls the avalanche statistics. They are numerical evidence, not precise determinations: the runs are limited to $t \leq 10^3$, the critical points carry an uncertainty of order 10^{-3}, and the two estimates of τ_S differ by up to 0.06. The simulation also shows that the critical point is shifted well above its mean-field value, $b_c > 1$, as the one-loop estimate of Appendix A 5 indicates qualitatively. The shift is large here because a single spike already produces an input comparable to s_0 at its closest neighbors. We also simulated, with the same rules and the same procedure, a Euclidean chain with power-law coupling $w(r) \propto r^{-1-\sigma}$, whose critical point is directed percolation with long-range spreading [44, 49]. For $\sigma \leq 1.5$ its avalanche exponents coincide with those of the hierarchical network within numerical accuracy (the fitted values of τ_S differ by at most 0.005 for $\sigma \leq 1$ and by 0.02 at $\sigma = 1.5$), while the critical points differ. Whether the two networks differ at larger σ, where the Euclidean theory crosses over to ordinary directed percolation, is an open question.

Comparison with the Euclidean model. The chain is more than a control: it is the Euclidean counterpart of the tree, with the same number of dimensions ($N = 1$), the same local rules and the same total coupling. The two networks behave differently for a simple reason. On the chain a spike excites only its nearest neighbors, so the activity spreads as a compact patch and new spikes keep landing on neurons that already receive input. This crowding is what the saturation of f penalizes, and it drives the exponents away from their mean-field values: $\tau_S \simeq 1.11$, far from $3/2$. On the tree with small σ the coupling decays slowly with the hierarchical distance, most new spikes land far from the active neurons, crowding is rare, and the avalanches are those of a branching process. As σ grows the activity stays inside small balls, crowding returns, and τ_S falls toward the Euclidean value. Hence in the Euclidean theory with short-range coupling the dimension of space decides whether the avalanches are mean-field, and the Beggs–Plenz exponent $3/2$ requires $N \geq 4$; in the p-adic theory the decay exponent σ decides, and a network with $N = 1$ gives $3/2$ when $\sigma \leq 1/2$. The shift of the critical point shows the same ordering: b_c grows from 1.11 at $\sigma = 0.25$ to 3.16 at $\sigma = 1.5$, and is largest for the chain, $b_c = 4.85$. Mean-field theory is therefore a better guide for a hierarchical network with slowly decaying coupling than for the Euclidean chain. Section IX relates this difference to the small-world property. This comparison is limited to one tree ($p = 2$, $N = 1$) and one value of s_0; other values of p and N were not simulated.

Thus an estimate of σ from connectivity data, from how the coupling decays with hierarchical distance, predicts whether the avalanche exponents are mean-field ($\sigma \leq N/2$) or not ($\sigma > N/2$); the size of the departure is, at present, known only from simulations.

IX. THE SMALL-WORLD PROPERTY

A network has the small-world property if it combines a high clustering of connections with short paths: the mean number of steps between two nodes grows as the logarithm of the number of nodes, as in a random graph, and not as a power of it, as in a lattice [10]. Cortical networks have this property [11]. In this section we ask whether the networks described by the p-adic theory and by the B-C theory have it.

A. Two tests for a network with weighted coupling

In the effective spike model every neuron is coupled to every other neuron, with a weight w that depends on the distance. Path lengths in the usual sense are then trivially equal to one, so the definition has to be adapted. We use two tests.

1. *Sparse network.* Each neuron draws k connections, and the probability of a connection to a given neuron is proportional to w. On the resulting network we measure the clustering coefficient C (the probability that two neighbors of a neuron are connected to each other) and the mean path length ℓ.

2. *Spreading time.* We simulate the effective spike model itself, with the full weighted coupling, in the active phase. One spike is placed in a network of n neurons, and $T_{1/2}(n)$ is the mean time at which half of the neurons have fired at least once. Runs in which the activity dies out are discarded.

The network is small-world if C stays finite while ℓ and $T_{1/2}$ grow as $\ln n$, or as a power of $\ln n$.

B. A simple estimate

Consider the kernel (53), $w \propto \|x\|_p^{-N-\sigma}$, and a ball of level m, which has p^{mN} neurons. The total coupling of one neuron to the neurons outside the ball is proportional to $p^{-m\sigma}$. The coupling of the whole ball to its exterior is therefore proportional to

$$p^{mN}\, p^{-m\sigma} = p^{m(N-\sigma)}. \tag{85}$$

If $\sigma < N$ this grows with m: the larger the structure, the more strongly it is connected to the rest of the network, at every level. Once a ball is active, its activity passes to the next level in a time that does not grow with m, and the time to reach $n = p^{lN}$ neurons is at most of order $l \propto \ln n$. If $\sigma > N$ the quantity (85) decreases with m: large structures are almost isolated, activity must wait a time of order $p^{m(\sigma-N)}$ to leave a ball of level m, and the spreading time grows as the power $n^{(\sigma-N)/N}$. The clustering is high in both cases, since most of the coupling

of a neuron stays inside the smallest balls that contain it. This estimate gives the threshold

$$\sigma < N \tag{86}$$

for the small-world property. It is the same threshold that separates the two regimes of long-range percolation on a hierarchical lattice [65] and on a Euclidean lattice [66, 67]. It is also the condition under which the random walk generated by the Taibleson–Vladimirov operator is transient, since $\int \|k\|_p^{-\sigma} d^N k$ converges at $k = 0$ exactly when $\sigma < N$. (Heuristic.)

C. Numerical results

We took $p = 2$ and $N = 1$, as in Sec. VIII A, with finite trees of $n = 2^l$ neurons, and compared them with Euclidean rings of the same size. The sparse networks have $k = 6$ and n up to 2^{18}. The spreading times were measured with the simulator of Sec. VIII A, with $s_0 = 0.25$ and $b = 6$, which is above the critical point of every network considered, with 5 to 37 surviving runs per point. For the Euclidean ring with power-law coupling, $w(r) \propto r^{-1-\sigma}$ with r the distance along the ring, we wrote a separate program with the same rules. The results are in Fig. 6 and Table III.

Hierarchical network. For $\sigma = 0.25$ and $\sigma = 0.5$ the path length and the spreading time increase by an approximately constant amount each time the network is enlarged by a factor of four: they grow as $\ln n$. The clustering coefficient stays between 0.08 and 0.18, of order a thousand times that of a random graph. These networks are small-world. For $\sigma = 0.75$ the growth is faster than $\ln n$ but slower than a power of n. The case $\sigma = 1 = N$ is marginal: over the range studied, ℓ and $T_{1/2}$ grow approximately as $n^{0.3}$. For $\sigma = 1.5$ the sparse network falls apart into small pieces (the largest contains less than 1% of the neurons at $n = 2^{16}$), and $T_{1/2}$ grows as $n^{0.55}$, close to the estimate $n^{\sigma-N} = n^{0.5}$. A tree is therefore not automatically a small-world network: the property holds for $\sigma < N$, in agreement with Eq. (86).

Euclidean networks. On the ring with short-range coupling both ℓ and $T_{1/2}$ are proportional to n: activity advances as a front with constant velocity, as in the B-C theory. At $n = 2^{14}$ the spreading time is 7186, against 4.2 on the tree with $\sigma = 0.5$. The ring with power-law coupling behaves like the tree with the same exponent. For $\sigma = 0.5$ the two networks have almost the same clustering, path length and spreading time. For $\sigma = 1.5$ neither is small-world.

TABLE III. Small-world tests for $N = 1$. C and ℓ: clustering coefficient and mean path length of the sparse network with $n = 2^{16}$ neurons. $T_{1/2}$: spreading time of the spike model with $n = 2^{12}$ and $n = 2^{14}$ neurons, in units of $1/\alpha$. The sparse tree network with $\sigma = 1.5$ is not connected. A random graph of the same size and degree has $C \approx 10^{-4}$.

	C	ℓ	$T_{1/2}(2^{12})$	$T_{1/2}(2^{14})$
Tree, $\sigma = 0.25$	0.08	6.0	2.4	2.9
Tree, $\sigma = 0.5$	0.18	8.4	3.5	4.2
Tree, $\sigma = 0.75$	0.27	14.1	5.6	7.5
Tree, $\sigma = 1$	0.34	32.6	10.8	15.7
Tree, $\sigma = 1.5$	0.45	—	57	124
Ring, power law, $\sigma = 0.5$	0.17	8.1	3.2	4.0
Ring, power law, $\sigma = 1.5$	0.40	921	33	65
Ring, short range	0.67	3277	1783	7186

D. Comparison with the B-C theory

The master equation and the action of B-C are written for a general weight function, so the formalism does not exclude a small-world network. The restriction enters in the continuum analysis of the critical point. There the connectivity is assumed to be short range, its Fourier transform is expanded as $\hat{w}(q) \simeq w_0 - \frac{1}{2} w_2 q^2$, which gives the diffusion term $D\nabla^2$, and the Ginzburg criterion uses the moments $w_2, w_4, \ldots$ [Eqs. (50)–(52) of Ref. [13]]. These hypotheses exclude the small-world property. A kernel with power-law tail $|x|^{-N-\sigma}$ produces a small-world network only for $\sigma < N$ [66, 67], and then its moment of order N is infinite. A finite fourth moment forces $\sigma > 4 > N$ for $N \leq 3$. (A finite second moment alone does not suffice when $N = 3$: a tail with $2 < \sigma < 3$ has a finite second moment and gives a small-world network, while the critical behavior remains that of DP.) Under the hypotheses of B-C, activity passes only to nearby neurons, the time to reach n neurons grows as the power $n^{1/N}$, and the network is not small-world. This is the case represented by our short-range ring.

By our tests, which concern finite networks, the property is recovered within the B-C formalism if the hypothesis is dropped and the kernel is taken to have a power-law tail: w bounded for $|x| < 1$ and $w(x) \propto |x|^{-N-\sigma}$ for $|x| \geq 1$, with $\sigma < \min(N, 2)$. The cutoff at short distances is necessary, because the pure power law is not integrable at the origin. With it, w is integrable for every $\sigma > 0$, although its second moment is infinite for $\sigma \leq 2$. Integrability is what the mean-field theory requires. The Wilson–Cowan model on a general locally compact group was studied in Ref. [12] under the hypothesis that the kernels are integrable. The existence proof uses only the inequality $\|w \star a\|_\infty \leq \|w\|_1 \|a\|_\infty$, so that it holds on $\mathbb{R}^N$ as well: for an integrable kernel and a Lipschitz firing-rate function the Wilson–Cowan equation has a unique global solution in L^∞. Hence the long-range Euclidean model is well defined at the mean-field level. In our simulations the neurons are on a lattice, $r \geq 1$, and the kernel is summable. The diffusion term is then replaced by the fractional Laplacian $(-\nabla^2)^{\sigma/2}$, and the critical behavior is that of directed percolation with long-range spreading [44, 49], whose upper critical dimension is also

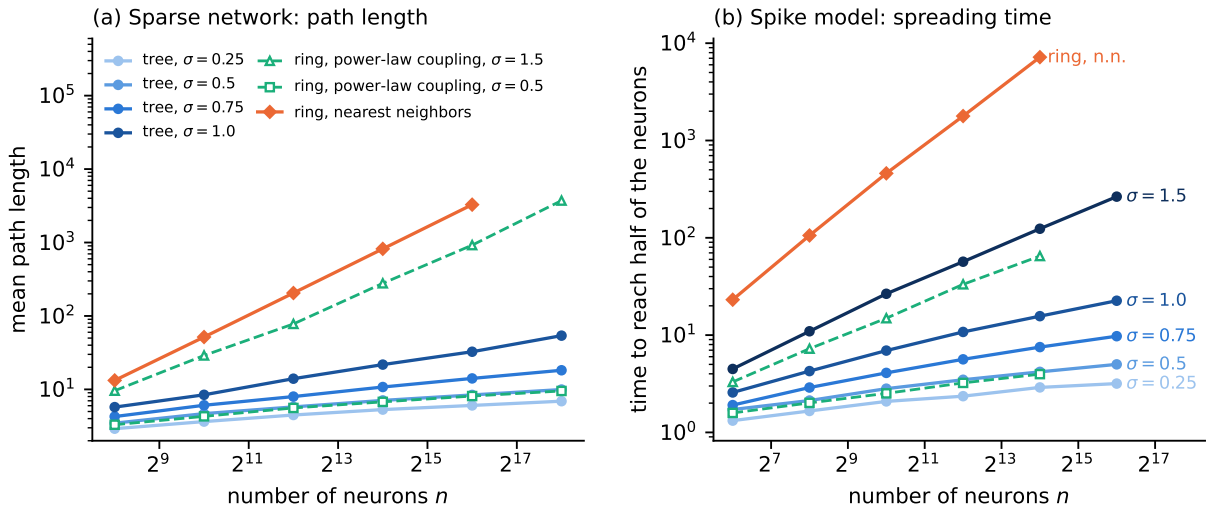


FIG. 6. Small-world tests. (a) Mean path length of sparse networks with n neurons, in which each neuron draws $k = 6$ connections with probabilities proportional to the coupling. Filled circles: tree with $p = 2$, $N = 1$ (the network with $\sigma = 1.5$ is not connected and is not shown). Open symbols: Euclidean ring with power-law coupling. Diamonds: Euclidean ring in which each neuron is connected to its five nearest neighbors on each side. (b) Spike model in the active phase, $b = 6$: mean time at which half of the n neurons have fired at least once, for the same networks; here the short-range ring has nearest-neighbor coupling. Error bars are standard errors and are mostly smaller than the symbols.

2σ. Our simulations of the ring with power-law coupling show that this network is small-world for $\sigma = 0.5$.

The conclusion is the following. The B-C theory in its diffusion limit describes networks that are not small-world. The small-world property requires long-range coupling. In the Euclidean setting this coupling must be imposed as an additional hypothesis, on a theory whose generic limit is local diffusion. In the p-adic setting all diffusion operators are nonlocal (Sec. IV A), so long-range coupling is the only possibility, and the property holds for $\sigma < N$. What distinguishes the p-adic theory from a finite Euclidean network with a power-law coupling is the hierarchy: nested balls, discrete scale invariance and log-periodic corrections, none of which the Euclidean kernel has.

Relation to the compactness argument of Ref. [12]. Reference [12] treats the small-world property of the Wilson–Cowan model in the continuum, with a different and stronger criterion: the interaction between any two groups of neurons must be non-negligible, $w(x) \geq \varepsilon > 0$ for all x. The kernels must also be integrable, since this is the hypothesis under which the Wilson–Cowan equations are well posed on a general group. The two conditions together force the group to be compact. The group $(\mathbb{R}^N, +)$ has no nontrivial compact subgroups, which is a consequence of the Archimedean axiom. Hence the classical model, whose Gaussian or exponential kernels have compact support for practical purposes, is incompatible with the small-world property, and by this criterion so is every integrable kernel on $\mathbb{R}^N$, including one with a power-law tail. In $\mathbb{Q}_p^N$ the balls are compact subgroups. On a ball every positive radial kernel satisfies the criterion, and the theory restricted to a ball is again a theory of the same type (Proposition 1).

The two approaches answer different questions and are consistent. The criterion of Ref. [12] concerns the continuum model on the whole space. Our tests concern finite networks of increasing size, and they separate the kernels on a ball further: path lengths and spreading times grow logarithmically only for $\sigma < N$. The Euclidean ring with power-law coupling used in our simulations does not contradict the compactness argument. A ring is a compact group, but it is not a subgroup of $\mathbb{R}^N$: it is obtained by imposing periodic boundary conditions, and the power-law coupling is a further hypothesis foreign to the classical model. In the p-adic setting the space does not have to be changed: every finite network is a ball, and every ball is a compact subgroup.

Small-world property and avalanches. The threshold $\sigma = N$ is not the threshold $\sigma = N/2$ of Sec. VIII. There are three regimes. For $\sigma \leq N/2$ the network is small-world and the avalanche exponents are mean-field. For $N/2 < \sigma < N$ the network is still small-world, but the avalanche exponents are not mean-field: at $\sigma = 0.75$ we found $\tau_S \simeq 1.37$ (Table II). For $\sigma > N$ the network is not small-world and the exponents approach the short-range values. In our simulations the small-world property is thus necessary for the exponent $3/2$, but not sufficient, and the avalanche exponents alone cannot be used as a test of it.

Relation to fast transition modes. In ultrametric models of transition networks, the hierarchical organi-

zation of the states produces fast transition modes that are absent in an ordinary Markov chain [43]. The logarithmic spreading time found here for $\sigma < N$ is an effect of the same kind, since activity leaves each ball at a rate that does not decrease with its size. We have not made this connection quantitative.

Limitations. The tests were made for one prime, $N = 1$, one value of b and network sizes up to 2^{18}. The laws quoted above describe the range studied; in particular, the data do not distinguish $\ln n$ from a power of $\ln n$ for σ close to N.

X. PHASE TRANSITION AND SELF-ORGANIZATION

The review [25] develops the physical consequences of the B-C theory: the neocortical phase transition (Sec. 10 there), and large avalanches, rhythms, correlations, power spectra and self-organized criticality (Sec. 12 there). In this section we give the p-adic counterparts. We label each statement as *exact* (it follows from the p-adic setting at zero-loop order, with no further approximation) or *heuristic*. Equation numbers of the form "R(...)" refer to Ref. [25].

A. The control parameter, layer by layer

Let

$$W_k = \int_{\|x\|_p = p^k} w(\|x\|_p)\, d^N x = (1 - p^{-N})\, p^{kN}\, w(p^k), \tag{87}$$

$k \in \mathbb{Z}$, be the total coupling that a neuron receives from the neurons at hierarchical distance p^k. Then $w_0 = \hat{w}(0) = \sum_k W_k$, and the control parameter of the transition [R(137)] is

$$\mu = \alpha - f'(0) \sum_{k \in \mathbb{Z}} W_k. \tag{88}$$

The balance condition $\mu = 0$ constrains only the total, so the balance between branching ($f'(0)\hat{w}(0)$) and decay (α) can be distributed across layers in many ways. (Exact.) If negative weights are allowed, so that excitation within small balls is compensated by inhibition at larger hierarchical distances, f must be extended to $s < 0$ as in Remark 1.

B. Level-resolved linear spectrum

Proposition 2. *Let w be radial and integrable, and let $\|\xi\|_p = p^{1-\gamma}$, i.e. ξ labels the wavelets supported on balls of radius p^γ. Then*

$$\hat{w}(\xi) = \sum_{j \le \gamma - 1} W_j - \frac{W_\gamma}{p^N - 1}, \tag{89}$$

and the linearized growth rate of the level-γ modes is

$$\varrho_\gamma = -\lambda(p^{1-\gamma}) = -\alpha + f'(0)\Big[\sum_{j<\gamma} W_j - \frac{W_\gamma}{p^N - 1}\Big]. \tag{90}$$

Proof. Using Eq. (2), for $\|\xi\|_p = p^m$, $\int_{\|x\|_p = p^j} \chi_p(\xi \cdot x)\, d^N x$ equals $(1 - p^{-N}) p^{jN}$ if $j \le -m$, equals $-p^{-mN}$ if $j = 1 - m$, and vanishes otherwise. Sum over spheres and set $m = 1 - \gamma$. □

A mode supported on a ball of radius p^γ averages to zero over that ball and is constant on each of the p^N balls of radius $p^{\gamma-1}$ into which it splits (its children in the tree); these are the sub-balls meant below. It receives the full coupling from inside these sub-balls (levels $j < \gamma$). The coupling between different sub-balls (level γ) enters with weight $-1/(p^N - 1)$. For power-law tails, Eq. (89) reproduces Lemma 2. Since $\hat{w}(p^{-\gamma}) - \hat{w}(p^{1-\gamma}) = (p^N W_\gamma - W_{\gamma+1})/(p^N - 1)$ and $W_{\gamma+1}/W_\gamma = p^N\, w(p^{\gamma+1})/w(p^\gamma)$, $\hat{w}$ increases towards coarser levels exactly when w decreases with hierarchical distance (Fig. 7). (Exact.)

Pattern-forming instabilities and large avalanches. If $w \ge 0$, then $|\hat{w}(\xi)| \le \hat{w}(0) = w_0$, so the homogeneous DP transition always comes first, even for non-monotone w. A pattern-forming instability requires inhibition (negative coupling at some hierarchical distances, with the extension of f of Remark 1), as in the Euclidean theory, where the band-pass kernel is a difference of Gaussians. The first instability then occurs at the level γ_c that maximizes the bracket in Eq. (90), and all wavelets of that level, one set per ball of radius p^{γ_c}, become unstable simultaneously, a degeneracy much larger than that of the Euclidean ring $|q| = q_c$. When inhibition is blocked pharmacologically in cortical cultures (e.g. by picrotoxin), the avalanche-size distribution develops a second peak at large sizes, and the review attributes this peak to such an instability (its Sec. 12.5). In the p-adic theory the peak should sit at sizes proportional to $p^{N\gamma_c}$, and several destabilized levels give several peaks at geometrically spaced sizes. (Exact for the linear instability; the avalanche interpretation is heuristic.)

C. Propagator, scaling and exponents

Below criticality the review's propagator R(134) is a Gaussian times $e^{-\Delta t}$. In the p-adic theory,

$$G(x,t) = \theta(t)\, e^{-\mu t}\, Z_t(x),$$
$$Z_t(x) := Z_t^W(x) = \int_{\mathbb{Q}_p^N} \chi_p(-\xi \cdot x)\, e^{-t A_w(\xi)}\, d^N \xi, \tag{91}$$

where $Z_t = Z_t^W$ is the heat kernel of the W-operator of Sec. IV A, understood, as in Eq. (31), as an inverse Fourier transform in $\mathcal{D}'(\mathbb{Q}_p^N)$. It depends on the whole connectivity kernel w through A_w, not only on σ. Since

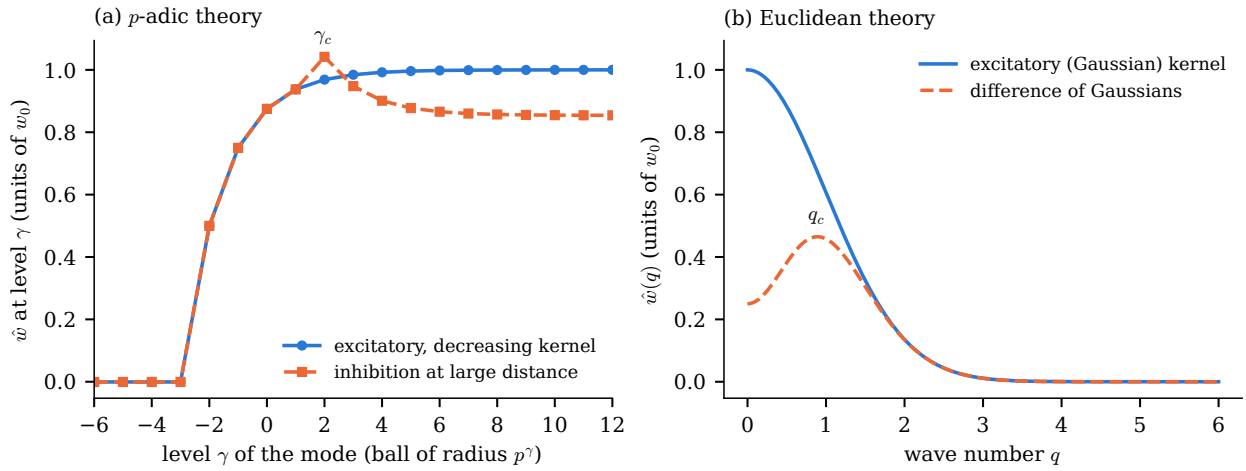


FIG. 7. Spectrum of the linearized dynamics (Sec. X B). Linearizing the Wilson–Cowan equation about the quiescent state gives $\partial_t u = -\alpha u + f'(0)\, w \star u$, whose eigenfunctions are the wavelets; a mode supported on balls of radius p^γ grows at the rate $\varrho_\gamma = -\alpha + f'(0)\hat{w}(p^{1-\gamma})$ [Eq. (90)], so the first instability occurs where $\hat{w}$ is largest. (a) p-adic theory: $\hat{w}$ at each level γ, from Eq. (89), for $p = 2$, $N = 1$. Solid: a nonnegative kernel, constant for $|x|_p \le p^{-3}$ and decaying as $|x|_p^{-2}$ beyond; $\hat{w}$ increases monotonically to its value w_0 at the homogeneous mode ($\gamma \to \infty$). Dashed: the same kernel with inhibitory (negative) coupling for $|x|_p \ge p^2$; the maximum is at a finite level γ_c. (b) Euclidean theory: $\hat{w}(q)$ for a Gaussian kernel and for a difference of Gaussians, whose maximum is at a nonzero wave number q_c. In both theories a pattern-forming instability requires inhibition.

A_w is bounded [$A_w(\xi) \to f'(0)w_0$ as $\|\xi\|_p \to \infty$], Z_t is not a function. By Eq. (32),

$$Z_t = e^{-tf'(0)w_0}\big[\delta + K_t\big], \tag{92}$$

and by Proposition 4, for nonnegative, radially nonincreasing w,

$$G(x,t) \simeq \theta(t)\, f'(0)\, t\, e^{-\mu t}\, w(\|x\|_p), \qquad \|x\|_p \to \infty. \tag{93}$$

For kernels with the power-law tail of Lemma 2, $G(x,t) \simeq \theta(t)\, f'(0) C_w\, t\, e^{-\mu t}\|x\|_p^{-N-\sigma}$. The propagator therefore has *power-law* tails in hierarchical distance for every μ, not Gaussian ones. (Exact.)

The representative $\kappa_W \boldsymbol{D}^\sigma$ of the universality class has an unbounded symbol. Its heat kernel, which we denote by $Z_t^{(\sigma)}$ (it depends only on σ and κ_W), is a genuine function, with no δ part. In Ref. [68], $\boldsymbol{D}^\sigma$ is the Taibleson operator T_σ (Definition 5.16 there, with the same norm $\|x\|_p = \max_i |x_i|_p$), and it is shown there to be an isotropic Laplacian on the ultrametric space $(\mathbb{Q}_p^N, \|\cdot\|_p)$. Theorem 5.17 of that reference gives the two-sided estimate

$$c\, F_t(x) \le Z_t^{(\sigma)}(x) \le C\, F_t(x), \tag{94}$$

with

$$F_t(x) = \kappa_W t \left((\kappa_W t)^{1/\sigma} + \|x\|_p\right)^{-N-\sigma},$$

for all $t > 0$ and $x \in \mathbb{Q}_p^N$, where $0 < c \le C$ depend only on p, N, and σ. The upper bound was obtained earlier in Ref. [69] (see also Ref. [34, Chap. 4] for $N = 1$). The factors of κ_W follow from $Z_t^{(\sigma)} = p_\sigma(\kappa_W t, \cdot)$, where p_σ is the heat kernel of T_σ. Explicit admissible values of c and C are derived in Appendix D 7, Eq. (D20).

Spectral dimension. Let X_t be the Markov process generated by $\kappa_W \boldsymbol{D}^\sigma$ (the p-adic diffusion), started at $X_0 = y$. The function $u(x,t) = Z_t^{(\sigma)}(x-y)$ solves the heat equation $\partial_t u + \kappa_W \boldsymbol{D}^\sigma u = 0$ with initial condition $u(x,0) = \delta(x-y)$, and it is the probability density of X_t. Its value at the starting point, $Z_t^{(\sigma)}(0)$, is the *return probability* (density) of the diffusion; by translation invariance it does not depend on y. For a ball B of radius $r \le (\kappa_W t)^{1/\sigma}$ centered at y, the probability that the walker is back in B at time t is $P(X_t \in B) \asymp Z_t^{(\sigma)}(0)\, r^N$, since $F_t(x) \asymp F_t(0)$ for $\|x\|_p \le (\kappa_W t)^{1/\sigma}$.

The *spectral dimension* d_s of a diffusion is defined by the decay of its return probability at large times: if h_t denotes its heat kernel, $h_t(0) \asymp t^{-d_s/2}$. For Brownian motion in $\mathbb{R}^N$ with diffusion constant D, $h_t(0) = (4\pi D t)^{-N/2}$, so $d_s = N$: the spectral dimension coincides with the dimension of space. On fractals, graphs and ultrametric spaces the two can differ. The name refers to the spectrum of the generator. Writing $h_t(0) = \int e^{-tA(\xi)}\, d^N\xi = \int_0^\infty e^{-t\lambda}\, d\mathcal{N}(\lambda)$, where A is the symbol of the generator and $\mathcal{N}(\lambda) = \mathrm{vol}\{\xi : A(\xi) \le \lambda\}$ counts the modes with relaxation rate at most λ, the decay $t^{-d_s/2}$ corresponds to $\mathcal{N}(\lambda) \asymp \lambda^{d_s/2}$ as $\lambda \to 0$: a Weyl law in which d_s plays the role of the dimension. For the p-adic diffusion generated by $\kappa_W \boldsymbol{D}^\sigma$, $\mathcal{N}(\lambda) = \mathrm{vol}\{\xi : \kappa_W \|\xi\|_p^\sigma \le \lambda\} \asymp (\lambda/\kappa_W)^{N/\sigma}$ (compare the spectral distribution function of Ref. [68], Sec. 5.3.2). Directly, at $x = 0$, Eq. (94) gives

$$Z_t^{(\sigma)}(0) \asymp (\kappa_W t)^{-N/\sigma} = (\kappa_W t)^{-d_s/2}, \qquad d_s = \frac{2N}{\sigma}, \tag{95}$$

so the spectral dimension of the p-adic diffusion gener-

ated by $\kappa_W \boldsymbol{D}^\sigma$ is $d_s = 2N/\sigma$. The neuron space has dimension N, the Haar volume of a ball of radius r being r^N, and the hierarchy enters only through σ. The mean-field condition $N \geq 2\sigma$ of Sec. V is exactly $d_s \geq 4$, the same criterion as in the Euclidean theory with local coupling, where $d_s = N$ and the condition reads $N \geq 4$. Likewise, the transience criterion $\sigma < N$ of the random walk generated by $\boldsymbol{D}^\sigma$, which is also the small-world threshold (86) of Sec. IX, is $d_s > 2$, as for Euclidean diffusion. Thus the hierarchy does not change the dimension of the neuron space; it changes the spectral dimension of the diffusion that the connectivity supports, and this is what decides whether fluctuations are relevant. For the W-operator of the B-C theory, Eq. (94) is expected to describe the regular part K_t in the scaling regime of large t and $\|x\|_p$, where only the small-ξ behavior $A_w \simeq \kappa_W \|\xi\|_p^\sigma$ matters. Its tail agrees in form with Eq. (93). Proving this is an open problem; see Remark 3.

Remark 3 (Open problem: heat-kernel estimate for the regular part K_t). Let A_w be radial and satisfy Eq. (60), and write $Z_t = e^{-tf'(0)w_0}\big[\delta + K_t\big]$. The δ part carries the factor $e^{-tf'(0)w_0}$ and is negligible at large t. The problem is to show that the regular part $e^{-tf'(0)w_0}K_t$ obeys the estimate (94) in the scaling regime: there are $t_0 > 0$ and constants $0 < c' \leq C'$ such that

$$c'\, F_t(x) \leq e^{-tf'(0)w_0} K_t(x) \leq C'\, F_t(x), \qquad t \geq t_0,\ x \neq 0, \tag{96}$$

and, more precisely, that $e^{-tf'(0)w_0}K_t(x)/Z_t^{(\sigma)}(x) \to 1$ as $t \to \infty$, uniformly in $x \neq 0$. Theorem 5.17 of Ref. [68] does not apply directly: A_w is bounded, so the W-operator is not of the form $\psi(T_1)$ with ψ a bijection of $[0,\infty)$ (Theorem 5.18 of Ref. [68]), and Z_t has a δ part. A possible route is the series of Appendix D 7. Because A_w is radial, the same summation by parts gives, for $x \neq 0$,

$$Z_t(x) = \sum_{k\in\mathbb{Z}} p^{kN}\, \Omega(p^k\|x\|_p)\, \big(e^{-tA_k} - e^{-tA_{k+1}}\big), \tag{97}$$

where A_k is the value of A_w on the sphere $\|\xi\|_p = p^k$. The terms are nonnegative when A_k is nondecreasing in k. For large t, the dominant terms have $tA_k = O(1)$, hence small p^k, where Eq. (60) gives $A_k = \kappa_W p^{k\sigma}(1 + O(p^{k\delta}))$, and the bounds of Appendix D 7 should carry over. The remaining work is to control the terms with large p^k, where A_k saturates at $f'(0)w_0$, uniformly in $\|x\|_p$.

The review's critical scaling form R(143) holds with $z = \sigma$, but only under the discrete dilations $x \mapsto p^{-1}x$, $t \mapsto p^\sigma t$. Under them $Z^{(\sigma)}_{p^\sigma t}(p^{-1}x) = p^{-N} Z_t^{(\sigma)}(x)$ exactly, and the B-C propagator inherits this covariance asymptotically in the scaling regime. The scaling function therefore carries a periodic dependence on $\sigma^{-1}\log_p t$, the same discrete scale invariance found in Sec. VI.

With the mean-field values (66), the exponent used in the review, $\vartheta = N/z - 2\beta/\nu_\parallel$ [R(136)], where it is denoted θ, becomes $\vartheta = N/\sigma - 2$. The scaling relation $\beta = \frac{1}{2}\nu_\parallel(N/z - \vartheta)$ gives $\beta = 1$, as it should. The susceptibility R(136) is $\chi = \int_0^\infty dt\, \hat{G}(0,t) = 1/\mu$, unchanged: all quantities of the homogeneous ($\xi = 0$) mode coincide with their Euclidean counterparts. (Exact at zero-loop order.)

Active phase. The coherent-state description of the active phase and the time-reversal argument [R(138)–R(140)] do not depend on the geometry and carry over unchanged, with $a \propto \mu^\beta$. The ballistic front of R(141) does not. For $\mu < 0$, combining the growth $e^{|\mu|t}$ with the power-law tail of Eq. (94), the activity reaches the saturation level at $\log_p \|x\|_p \simeq |\mu|t/\big((N+\sigma)\ln p\big)$: it climbs the hierarchy at a constant number of levels per unit time, so recruiting $\mathcal{N}$ neurons takes a time of order $\ln \mathcal{N}$ rather than $\mathcal{N}^{1/N}$, as for Fisher–KPP fronts with fractional diffusion [70]. This heuristic estimate is the small-world spreading studied, with simulations, in Sec. IX.

D. Correlations

The review's correlation lengths and susceptibility (Sec. 12.6 there) are computed in the absorbing state perturbed by a weak background input. In this subsection we therefore relax the assumption $I = 0$ made in Sec. III C. The reason is that with $I = 0$ and $\mu > 0$ the only stationary state is the absorbing one, $a^* = 0$, in which there is no activity and hence no stationary correlations to compute. A weak constant input $I > 0$ produces a small spontaneous firing rate $f(I) > 0$, which maintains a stationary state with low activity, and the correlations of this state are what the review computes. We take $f(I) > 0$ small, so that the homogeneous state is slightly subcritical with $a^* \simeq f(I)/\mu$. The absorbing state is recovered in the limit $I \to 0$.

In the stationary, homogeneous state, $a(x,t) = a^*$ for all x and t, so the input is the same at every neuron, $s(x) = (w \star a^*)(x) + I = \hat{w}(0)\, a^* + I =: s^*$, and, by translation invariance, $C(x,y) = C(x-y)$. Since w is radial, both convolutions in Eq. (83) reduce to $(w \star C)(x-y)$; for instance, $(w \star_2 C)(x,y) = \int w(\|y-u\|_p)\, C(x-u)\, d^N u = (w \star C)(x-y)$. Setting $\partial_t C = 0$, Eq. (83) becomes

$$0 = -2\alpha\, C + 2f'(s^*)\, w \star C + 2f'(s^*)\, a^*\, w,$$

and its Fourier transform in $x - y$ gives

$$\hat{C}(\xi) = \frac{f'(s^*)\, a^*\, \hat{w}(\xi)}{\alpha - f'(s^*)\, \hat{w}(\xi)} = \frac{f'(s^*)\, a^*\, \hat{w}(\xi)}{\lambda(\xi)}, \tag{98}$$

with $\lambda(\xi) = \mu + A_w(\xi)$. Here $\mu = \alpha - f'(s^*)\,\hat{w}(0)$, and $A_w(\xi) = f'(s^*)\big(\hat{w}(0) - \hat{w}(\xi)\big)$ is the symbol (46) with $f'(0)$ replaced by $f'(s^*)$. The value of the inverse Fourier transform of a radial function at $\|x\|_p = p^k$ depends only on its restriction to $\|\xi\|_p \leq p^{1-k}$. On that ball, $A_w(\xi) \simeq \kappa_W \|\xi\|_p^\sigma \leq p^\sigma \mu\, (\xi_\perp/\|x\|_p)^\sigma$, so the expansion parameter A_w/μ is small precisely when $\|x\|_p \gg \xi_\perp$. Expanding Eq. (98) to first order in A_w/μ, and using $\hat{w}(\xi) = \hat{w}(0) - A_w(\xi)/f'(s^*)$ and $\mu + f'(s^*)\, w_0 = \alpha$, gives $\hat{C}(\xi) \simeq \big(f'(s^*)a^*/\mu\big)\big[\hat{w}(0) - \alpha A_w(\xi)/(f'(s^*)\mu)\big]$.

For $x \neq 0$ the constant term does not contribute, and the inverse Fourier transform of $-A_w$ is $f'(s^*)\,w(\|x\|_p)$; hence

$$C(x) \simeq \frac{\alpha f'(s^*)\,a^*}{\mu^2}\; w(\|x\|_p) \tag{99}$$

for $\|x\|_p \gg \xi_\perp = (\kappa_W/\mu)^{1/\sigma}$. At $\mu = 0$, $\hat{C} \propto \|\xi\|_p^{-\sigma}$ at small ξ, and $C(x) \propto \|x\|_p^{\sigma-N}$ for $N > \sigma$. The review does not give the critical form of C. Its Euclidean counterpart follows from the denominator $\mu + \frac{1}{2}Dq^2$ of R(150) (where the momentum is denoted p): $\hat{C} \propto q^{-2}$ and $C(r) \propto r^{2-N}$ for $N > 2$, the Ornstein–Zernike form [52, 71], which our result reproduces at $\sigma = 2$. Three consequences follow.

- Away from criticality, spatial correlations follow the connectivity and remain power laws in hierarchical distance. They cross over from $\|x\|_p^{\sigma-N}$ to $\|x\|_p^{-N-\sigma}$ at $\xi_\perp$, but they are *not* exponentially truncated. Temporal correlations are still cut off by $e^{-\mu t}$, with $\xi_\parallel = 1/\mu$.
- The review predicts that power laws become exponentially truncated as one moves away from criticality [its Sec. 12.12.2]. The p-adic theory predicts exponential truncation in time only.
- The correlation length is a level of the hierarchy, $k_\xi = \log_p \xi_\perp = \sigma^{-1}\log_p(\kappa_W/\mu)$, the same as k^* in Sec. VI. It grows only logarithmically as $\mu \to 0$.

(Exact to leading order in A_w/μ.)

E. Power spectra of local field potentials

The review derives the $1/\omega^2$ spectrum of local field potentials (LFP) and of the electroencephalogram (EEG) from the propagator of the homogeneous mode, i.e. at zero momentum [R(150)–R(152); there the momentum is denoted p, which in this paper is the prime]. An electrode averages activity over a region. In the p-adic model we take that region to be a ball B of radius p^k. The Fourier transform of its indicator is exactly $p^{kN}\,\Omega(p^k\|\xi\|_p)$, so the ball average retains precisely the modes with $\|\xi\|_p \le p^{-k}$, the levels coarser than the electrode, with no leakage.

Gaussian approximation. We treat the fluctuation $\delta\varphi = \varphi - a^*$ about the stationary state of Sec. X D as a Gaussian random field. This is the linear noise approximation of van Kampen's system-size expansion [72]: the action is expanded about a^* and only the terms quadratic in $\delta\varphi$ are kept, so the fluctuations obey a linear equation driven by noise whose covariance is evaluated at a^*, and are therefore Gaussian. The neglected cubic terms are the vertices whose one-loop effect is measured by the Ginzburg criterion of Sec. VI, so the approximation is controlled where that criterion holds: for $N > 2\sigma$ up to the critical point, and for $N < 2\sigma$ when μ is not too small. In addition, the electrode signal is an average over many neurons; when the ball B contains many correlation volumes (radius of B much larger than $\xi_\perp$), it is a sum of many weakly dependent contributions and is close to Gaussian by the central limit theorem. Close to $\mu = 0$ with $N < 2\sigma$ the fluctuations are not Gaussian, and the results below do not apply.

Power spectrum. In this approximation each Fourier mode of $\delta\varphi$ obeys an Ornstein–Uhlenbeck equation,

$$\partial_t \delta\hat{\varphi}(\xi,t) = -\lambda(\xi)\,\delta\hat{\varphi}(\xi,t) + \hat{\eta}(\xi,t),$$

with $\lambda(\xi) = \mu + A_w(\xi)$ as in Eq. (98). The noise $\hat{\eta}$ is uncorrelated in time, and its covariance in momentum space is $\hat{Q}(\xi) = 2f'(s^*)\,a^*\hat{w}(\xi)$, the source term of Eq. (83); indeed, the stationary condition $2\lambda\hat{C} = \hat{Q}$ reproduces Eq. (98). At small ξ, $\hat{Q}(\xi) \simeq 2f'(s^*)\,a^* w_0$ is approximately constant: the noise is white in space as well as in time, and the ξ-dependence of the spectrum comes only from $\lambda(\xi)$. Each mode relaxes at its own rate, so the two-time correlation is $\hat{C}(\xi,\tau) = \hat{C}(\xi)\,e^{-\lambda(\xi)|\tau|}$, and its Fourier transform in τ is $\hat{Q}(\xi)/\big(\lambda(\xi)^2 + \omega^2\big)$, which is flat for $\omega \ll \lambda(\xi)$ and decays as ω^{-2} for $\omega \gg \lambda(\xi)$; integrating it over ω returns the equal-time correlation (98). The electrode records the ball average $X_B(t) = \mathrm{vol}(B)^{-1}\int_B \delta\varphi(x,t)\,d^N x$. Its power spectrum is

$$\begin{aligned} S_B(\omega) &= \int_{\mathbb{R}} e^{i\omega\tau}\,\langle X_B(t+\tau)X_B(t)\rangle\,d\tau \\ &\propto \int_{\|\xi\|_p \le p^{-k}} \frac{\hat{Q}(\xi)}{\lambda(\xi)^2+\omega^2}\,d^N\xi, \end{aligned}$$

the sum of the spectra of the modes retained by the ball. At $\mu = 0$ (in practice, for $\omega \gg \mu$), summing over spheres gives

$$\begin{aligned} S_B(\omega) &\propto \sum_{j \le -k} \frac{p^{jN}}{\kappa_W^2 p^{2j\sigma} + \omega^2} \\ &\propto \begin{cases} \omega^{-(2-N/\sigma)}, & \omega \ll \omega_k, \\ \omega^{-2}, & \omega \gg \omega_k, \end{cases} \end{aligned} \tag{100}$$

with $\omega_k = \kappa_W p^{-k\sigma}$, for $N < 2\sigma$, with log-periodic ripple. The high-frequency regime reproduces the review's $1/\omega^2$ law. Below the crossover ω_k, set by the electrode's level, the exponent $2 - N/\sigma$ depends on the hierarchy; $1/f$ noise corresponds to $\sigma = N$. For $N > 2\sigma$ the low-frequency spectrum saturates. For $\mu > 0$ it saturates below $\omega \sim \mu$, as in R(152). (Exact within the Gaussian approximation.)

F. Rhythms and finite size

In the review (Sec. 12.5.1 and Figs. 27–29 there), the periods of large-scale rhythms are related to the number $\mathcal{N}$ of neurons recruited. In the power-law class at $\mu = 0$, the relaxation rate of the level-γ modes is $\lambda(p^{1-\gamma}) = \kappa_W p^{(1-\gamma)\sigma} \propto p^{-\gamma\sigma}$.

- *Geometric progression.* The characteristic time scales of successive levels form a geometric progression with ratio p^σ. The EEG frequency bands form a geometric progression with ratio close to e [73] (Fig. 28 of Ref. [25]).

- *Scaling with size.* A ball of level γ contains $\mathcal{N} \propto p^{N\gamma}$ neurons, so $f \propto \mathcal{N}^{-\sigma/N}$. Figure 29 of Ref. [25] is fitted by $f \propto \mathcal{N}^{-1/2}$, which corresponds to $\sigma = N/2$, i.e. $N = N_c$. This is marginal mean-field behavior, consistent with B-C's claim that cortex is mean-field.

- *Caveat.* These are relaxation rates of overdamped modes, not oscillation frequencies. Turning them into rhythms requires a second variable at each level: inhibition (Sec. 12.5.2 of Ref. [25]), refractoriness, or synaptic depression (Sec. X G below).

(Heuristic and speculative.)

G. Self-organized criticality

In the review (Sec. 12.9 there), the homogeneous equilibrium and its stability are $\alpha\varphi_0 = f(w_0\varphi_0 + h_0)$ and $\mu = \alpha - f'(w_0\varphi_0 + h_0)\, w_0$ [R(148)–R(149)]; in our notation, $\varphi_0 = a^*$ and $h_0 = I$. The control parameters are $w_0 = \sum w$ and the background input I. Candidate mechanisms are synaptic depression, spike-timing-dependent plasticity, thalamocortical control of I, and stochastic resonance.

Plasticity. In neuroscience, *synaptic plasticity* is the ability of synapses to change their strength in response to the activity of the neurons they connect [74]. It acts on several time scales. Short-term plasticity, such as synaptic depression, lasts from milliseconds to seconds: repeated firing temporarily depletes the resources (neurotransmitter vesicles) that a synapse can release, and weakens it until they recover. Long-term plasticity lasts from minutes to much longer. It includes Hebbian rules, in which a synapse is strengthened when the two neurons it connects are active together, and spike-timing-dependent plasticity, in which the sign of the change depends on the order of the pre- and postsynaptic spikes. Homeostatic plasticity slowly rescales synaptic strengths to keep firing rates within a range. In the model, plasticity is a slow dynamics of the connectivity w, and hence of μ, driven by the activity a or by the correlations C; the mechanisms below are of this kind.

What must be tuned. By Eq. (88), criticality ($\mu = 0$) is a single scalar condition on the layer profile $\{W_k\}$. It is not sufficient on its own. Every level must also remain stable, $\varrho_\gamma \le 0$ in Eq. (90). Otherwise a hierarchical pattern instability precedes the DP transition.

Synaptic depression. Section VIII argues that only a slow, activity-dependent plasticity can bring the network to the critical point. The standard example is short-term synaptic depression: Levina, Herrmann and Geisel [75] showed that networks with depressing synapses self-organize to criticality, with the avalanche exponent 3/2, over a wide range of parameters. The mechanism is a negative feedback on μ: high activity depletes the synaptic resources and weakens the effective coupling, which raises μ, and low activity lets them recover, which lowers μ. We ask what this mechanism does in a hierarchical network. Let $u(x,t) \in [0,1]$ be the fraction of synaptic resources available at the outgoing synapses of neuron x; each spike uses a fraction $v \in (0,1]$ of them, and they recover toward 1 with time constant τ_R. In mean-field theory, with $a(x,t)$ the mean activity of Sec. III C, the dynamics is $\partial_t a = -\alpha a + f(s)$, with

$$s(x,t) = \int_{\mathbb{Q}_p^N} w(\|x-y\|_p)\, u(y,t)\, a(y,t)\, d^N y + I,$$
$$\partial_t u(x,t) = \frac{1-u(x,t)}{\tau_R} - v\, u(x,t)\, a(x,t). \tag{101}$$

The second equation is nonlinear: the depletion rate is proportional to the product ua. In the homogeneous state $u^* = 1/(1+v\tau_R a^*)$, so depression rescales the connectivity *uniformly across all layers*: $w_{\text{eff}} = u^* w$. It tunes μ and leaves the shape of $\{W_k\}$, hence σ and the universality class, unchanged. Linear stability is decided level by level through the 2×2 matrix

$$J(\xi) = \begin{pmatrix} -\alpha + f'(s^*)\, u^* \hat{w}(\xi) & f'(s^*)\, a^* \hat{w}(\xi) \\ -v\, u^* & -\big(\tau_R^{-1} + v\, a^*\big) \end{pmatrix}, \tag{102}$$

in which $s^* = u^* \hat{w}(0) a^* + I$ is now the stationary input and $\hat{w}(\xi)$ takes one value per level [Eq. (89)]. A Hopf instability, if present, appears first at the level maximizing $\hat{w}$. This is one mechanism for level-specific oscillations (Sec. X F). (Exact within mean field.)

Correlation-based plasticity. By Eq. (99), $C \propto w$ at large hierarchical distance. Any rule of the form $\partial_t w = \eta\, C - \varepsilon\, w$ therefore has a growth rate $\eta\alpha f'(\hat{w}(0)a^* + I)\, a^*/\mu^2 - \varepsilon$ that is the same for every layer, and it also preserves σ. On its own such a Hebbian rule is unstable: the fixed point $\mu^* = \big(\eta\alpha f'(\hat{w}(0)a^* + I)\, a^*/\varepsilon\big)^{1/2}$ repels, because growth of w lowers μ and increases C. It must be combined with depression or homeostasis. (Exact to leading order.)

Conclusion. In the p-adic B-C theory, the standard self-organizing mechanisms tune the distance to criticality μ but leave the hierarchy exponent σ untouched. This matters because μ and σ control different things: μ decides how close the network is to the critical point, while σ decides which critical behavior it shows there, mean-field avalanches ($\tau_S = 3/2$) for $\sigma \le N/2$ and anomalous ones for $\sigma > N/2$. Plasticity can therefore explain why cortex is near criticality, but not which exponents it displays: these are fixed by the architecture. Experiments that alter plasticity should change the distance to criticality, and hence the cutoff and the mean avalanche size, but not the critical exponents. Changing σ requires a mechanism that depends explicitly on the level, such as

a wiring cost or conduction delay that grows with hierarchical distance. Whether σ is tuned dynamically by such a mechanism, or fixed by development, is an open problem (Sec. XI).

XI. DISCUSSION AND OPEN PROBLEMS

The p-adic theory as an extension of the B-C picture. Buice and Cowan argued, using field-theoretic methods, that their effective spike model with an absorbing state has a phase transition in the directed-percolation universality class, and that the avalanche-size exponent takes its mean-field value 3/2, the value observed by Beggs and Plenz [3, 13], when fluctuations are negligible: exactly for $N \geq 4$, the upper critical dimension of DP, and approximately in cortex, by their Ginzburg argument (Sec. V). In the p-adic model the upper critical dimension is $N_c = 2\sigma$, and the exponent 3/2 holds for $N \geq 2\sigma$. Since the Taibleson–Vladimirov operator is defined for every $\sigma > 0$, the p-adic theory gives a one-parameter family of ultrametric DP classes, with N_c ranging over $(0, \infty)$. At $\sigma = 2$ its power counting and mean-field exponents coincide with those of the B-C model. In this sense the p-adic theory extends the B-C picture, and the 3/2 law no longer requires $N \geq 4$: on a hierarchical network it holds even for $N = 1$ when $\sigma \leq 1/2$. In Euclidean Lévy-flight DP, by contrast, $N_c = 2\sigma$ only for $\sigma < 2$, since ∇^2 takes over for $\sigma \geq 2$ [44] (Sec. V). More generally, the p-adic B-C theory provides a framework to study criticality in large deep neural networks for which a thermodynamic-continuous limit is available: in hierarchical architectures this limit exists and is exact, and the criterion $\sigma \leq N/2$ and the Ginzburg number then decide whether the critical behavior of the network is mean-field.

General neuron spaces. The construction of Sec. III does not depend on the geometry of the neuron space. Let Ω be a neuron space with a measure μ, and let $w(x, y)$ be a connectivity kernel on $\Omega \times \Omega$. Replacing $\int_{\mathbb{Q}_p^N} d^N x$ by $\int_\Omega d\mu(x)$ and $w \star g$ by $\int_\Omega w(x, y)\, g(y)\, d\mu(y)$ in Eq. (18), one obtains the action of the effective spike model on Ω. It is the limit of the discrete actions S_Λ of finite networks Λ that fill up Ω. Its mean-field theory is the Wilson–Cowan equation

$$\begin{aligned} \partial_t a(x,t) = & -\alpha\, a(x,t) + \bar{n}(x)\, \delta(t) \\ & + f\Big(\int_\Omega w(x,y)\, a(y,t)\, d\mu(y) + I \Big), \end{aligned} \tag{103}$$

and the generating functional and the loop expansion of Ref. [13] carry over formally. In the original formulation of B-C, $\Omega = \mathbb{R}^N$ and $w = w(|x - y|)$. Here Ω is $\mathbb{Q}_p^N$, or a finite disjoint union of balls of $\mathbb{Q}_p^N$, and $w = w(\|x - y\|_p)$. In the p-adic theory the hierarchy hypothesis of Sporns and co-workers [7–9] is implemented naturally, because the p-adic numbers are organized in a tree-like structure (Sec. III A). The study of more general architectures, such as

$$\Omega = \mathbb{R}^{N_0} \times \mathbb{Q}_{p_1}^{N_1} \times \cdots \times \mathbb{Q}_{p_k}^{N_k}, \tag{104}$$

which combine Euclidean and hierarchical directions, is an open problem. Cellular neural networks are bio-inspired by the Wilson–Cowan equation, and they have been applied to several practical tasks, such as image processing; see Ref. [42] and the references therein.

Not DP on a Cayley tree. The tree enters the two models in different ways. In DP on a Cayley tree (Bethe lattice), the units sit on all the vertices of the tree, internal ones included, and activity spreads only along its edges. Distance is graph distance, and the number of units within distance d grows exponentially in d; the lattice is effectively infinite-dimensional, and DP on it is mean-field. In the p-adic model the tree is not the graph along which activity spreads. Only its leaves are neurons, and the internal vertices label the modules (balls) of the hierarchy. Every pair of neurons is coupled directly, with a strength fixed by the level of their last common ancestor, and a ball of radius r contains a number of neurons proportional to r^N, a power of the radius. The neuron space therefore has finite dimension N, and the theory has a finite upper critical dimension $N_c = 2\sigma$, like Dyson's hierarchical model [53, 54]. The simulations of Sec. VIII A show branching-process avalanches at small σ and departures from them as σ grows.

Relation to Ref. [24]. That work treats Gaussian-noise-driven continuous networks (Martin–Siggia–Rose type). There, criticality is defined through the largest Lyapunov exponent in a double-copy system, on both real and p-adic neuron spaces. The present theory is Poissonian (Doi–Peliti) and has an absorbing state, and its criticality is DP-like. Comparing the two notions of criticality on the same p-adic architecture is a natural next step: when do the edge of chaos and the absorbing-state transition coincide?

Disorder and Griffiths phases. Hierarchical networks with quenched structural heterogeneity display extended critical (Griffiths) regions [76]. The p-adic model is translation-invariant under $\mathbb{Z}_p^N$ and has a sharp transition. Level-dependent quenched disorder in w or α interpolates between the two pictures. B-C's remark that spatiotemporal disorder preserves DP [19] should be revisited in the ultrametric setting.

Finite size. B-C leave finite-size effects open. On G_l^N the tree depth l is the natural finite-size parameter, and scaling forms in p^l follow from Sec. V.

Refractoriness. The two- and three-state Cowan actions (B-C Appendix C) transfer directly to $\mathbb{Q}_p^N$. Whether refractoriness drives the model toward an ultrametric version of dynamic isotropic percolation is open.

Pathology. In B-C's reading, seizures correspond to $\hat{w}(0)$ exceeding threshold. Hierarchically, one can ask at which level γ the effective rate $\lambda(p^{1-\gamma})$ first changes sign. Instability confined to a small ball versus instability of the coarsest modes offers a caricature of focal versus generalized seizures.

Summary of open technical tasks.

1. Rigorous $l \to \infty$ limit of the Doi–Peliti integral (Sec. III).
2. Rigorous proof of the mean-field critical behavior of regional networks (Lemma 1).
3. Critical exponents for $N < 2\sigma$, using the p-adic renormalization group [55, 56] (Sec. V).
4. Amplitude of log-periodic terms in observables other than I_2, such as avalanche statistics (Sec. VI).
5. Analysis of Eqs. (82)–(83) (Sec. VII).
6. Rigorous version of the logarithmic spreading law of the active phase (Sec. X), and nonlinear selection of hierarchical patterns (Sec. X).
7. Whether σ is fixed by development or tuned by a level-dependent self-organizing mechanism, e.g. toward the marginal value $\sigma = N/2$ (Sec. X G).
8. Fluctuation effects in regional networks (Lemma 1 and Appendix E treat only the linear dynamics).

ACKNOWLEDGMENTS

During the preparation of this work, the author used Claude (Anthropic; model claude-opus-5-5) to improve the clarity and readability of the language of select sections. After using this tool, the author reviewed and edited the content as needed and took full responsibility for the publication. The author was partially supported by the Debnath Endowment.

Appendix A: Field theory of branching processes: a primer

This appendix explains the field-theoretic tools used in the main text. It is written for readers from neuroscience or machine learning who are not familiar with quantum field theory (QFT). We work with a simple branching process on a network that contains all the ingredients of the effective spike model of Sec. III. At each step we indicate the corresponding objects of the main text. Standard references are Refs. [14, 15, 19, 52, 77, 78].

1. Core concepts

What a branching process is. A branching process is a model of a population in which each individual (a "particle"), independently of the others, disappears or produces new individuals at random times. The classical example is the Galton–Watson process of population genetics [79]. In our setting a particle is an *effective spike*, i.e. a spike that still influences the network [13], and particles live on the nodes i of a network (neurons, or small groups of neurons). We use a single branching process throughout this appendix. Its dynamics consist of three kinds of elementary random events:

$$\begin{array}{lll} \text{decay:} & A_i \to \varnothing & \text{rate } \alpha, \\ \text{branching:} & A_j \to A_j + A_i & \text{rate } \beta K_{ij}, \\ \text{saturation:} & A_i + A_i \to A_i & \text{rate } \lambda. \end{array} \tag{A1}$$

Here A_i denotes one effective spike at node i, and the rates are per particle (for saturation, the total rate at node i is $\lambda n_i(n_i - 1)$).

- *Decay.* An effective spike stops influencing the network. Each spike does so at rate α, so its influence lasts on average $1/\alpha$.
- *Branching.* An effective spike at node j causes a new spike at node i. For each spike at j this happens at rate βK_{ij}, where $K_{ij} \geq 0$ is the strength of the connection from j to i. This is the only event that increases activity.
- *Saturation.* When two effective spikes are present at the same node, one of them is removed, at total rate $\lambda n_i(n_i - 1)$. This limits growth.

With decay and branching only, one obtains the classical, linear branching process: each spike independently has on average $m = \beta \sum_i K_{ij}/\alpha$ offspring during its lifetime, and in the supercritical regime ($m > 1$) activity grows exponentially without bound. Saturation is the simplest way to keep activity bounded, as real firing rates are.

Relation to the effective spike model. The effective spike model of Sec. III has two kinds of events: the decay of an effective spike at rate α, and the emission of a new spike at neuron i at rate $f(s_i)$, where $s_i = \sum_j w_{ij} n_j$ is the input from the other neurons. Expanding f for small input, with $f(0) = 0$ [Eq. (24)],

$$f(s_i) \simeq f'(0)\, s_i + \tfrac{1}{2} f''(0)\, s_i^2. \tag{A2}$$

The linear term makes the emission rate at i proportional to the number of spikes n_j at the presynaptic neurons (i.e., the neurons j that send connections to neuron i, those with $w_{ij} \neq 0$). This is exactly branching, with $\beta K_{ij} = f'(0) w_{ij}$. The quadratic term is negative, since $f''(0) < 0$, so the emission rate grows more slowly than linearly when many spikes are present. This is the effect of saturation. In the effective spike model it acts on the total input s_i rather than on pairs of spikes at one node; this difference does not change the large-scale behavior (Appendix B). The process (A1) is therefore a simplified version of the effective spike model with the same three ingredients: decay, branching and saturation. We use it because every step below can be written out explicitly.

Absorbing state and criticality. If there are no particles, nothing can happen: the empty network is an *absorbing state*. Whether activity survives depends on the *branching ratio* $m = \beta \sum_i K_{ij}/\alpha$, the mean number of

offspring of one particle during its lifetime. For $m < 1$ (subcritical) activity dies out quickly. For $m > 1$ (supercritical) it can grow and persist. The *critical point* $m = 1$ separates the two phases. There, activity neither dies out quickly nor explodes, perturbations spread over all scales, and statistical quantities follow power laws. This is the non-equilibrium phase transition behind the critical-brain hypothesis. In machine learning, the analogous "edge of chaos" separates layers that damp and layers that amplify signals in deep networks [5, 80]. In the main text the distance to criticality is $\mu = \alpha - f'(0)\hat{w}(0)$, and $\mu = 0$ corresponds to $m = 1$.

Why fluctuations matter. The simplest description of the process is a deterministic *rate equation* for the mean activity, derived below; see Eq. (A5). It ignores fluctuations. The number of events in a short time is random, with a variance proportional to the number of particles. Relative fluctuations are therefore large when activity is low, which is exactly the situation near an absorbing state. A systematic theory of fluctuations is needed, and field theory provides one.

2. The master equation

Let n_i be the number of particles at node i, let $\mathbf{n} = (n_i)$ be the state (or configuration) of the network, which varies with time, and let $P(\mathbf{n}, t)$ be the probability of configuration $\mathbf{n}$ at time t. The dynamics is modeled by a continuous-time Markov chain, so P obeys a *master equation*: its rate of change is the probability flowing in from other configurations minus the probability flowing out. For the events (A1),

$$\begin{aligned}\partial_t P(\mathbf{n},t) = &\sum_i \alpha\big[(n_i+1)P(\mathbf{n}_i^+) - n_i P(\mathbf{n})\big] \\ &+ \sum_{i,j} \beta K_{ij}\big[n_j P(\mathbf{n}_i^-) - n_j P(\mathbf{n})\big] \\ &+ \sum_i \lambda\big[(n_i+1)n_i P(\mathbf{n}_i^+) - n_i(n_i-1)P(\mathbf{n})\big],\end{aligned} \tag{A3}$$

where $\mathbf{n}_i^{\pm}$ differs from $\mathbf{n}$ only by $n_i \to n_i \pm 1$, and we take $K_{ii} = 0$. The master equation of the effective spike model, Eq. (6), has the same structure, with the creation rate $f(s_i)$ in place of the branching and saturation terms; by Eq. (A2) the two agree to second order in the input.

Equation (A3) is exact but is a system of infinitely many coupled linear equations. Multiplying it by n_i and summing over $\mathbf{n}$ gives

$$\frac{d\langle n_i\rangle}{dt} = -\alpha\langle n_i\rangle + \beta\sum_j K_{ij}\langle n_j\rangle - \lambda\langle n_i(n_i-1)\rangle. \tag{A4}$$

The mean couples to the second moment, the second moment to the third, and so on. This is the *closure problem.* Mean-field theory closes the hierarchy by the approximation $\langle n_i(n_i-1)\rangle \approx \langle n_i\rangle^2$, which neglects fluctuations. Writing $a_i = \langle n_i\rangle$, one obtains the *rate equation*

$$\frac{da_i}{dt} = -\alpha a_i + \beta\sum_j K_{ij}a_j - \lambda a_i^2. \tag{A5}$$

Master equation versus rate equation. The master equation (A3) is an *exact* equation for the *probability distribution* $P(\mathbf{n}, t)$ of all configurations. It is linear in P, but it involves infinitely many unknowns (one for each configuration), and it contains the full statistics: means, variances, correlations, avalanche distributions. The rate equation (A5) is an *approximate* equation for the *means* a_i only. It is a closed, finite system of deterministic ordinary differential equations, but it is nonlinear and it contains no information about fluctuations. It is the analogue for this process of the Wilson–Cowan equation (21) of the effective spike model. The field theory replaces the ad hoc closure leading from Eq. (A4) to Eq. (A5) by a controlled expansion, in which the rate equation is the lowest order (Appendix A 5).

3. The Doi–Peliti formulation

The Doi–Peliti method [14, 15] rewrites the master equation as a linear evolution equation that can be turned into a path integral.

Occupation-number states. At each node introduce a vector $|n\rangle$ for each $n = 0, 1, 2, \ldots$, and two linear operators a and $a^\dagger$ ("annihilation" and "creation") acting as

$$a^\dagger|n\rangle = |n+1\rangle, \qquad a|n\rangle = n\,|n-1\rangle. \tag{A6}$$

Then $a^\dagger a|n\rangle = n|n\rangle$: the operator $a^\dagger a$ counts particles. One checks $aa^\dagger - a^\dagger a = 1$. These are the same algebraic rules as for the quantum harmonic oscillator, but nothing quantum is involved; they are bookkeeping devices. For a network, each node i carries its own pair of operators $a_i, a_i^\dagger$, which act on the number of particles at that node; operators associated with different nodes commute, so that $a_i a_j^\dagger - a_j^\dagger a_i = \delta_{ij}$.

The state vector. The probability distribution is encoded in $|\Psi(t)\rangle = \sum_{\mathbf{n}} P(\mathbf{n}, t)\,|\mathbf{n}\rangle$, with $|\mathbf{n}\rangle = \prod_i (a_i^\dagger)^{n_i}|0\rangle$. The master equation (A3) becomes

$$\partial_t|\Psi(t)\rangle = -H\,|\Psi(t)\rangle, \tag{A7}$$

with the *Liouvillian* ("Hamiltonian") H obtained from a simple rule: an event that removes k particles and puts back l particles at rate r contributes $-r\,[(a^\dagger)^l - (a^\dagger)^k]\,a^k$. For Eq. (A1),

$$\begin{aligned}H = &\sum_i \alpha\,(a_i^\dagger - 1)\,a_i - \sum_{i,j}\beta K_{ij}\,(a_i^\dagger - 1)\,a_j^\dagger a_j \\ &+ \sum_i \lambda\,(a_i^\dagger - 1)\,a_i^\dagger a_i^2.\end{aligned} \tag{A8}$$

For example, applying $\alpha(a_i - a_i^\dagger a_i)$ to $|\mathbf{n}\rangle$ gives $\alpha n_i(|\mathbf{n}_i^-\rangle - |\mathbf{n}\rangle)$, which reproduces the decay terms of Eq. (A3).

Averages and the Doi shift. Averages are computed with the "projection" vector $\langle\mathcal{P}| = \langle 0|\prod_i e^{a_i}$, which satisfies $\langle\mathcal{P}|\mathbf{n}\rangle = 1$ for every $\mathbf{n}$. Then $\langle\mathcal{P}|\Psi\rangle = \sum_{\mathbf{n}} P(\mathbf{n}) = 1$, and $\langle n_i\rangle = \langle\mathcal{P}|a_i|\Psi\rangle$. Since $\langle\mathcal{P}|a_i^\dagger = \langle\mathcal{P}|$, every term of H vanishes when $a_i^\dagger$ is replaced by 1. This is conservation of probability. It suggests the *Doi shift* $a_i^\dagger = 1 + \tilde{a}_i^\dagger$, which moves the projection vector into the operators. After the shift, every term of H contains at least one factor $\tilde{a}^\dagger$. This is the origin of the property $S[\varphi, 0] = 0$ used in Sec. III D.

Initial condition. An initial state in which each n_i is independently Poisson distributed with mean $\bar{n}_i$ is $|\Psi(0)\rangle = \prod_i e^{\bar{n}_i(a_i^\dagger - 1)}|0\rangle$. After the Doi shift it becomes $\prod_i e^{\bar{n}_i\tilde{a}_i^\dagger}|0\rangle$. This produces the initial-state term $-\sum_i \bar{n}_i\tilde{\varphi}_i(0)$ of Eq. (7).

4. The action

From operators to a path integral. The formal solution of Eq. (A7) is $|\Psi(t)\rangle = e^{-Ht}|\Psi(0)\rangle$. Divide $[0, t]$ into many small steps and insert, at each step, a resolution of the identity in terms of *coherent states*, the eigenvectors of a_i: $a_i|\varphi\rangle = \varphi_i|\varphi\rangle$. In the limit of infinitely many steps, the matrix elements of e^{-Ht} become an integral over the trajectories of two sets of variables: $\varphi_i(t)$, the eigenvalue of a_i, and $\tilde{\varphi}_i(t)$, the variable associated with $\tilde{a}_i^\dagger$. The result is the generating functional

$$Z[J, \tilde{J}] = \int \mathcal{D}\varphi\, \mathcal{D}\tilde{\varphi}\; e^{-S[\varphi,\tilde{\varphi}] + \int_0^\infty dt \sum_i (J_i\tilde{\varphi}_i + \tilde{J}_i\varphi_i)}, \quad \text{(A9)}$$

whose derivatives with respect to the "sources" $J, \tilde{J}$ at $J = \tilde{J} = 0$ give all moments. The *action* is obtained by replacing, in the normal-ordered operator H, $a_i \to \varphi_i$ and $\tilde{a}_i^\dagger \to \tilde{\varphi}_i$:

$$S = \int_0^\infty dt \sum_i \left[\tilde{\varphi}_i\, \partial_t\varphi_i + H(\tilde{\varphi}_i, \varphi_i)\right] - \sum_i \bar{n}_i\, \tilde{\varphi}_i(0). \quad \text{(A10)}$$

The derivation is standard [52, 77, 78]. The only technical point is that, after the Doi shift, $\tilde{\varphi}$ is not the complex conjugate of φ. The two are treated as independent variables, with $\tilde{\varphi}$ integrated along the imaginary axis.

The action of the branching process. For Eq. (A8), after the shift and in the continuum (sums over nodes become integrals), and assuming that K_{ij} depends only on the difference $i - j$, so that $K_{ij} \to K(x - y)$ and the sums $\sum_j K_{ij}(\cdot)_j$ become convolutions $(K \star \cdot)(x)$,

$$S = \int_0^\infty dt \int d^N x \Big\{\tilde{\varphi}\, \partial_t\varphi + \alpha\tilde{\varphi}\varphi - \beta\, \tilde{\varphi}\, K \star \varphi - \beta\, \tilde{\varphi}\, K \star (\tilde{\varphi}\varphi) + \lambda\tilde{\varphi}\varphi^2 + \lambda\tilde{\varphi}^2\varphi^2\Big\} - \int d^N x\, \bar{n}\, \tilde{\varphi}(x, 0). \quad \text{(A11)}$$

For a general kernel $K(x, y)$, not translation invariant, $K \star \varphi$ and $K \star (\tilde{\varphi}\varphi)$ are replaced by the integrals $\int K(x, y)\, \varphi(y, t)\, d^N y$ and $\int K(x, y)\, \tilde{\varphi}(y, t)\varphi(y, t)\, d^N y$. Compare with the B-C action, Eq. (18), expanded to second order in f: $-\tilde{\varphi} f\big(w \star (\tilde{\varphi}\varphi + \varphi)\big) \simeq -f'(0)\, \tilde{\varphi}\, w \star \varphi - f'(0)\, \tilde{\varphi}\, w \star (\tilde{\varphi}\varphi) - \frac{1}{2} f''(0)\, \tilde{\varphi}\, (w \star \varphi)^2 + \ldots$. The first two terms are exactly the branching terms of Eq. (A11) with $\beta K = f'(0)w$. The third is a (nonlocal) saturation term with positive coefficient $\frac{1}{2}|f''(0)|$, playing the role of $\lambda\tilde{\varphi}\varphi^2$.

Meaning of the two fields. The field φ is the "density" field: $\langle\varphi(x, t)\rangle = \langle n(x, t)\rangle$ exactly, and higher moments of φ are the factorial moments of n (e.g. $\langle\varphi^2\rangle = \langle n(n-1)\rangle$ at a single node). The field $\tilde{\varphi}$ is the *response field*. Adding a source $h\,\tilde{\varphi}$ to the action means injecting particles at rate h, and correlations with $\tilde{\varphi}$ measure the response to such injections. The name should not be read literally: $\tilde{\varphi}$ sits at the earlier point, where the injection acts, and φ at the later point, where the response is measured (see the note after Eq. (30)).

Connection with stochastic differential equations. Readers familiar with stochastic differential equations (SDEs) may find the following useful. If the action is truncated at second order in $\tilde{\varphi}$, it takes the form $S = \int \tilde{\varphi}[\partial_t\varphi - F(\varphi)] - \frac{1}{2}\int \tilde{\varphi}\, Q(\varphi)\, \tilde{\varphi}$. This is the Martin–Siggia–Rose–Janssen–De Dominicis (MSRJD) action [81–83] of the Langevin equation

$$\partial_t\varphi = F(\varphi) + \eta, \qquad F(\varphi) = -\alpha\varphi + \beta K \star \varphi - \lambda\varphi^2, \quad \text{(A12)}$$

with Gaussian noise η whose covariance $Q(\varphi)$ is read off from the terms $\beta\tilde{\varphi}K \star (\tilde{\varphi}\varphi)$ and $\lambda\tilde{\varphi}^2\varphi^2$. The noise covariance is proportional to φ, so it vanishes in the absorbing state: fluctuations are multiplicative. The response field plays the role of a Lagrange multiplier enforcing the equation of motion. (The saturation term contributes a negative piece to Q; this known subtlety of the Doi–Peliti Langevin representation does not affect the perturbative expansion below.)

Mean field from the action. The classical equations of motion, $\delta S/\delta\tilde{\varphi} = 0$ and $\delta S/\delta\varphi = 0$, admit the solution $\tilde{\varphi} = 0$, and then $\delta S/\delta\tilde{\varphi} = 0$ gives

$$\partial_t\varphi = -\alpha\varphi + \beta K \star \varphi - \lambda\varphi^2 + \bar{n}\, \delta(t), \quad \text{(A13)}$$

which is the rate equation (A5) with its initial condition. For the B-C action the same computation gives the Wilson–Cowan equation (20) (Sec. III C). Mean-field theory is the lowest order of a systematic expansion, to which we now turn.

5. Perturbation theory and Feynman diagrams

Free part and propagator. Split the action into a *free* (bilinear) part and an *interacting* part, $S = S_0 + S_{\text{int}}$, with

$$S_0 = \int_0^\infty dt \int d^N x\; \tilde{\varphi}\, \big(\partial_t + \alpha - \beta K \star\big)\varphi. \quad \text{(A14)}$$

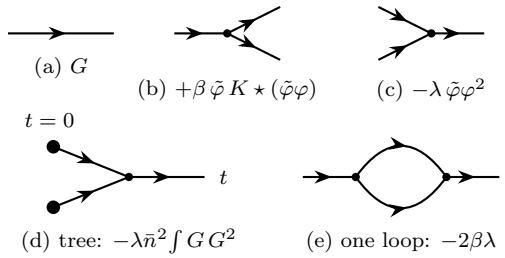


FIG. 8. Feynman diagrams for the branching process of Appendix A. Time runs from left to right. (a) The propagator G: the mean response to one injected particle. (b) Branching vertex: one particle becomes two. (c) Saturation vertex: two particles become one. (d) A zero-loop diagram contributing to the mean activity: two particles from the initial state (dots) merge; its value is the second term of Eq. (A16). (e) The one-loop correction to the propagator, Eq. (A17): a fluctuation in which a particle splits into two that later merge again.

With S_0 alone, the path integral is Gaussian and can be computed exactly. Its only nontrivial moment is the *propagator* $G(x,t) = \langle\varphi(x,t)\tilde\varphi(0,0)\rangle_0$. It is the Green function of the linearized dynamics, i.e. the mean response at (x,t) to one particle injected at $(0,0)$. In Fourier variables,

$$\hat G(q,t) = \theta(t)\,e^{-\mu(q)t}, \qquad \mu(q) = \alpha - \beta\hat K(q), \tag{A15}$$

where θ is the Heaviside function with $\theta(0)=0$ (causality: a response cannot precede its cause). The rate $\mu(0) = \alpha - \beta\hat K(0)$ is the distance to criticality; it vanishes at $m=1$. In the Euclidean case with a short-range kernel, $\hat K(q) \simeq \hat K(0) - \frac12 K_2 q^2$, so $\mu(q) \simeq \mu + Dq^2$: particles diffuse. In the p-adic case, $\mu(q) = \mu + A_w(q)$ with the W-operator of Sec. IV A.

Wick's theorem and diagrams. Expand $e^{-S_{\rm int}}$ in powers of the couplings. Each term is a Gaussian average of a product of fields, and Gaussian averages are sums over all ways of pairing each φ with a $\tilde\varphi$ (*Wick's theorem*), each pair contributing a propagator. These sums are organized by *Feynman diagrams* (Fig. 8):

- a *line* is a propagator G, drawn from an earlier to a later time;
- a *vertex* is a term of $-S_{\rm int}$, with one incoming line for each φ and one outgoing line for each $\tilde\varphi$; its position and time are integrated;
- an *initial-state dot* is a factor $\bar n$ with one outgoing line at $t=0$.

In the branching process, $+\beta\tilde\varphi K\star(\tilde\varphi\varphi)$ is a vertex with one incoming and two outgoing lines (one particle becomes two), and $-\lambda\tilde\varphi\varphi^2$ has two incoming lines and one outgoing line (two particles become one). The factor $\theta(0)=0$ forbids lines that start and end at the same time, so every diagram respects causality.

Zero-loop diagrams give mean-field theory. The *loop expansion* orders diagrams by their number of loops. Diagrams without closed loops form the zero-loop order. In field theory they are usually called *tree diagrams*, and this order *tree level*; as in Sec. III C, we avoid these names, because in this paper "tree" refers to the hierarchical organization of the neurons. Zero-loop order is mean-field theory, and each loop adds one order of fluctuation corrections. The sum of the zero-loop diagrams is exactly the solution of the classical equation (A13). For a single node with $\beta=0$, $\mu=\alpha$, the mean has the exact zero-loop expansion

$$\begin{aligned} a(t) &= \bar n\,e^{-\mu t} - \lambda\bar n^{\,2}\int_0^t ds\,e^{-\mu(t-s)}e^{-2\mu s} + \dots \\ &= \bar n\,e^{-\mu t}\Big[1 - \frac{\lambda\bar n}{\mu}\big(1-e^{-\mu t}\big) + \dots\Big], \end{aligned} \tag{A16}$$

where the second term is the diagram in which two lines from initial-state dots merge at one saturation vertex. The series is the expansion of $a(t) = \bar n e^{-\mu t}/[1 + (\lambda\bar n/\mu)(1-e^{-\mu t})]$, the exact solution of $\dot a = -\mu a - \lambda a^2$.

Loops are fluctuation corrections. Diagrams with closed loops describe fluctuations. The simplest one is a line that splits into two at a branching vertex, after which the two lines merge again at a saturation vertex. It corrects the propagator. Collecting the combinatorial factor (two ways to join the two lines),

$$\begin{aligned} \hat G^{-1}(q,\omega) = &-i\omega + \mu(q) \\ &+ 2\beta\lambda\int\frac{d^Nk}{-i\omega+\mu(k)+\mu(q-k)} + \dots \end{aligned} \tag{A17}$$

(for a local kernel, $\hat K \simeq \hat K(0)$ at the vertex; in the Euclidean case d^Nk stands for $d^Nk/(2\pi)^N$, in the p-adic case for the Haar measure). At $q=\omega=0$ the effective decay rate is $\mu_R = \mu + \beta\lambda\int d^Nk/\mu(k)$. It is larger than μ, so fluctuations shift the critical point to a larger branching rate than predicted by mean-field theory. The size of the loop relative to the zero-loop terms is controlled by

$$u(\mu) = \beta\lambda\int\frac{d^Nk}{\big(\mu+Dk^2\big)^2} \;\propto\; \mu^{N/2-2}. \tag{A18}$$

For $N>4$ this stays finite as $\mu\to0$, so mean-field theory is correct near the critical point. For $N<4$ it diverges, and fluctuations dominate. This is the Ginzburg-type criterion of Ref. [13] and of Sec. VI. In the p-adic theory, Dk^2 is replaced by $\kappa_W\|k\|_p^\sigma$, and the same argument gives $u\propto\mu^{N/\sigma-2}$, i.e. the upper critical dimension $N_c = 2\sigma$. Adding the loop corrections order by order is how the closure problem of Eq. (A4) is solved in a controlled way.

6. Universality

Coarse-graining and relevance. Near a critical point, correlations extend over lengths and times much larger

than the microscopic ones. The large-scale behavior can then be studied by coarse-graining: integrating out fluctuations on short scales and rescaling. This is the renormalization group (RG). Under this procedure each coupling in the action grows, shrinks or stays constant. Growing couplings are *relevant*, shrinking ones *irrelevant*, and the others *marginal*. Irrelevant couplings do not affect the large-scale behavior. Consequently, microscopically different models that share the same relevant couplings have the same critical exponents. They belong to the same *universality class*.

Power counting. Whether a coupling is relevant can be read off from dimensions. With diffusive scaling, $[x] = -1$ and $[t] = -2$ in momentum units. After a rescaling of φ and $\tilde{\varphi}$ that makes the branching and saturation couplings equal (both $= g/2$), $[\varphi] = [\tilde{\varphi}] = N/2$ and $[g] = 2 - N/2$. The cubic couplings are therefore relevant for $N < 4$, while the quartic term $\lambda\tilde{\varphi}^2\varphi^2$, with dimension $2-N$, is irrelevant near $N = 4$. The remaining action,

$$S_{\rm RFT} = \int_0^\infty dt \int d^N x \left\{ \tilde{\psi}\big(\partial_t + \mu - D\nabla^2\big)\psi + \frac{g}{2}\big(\tilde{\psi}\psi^2 - \tilde{\psi}^2\psi\big)\right\}, \qquad \text{(A19)}$$

is Reggeon field theory [18], discussed in detail in Appendix B. Its critical behavior defines the *directed percolation* (DP) universality class [16, 17, 19]. The action is invariant under the *rapidity reversal* $\psi(x,t) \leftrightarrow -\tilde{\psi}(x,-t)$. The p-adic counterpart, with $-D\nabla^2$ replaced by $\boldsymbol{W}$ and $[t] = -z$ with $z = \sigma$, is Eq. (59) of the main text.

The DP conjecture. Janssen and Grassberger conjectured [16, 17] that a stochastic process belongs to the DP class if (i) it has a unique absorbing state, (ii) its activity is described by a single non-negative order parameter, (iii) its interactions are short-ranged, and (iv) it has no additional symmetries, conservation laws or quenched disorder. The effective spike model with $f(0) = 0$, $f'(0) > 0$ and $f''(0) < 0$ satisfies (i), (ii) and (iv). Condition (iii) is where the geometry enters. In the p-adic theory, interactions are long-ranged in the sense of hierarchical distance, and the class is modified to the ultrametric DP class labelled by σ (Sec. V).

Critical exponents. The DP class is characterized by the exponents

$$\rho \propto (-\mu)^\beta, \quad \xi_\perp \propto |\mu|^{-\nu_\perp}, \quad \xi_\parallel \propto |\mu|^{-\nu_\parallel}, \quad z = \nu_\parallel/\nu_\perp, \qquad \text{(A20)}$$

for the stationary density of active sites, the spatial correlation length and the temporal correlation length. Rapidity reversal implies that the survival probability of activity started from a single seed has the same exponent $\beta' = \beta$. Mean-field values are $\beta = 1$, $\nu_\perp = \frac{1}{2}$, $\nu_\parallel = 1$, $z = 2$. Numerical estimates in low dimensions are (Table 2 of Ref. [19]; the digits in parentheses are the uncertainty in the last digits):

N	β	$\nu_\perp$	$\nu_\parallel$	z
1	0.276486(8)	1.096854(4)	1.733847(6)	1.580745(10)
2	0.584(4)	0.734(4)	1.295(6)	1.76(3)
3	0.81(1)	0.581(5)	1.105(5)	1.90(1)
≥ 4	1	1/2	1	2

Avalanches. In the mean-field regime, the process started from one particle is a critical Galton–Watson branching process. The total number of particles produced (the avalanche size S) and the duration T then have the distributions $P(S) \propto S^{-3/2}$ and $P(T) \propto T^{-2}$ [79]. The exponent $3/2$ is the one measured by Beggs and Plenz in cortical slices [3], and its p-adic counterpart is discussed in Sec. VIII.

Appendix B: Directed percolation and Reggeon field theory

This appendix explains the universality class to which the critical point of the effective spike model belongs. Directed percolation (DP) is a simple lattice model. Reggeon field theory (RFT) is the field theory that describes its large-scale behavior; see Refs. [19, 52, 78].

1. Directed percolation as a model of spreading

The lattice model. In this appendix, p denotes a probability, not a prime number, and N denotes the number of spatial dimensions of the lattice (the time direction is not counted); it plays the role of the dimension N of the neuron space in the main text. Place sites on a lattice in which one direction plays the role of *time* (Fig. 9). Each site is connected to two sites in the next time step, by bonds that are open with probability p and closed with probability $1 - p$, independently. A site at time $t + 1$ is *active* if at least one open bond connects it to an active site at time t. Starting from a single active site (the *seed*), the active sites form a *cluster* that spreads forward in time. Because bonds can only be followed forward in time, the percolation is *directed*.

Neural and machine-learning readings. For a neuroscientist, time steps are successive rounds of firing. Bonds are synapses that transmit a spike with probability p, and a cluster started by one spike is a *neuronal avalanche*. For a machine-learning reader, each time step is a layer of a deep feedforward network with random binary connections and "OR" units: a unit fires if at least one of its inputs fires. The question "does a signal injected at the input reach arbitrarily deep layers?" is exactly the question of whether a DP cluster survives.

The phase transition. If p is small, every cluster dies after a finite time: the empty configuration is an absorbing state, and the system is in the *absorbing phase*. If p is large, a cluster survives forever with positive probability and occupies a finite fraction of the sites: this

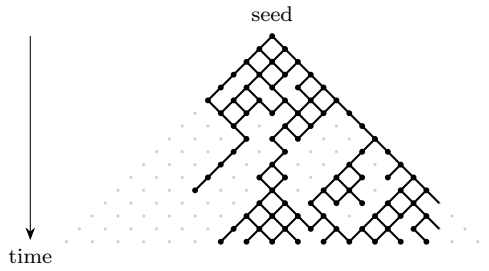


FIG. 9. A directed-percolation cluster in one spatial dimension (horizontal) grown from a single seed, for bond probability $p = 0.66$ (here p is a probability, not a prime), slightly above $p_c \simeq 0.6447$. Open bonds are drawn as lines; active sites as dots; grey dots are inactive sites. Bonds can only be followed downward, i.e. forward in time. In the neural reading, the cluster is an avalanche started by one spike.

is the *active phase*. The two phases are separated by a critical value p_c. For bond DP in one spatial dimension, $p_c \simeq 0.6447$ [19]. The value of p_c depends on the details of the lattice. The behavior near p_c does not.

The contact process. A continuous-time version is the *contact process*. Each active site becomes inactive at rate 1, and it activates each of its $2N$ nearest neighbors, if inactive, at rate $\lambda_{\rm cp}/(2N)$ (on $\mathbb{Z}^N$). It has a critical value $\lambda_{\rm cp,c} \simeq 3.2978$ in one dimension, and the same critical exponents as DP [19]. The branching process of Appendix A and the effective spike model are further members of the same family: decay plays the role of closed bonds, branching the role of open bonds, and saturation the role of the rule "a site is active if *at least one* input is active" (two inputs do not make a site twice as active).

2. What is measured

The critical behavior is described by a small number of quantities. Two initial conditions are natural.

- *Fully active start.* The *density* $\rho(t)$ of active sites decays at p_c as $\rho(t) \propto t^{-\delta}$. In the active phase it saturates at $\rho_{\rm st} \propto (p - p_c)^{\beta}$.
- *Single seed.* The *survival probability* $P(t)$ that the cluster is still alive at time t decays as $P(t) \propto t^{-\delta'}$. The mean number of active sites grows as $\mathcal{N}(t) \propto t^{\vartheta}$, and the mean squared spreading distance from the seed grows as $R^2(t) \propto t^{2/z}$.

Away from p_c, correlations decay over a spatial length $\xi_\perp \propto |p - p_c|^{-\nu_\perp}$ and a time $\xi_\parallel \propto |p - p_c|^{-\nu_\parallel}$. The exponents are related by

$$\delta = \frac{\beta}{\nu_\parallel}, \quad \delta' = \frac{\beta'}{\nu_\parallel}, \quad z = \frac{\nu_\parallel}{\nu_\perp}, \quad \vartheta = \frac{N\nu_\perp - \beta - \beta'}{\nu_\parallel}, \tag{B1}$$

the last one ("hyperscaling") holding below the upper critical dimension. In DP, $\beta = \beta'$ (Sec. B 4), so there are only three independent exponents, e.g. β, $\nu_\perp$, $\nu_\parallel$. Their values are listed in Appendix A 6. Avalanche statistics follow from the same exponents; for instance the avalanche-size exponent is $\tau_S = 1 + \beta'/(\nu_\parallel + N\nu_\perp - \beta)$ (Sec. VIII).

3. From DP to a field theory

A coarse-grained equation. On scales much larger than the lattice spacing, the state of the system is described by a coarse-grained density $\rho(x,t) \geq 0$. Janssen and Grassberger argued that its dynamics is captured by the Langevin equation [16, 17]

$$\begin{aligned} &\partial_t \rho(x,t) = a\rho(x,t) - b\rho(x,t)^2 + D\nabla^2\rho(x,t) + \eta(x,t), \\ &\langle \eta(x,t)\eta(x',t')\rangle = Q\,\rho(x,t)\,\delta(x-x')\,\delta(t-t'), \end{aligned} \tag{B2}$$

where $Q > 0$ is a constant. The noise is uncorrelated in space and time, and its variance at (x,t) is proportional to the local density $\rho(x,t)$; the product $\rho(x,t)\,\delta(x-x')$ means that the variance is evaluated at the point where the noise acts (with the Itô convention in time). Each term has a simple meaning: $a\rho$ is the net growth rate (branching minus decay), $-b\rho^2$ is saturation, $D\nabla^2\rho$ is spreading to neighbors, and η is the noise from the randomness of individual events. The noise variance is proportional to ρ, so it vanishes in the absorbing state $\rho = 0$. This is the defining feature of the class. The mean-field version ($\eta = 0$) is a rate equation of the same form as Eq. (A5).

The action. Applying the MSRJD construction of Appendix A 4 to Eq. (B2) gives the action

$$S = \int_0^\infty dt \int d^N x \left\{ \tilde\rho\big(\partial_t - a - D\nabla^2\big)\rho + b\,\tilde\rho\rho^2 - \frac{Q}{2}\,\tilde\rho^{\,2}\rho \right\}. \tag{B3}$$

Rescaling the fields as $\rho = c\,\psi$, $\tilde\rho = \tilde\psi/c$ with $c = \sqrt{Q/(2b)}$ makes the two cubic couplings equal. With $\mu = -a$ and $g = \sqrt{2bQ}$ one obtains

$$S_{\rm RFT} = \int_0^\infty dt \int d^N x \left\{ \tilde\psi\big(\partial_t + \mu - D\nabla^2\big)\psi + \frac{g}{2}\big(\tilde\psi\psi^2 - \tilde\psi^{\,2}\psi\big) \right\}. \tag{B4}$$

This is Reggeon field theory, Eq. (A19). The Doi–Peliti action of the branching process (A11) leads to the same result after the same rescaling (the quartic term $\lambda\tilde\varphi^2\varphi^2$ being irrelevant). So does the B-C action, whose rescaling is Eq. (58) of Ref. [13]. The *signs* matter. The saturation term $+\frac{g}{2}\tilde\psi\psi^2$ must enter the action with the sign shown, so that the mean-field equation $\partial_t\psi = -\mu\psi - \frac{g}{2}\psi^2 + D\nabla^2\psi$ saturates. In the effective spike model this sign is guaranteed by $f''(0) < 0$.

MSRJD and Doi–Peliti. The two constructions lead to actions of the same form but start from different places. The MSRJD construction [81–83] starts from a

Langevin equation for a continuous field, here the coarse-grained density ρ, with a postulated Gaussian noise. The response field $\tilde{\rho}$ enforces the equation of motion, and averaging over the noise produces the term $-\frac{Q}{2}\tilde{\rho}^2\rho$; the action is therefore exactly quadratic in $\tilde{\rho}$, and ρ is the density itself. The Doi–Peliti construction [14, 15] (Appendix A 3) starts from the master equation of the discrete particle numbers and involves no approximation. Its field φ is the eigenvalue of the annihilation operator, not the density: $\langle\varphi\rangle$ is the mean number, but higher moments of φ are factorial moments of the counts (Appendix A 4), and the action may contain any power of $\tilde{\varphi}$, such as the term $\lambda\tilde{\varphi}^2\varphi^2$ of Eq. (A11). Truncating the Doi–Peliti action at second order in $\tilde{\varphi}$ gives an action of MSRJD form, i.e. a Langevin equation (Appendix A 4, "Connection with stochastic differential equations"); this Langevin reading is formal, since its noise covariance can be negative. Near the critical point the two descriptions agree, because the higher powers of $\tilde{\varphi}$ are irrelevant and both actions reduce to Eq. (B4). The B-C theory uses the Doi–Peliti construction, so it is an exact rewriting of the effective spike model before any truncation; the Langevin equation (B2) of Janssen and Grassberger is a coarse-grained, phenomenological description of the same universality class.

Why "Reggeon"? The same action appeared in the 1970s in high-energy physics, in Gribov's description of hadron scattering at high energies in terms of "Reggeons". Grassberger and Sundermeyer, and Cardy and Sugar, showed that this theory is equivalent to DP [18, 84]. Only the name survives from that context.

4. Rapidity reversal

The action (B4) is invariant under

$$\psi(x,t) \to -\tilde{\psi}(x,-t), \qquad \tilde{\psi}(x,t) \to -\psi(x,-t), \quad \text{(B5)}$$

which exchanges the density field and the response field and reverses time. It is called *rapidity reversal*, and it is an emergent symmetry: the microscopic model need not have it. One consequence is that the density decay from a fully active start and the survival probability from a single seed have the same exponent, $\beta = \beta'$ and $\delta = \delta'$. In DP these are exactly the same statistics: on the lattice, the probability that a seed survives to time t equals the probability that a given site is active at time t when starting from a fully active configuration. In the neural reading, the exponent governing how long an avalanche lasts equals the one governing how fast activity decays after global stimulation.

5. Mean-field theory

Setting $\tilde{\psi} = 0$ in the equations of motion of Eq. (B4) gives $\partial_t\psi = -\mu\psi - \frac{g}{2}\psi^2 + D\nabla^2\psi$. Its homogeneous stationary solutions are $\psi = 0$ and, for $\mu < 0$, $\psi = 2|\mu|/g$. The density therefore vanishes linearly at the transition: $\beta = 1$. The linear response is $e^{-(\mu+Dq^2)t}$, which relaxes at rate μ and spreads diffusively. Hence $\nu_\parallel = 1$, $\nu_\perp = \frac{1}{2}$ and $z = 2$. At criticality, activity started from a single seed is a critical branching process. Its survival probability decays as $1/t$, and its avalanche sizes and durations are distributed as $S^{-3/2}$ and T^{-2} (Appendix A 6).

6. Beyond mean field: the ε expansion

Power counting (Appendix A 6) shows that the coupling g has dimension $2 - N/2$. Fluctuations are therefore negligible above $N_c = 4$ and change the exponents below it. Near four dimensions the RG can be carried out perturbatively in $\varepsilon = 4 - N$. The one-loop results are [16, 19, 52]

$$\beta = 1 - \frac{\varepsilon}{6}, \quad \nu_\perp = \frac{1}{2} + \frac{\varepsilon}{16}, \quad \nu_\parallel = 1 + \frac{\varepsilon}{12}, \quad z = 2 - \frac{\varepsilon}{12}. \quad \text{(B6)}$$

They reproduce the trend of the numerical values in $N = 1, 2, 3$ (Appendix A 6). These coefficients agree with Eqs. (208) and (210) and Table 2 of Ref. [85], where $1/\nu_\perp = 2 - \varepsilon/4$, $z = 2 - \varepsilon/12$ and $\beta = 1 - \varepsilon/6$, with $\nu_\parallel = z\nu_\perp = 1 + \varepsilon/12$ to this order. They also agree with the $d = 4 - \varepsilon$ column of Table 2 of Ref. [19].

Long-range spreading. If activity spreads by long jumps whose length distribution decays as $r^{-N-\sigma}$ ("Lévy flights"), the operator $-D\nabla^2$ is replaced by a fractional operator with symbol $|q|^\sigma$. Power counting then gives $N_c = 2\sigma$. In the Euclidean setting the diffusive term takes over again for $\sigma \geq 2$ [44]. In the p-adic theory of this paper there is no local diffusive term, the kinetic operator is the nonlocal diffusion operator (W-operator) of Sec. IV A, and every $\sigma > 0$ gives a distinct class with $N_c = 2\sigma$ (Sec. V).

7. From the effective spike model to Reggeon field theory

The following summary connects the main text with this appendix.

1. *Absorbing state:* $f(0) = 0$ with $I = 0$ (Sec. III D).
2. *Branching:* $f'(0) > 0$; the linear term $f'(0)\,\tilde{\varphi}\, w \star \varphi$.
3. *Saturation:* $f''(0) < 0$; it fixes the sign of $\frac{g}{2}\tilde{\psi}\psi^2$.
4. *Distance to criticality:* $\mu = \alpha - f'(0)\hat{w}(0)$.
5. *Spreading operator:* $-D\nabla^2$ in the Euclidean theory; the W-operator, with symbol $\simeq \kappa_W\|q\|_p^\sigma$, in the p-adic theory.
6. *Upper critical dimension:* $N_c = 4$ (Euclidean), $N_c = 2\sigma$ (p-adic).

Experimental status. Although DP is the generic class for absorbing-state transitions, clean experimental realizations are rare. A well-known example is the transition between turbulent states in liquid-crystal convection [86]. Neuronal avalanches, with size and duration exponents close to 3/2 and 2 [3], are consistent with the mean-field DP (critical branching) values. This is the regime predicted by B-C for highly connected cortical networks, and by the p-adic theory whenever $\sigma \leq N/2$.

Appendix C: Mean-field exponents of the p-adic theory

In this appendix we derive the mean-field exponents (66) from the action (59). Three of them (β, β', $\nu_\parallel$) come from the dynamics of the total activity alone and are the same as in B-C. The other two ($\nu_\perp$, z) come from how the relaxation rate depends on the length scale; this is where σ enters. Throughout, μ is the distance to criticality: the network is subcritical (absorbing phase) for $\mu > 0$, critical at $\mu = 0$, and supercritical (active phase) for $\mu < 0$.

1. Definitions

Near the critical point the characteristic quantities behave as powers of $|\mu|$, and the exponents are defined by these powers:

- β: the stationary density of activity, starting from a fully active network, $\rho \propto (-\mu)^{\beta}$ for $\mu < 0$ [Eq. (62)];
- β': the probability that the activity started by one spike never dies out, $P_\infty \propto (-\mu)^{\beta'}$ for $\mu < 0$ [Eq. (62)];
- $\nu_\parallel$: the correlation time, $\xi_\parallel \propto |\mu|^{-\nu_\parallel}$;
- $\nu_\perp$: the correlation length, $\xi_\perp \propto |\mu|^{-\nu_\perp}$;
- z: the dynamic exponent, $\xi_\parallel \propto \xi_\perp^{z}$, i.e. $z = \nu_\parallel/\nu_\perp$.

These are the standard exponents of directed percolation [Eq. (60) of Ref. [13]].

2. The mean-field equation

Mean-field theory is the saddle point of the action (59). The equation $\delta S^W_{\text{eff}}/\delta\tilde\psi = 0$ reads

$$(\partial_t + \mu + \boldsymbol{W})\psi + \frac{g}{2}\psi^2 - g\tilde\psi\psi = 0.$$

At the saddle point the response field vanishes, $\tilde\psi = 0$, by the same mechanism as $\tilde a = 0$ in Sec. III C (conservation of probability). The mean activity $\rho(x,t) = \langle\psi(x,t)\rangle$ therefore obeys

$$\partial_t\rho = -\mu\rho - \boldsymbol{W}\rho - \frac{g}{2}\rho^2. \tag{C1}$$

This is the hierarchical Wilson–Cowan equation near the critical point, written for the rescaled field: decay minus branching ($-\mu\rho$), spreading through the hierarchy ($-\boldsymbol{W}\rho$) and saturation ($-\frac{g}{2}\rho^2$).

3. $\beta = 1$

For a homogeneous state $\boldsymbol{W}\rho = 0$, since the symbol of $\boldsymbol{W}$ vanishes at $\xi = 0$. Equation (C1) becomes $d\rho/dt = -\rho(\mu + \frac{g}{2}\rho)$. Its stationary solutions are $\rho = 0$ (the absorbing state) and

$$\rho = -\frac{2\mu}{g} = \frac{2|\mu|}{g} \qquad (\mu < 0).$$

For $\mu > 0$ only $\rho = 0$ is admissible ($\rho \geq 0$), and it is stable. For $\mu < 0$ the state $\rho = 0$ is unstable, since $d\rho/dt \simeq |\mu|\rho > 0$ for small ρ, and the activity settles at $\rho = 2|\mu|/g$. The density grows linearly with the distance to criticality, so $\beta = 1$.

4. $\beta' = 1$

The survival probability is a property of a single avalanche. In mean-field theory an avalanche is a branching process (Appendix A): each active unit produces a new one at rate b and disappears at rate d, and $\mu \propto d - b$. For binary branching in continuous time, let q be the probability that the activity started by one unit eventually dies out. The first event is a birth with probability $b/(b+d)$, after which two independent units are present, or a death with probability $d/(b+d)$. Hence

$$q = \frac{d}{b+d} + \frac{b}{b+d}\,q^2, \qquad \text{i.e.} \qquad bq^2 - (b+d)q + d = 0,$$

whose roots are $q = 1$ and $q = d/b$. The extinction probability is the smallest root in $[0,1]$, so the probability that the activity never dies out is $P_\infty = 1 - d/b = (b-d)/b$ for $b > d$, and $P_\infty = 0$ for $b \leq d$. Near criticality $b - d \propto |\mu|$, so $P_\infty \propto |\mu|$ and $\beta' = 1$. The same result follows from the action: the saddle-point equation $\delta S^W_{\text{eff}}/\delta\psi = 0$ has a stationary homogeneous solution with $|\tilde\psi| = 2|\mu|/g$, and rapidity reversal (61), which exchanges ψ and $-\tilde\psi$, gives $\beta' = \beta$.

5. $\nu_\parallel = 1$

Linearize Eq. (C1) about its stable stationary state and expand the deviation in Fourier modes (equivalently,

in the wavelets $\psi_{\gamma nj}$). For $\mu > 0$ the stable state is $\rho = 0$, and each mode evolves independently, $\hat\rho(\xi,t) = \hat\rho(\xi,0)\, e^{-\lambda(\xi)t}$, with $\lambda(\xi) = \mu + A_w(\xi)$. For $\mu < 0$ the stable state is $\rho^* = 2|\mu|/g$; writing $\rho = \rho^* + \delta\rho$, the linear term of Eq. (C1) is $-(\mu + g\rho^*)\delta\rho = -|\mu|\,\delta\rho$, so the deviations relax with $\lambda(\xi) = |\mu| + A_w(\xi)$. On both sides of the transition

$$\lambda(\xi) = |\mu| + A_w(\xi).$$

The slowest mode is the homogeneous one, for which $A_w(0) = 0$; it relaxes on the time scale $\xi_\parallel = 1/|\mu|$. Hence $\nu_\parallel = 1$, with the same amplitude on both sides.

6. $\nu_\perp = 1/\sigma$

The relaxation rate has two parts: μ, the same for all modes, and $A_w(\xi) \simeq \kappa_W \|\xi\|_p^\sigma$, which is larger for modes supported on smaller balls. Modes with $\kappa_W\|\xi\|_p^\sigma \ll |\mu|$, supported on large balls, relax at essentially the same rate as the homogeneous mode: on these scales fluctuations are correlated. Modes with $\kappa_W\|\xi\|_p^\sigma \gg |\mu|$, supported on small balls, relax much faster, and fluctuations die out before they spread. The two regimes are separated by the scale at which $\kappa_W\|\xi\|_p^\sigma \approx |\mu|$, and the corresponding length is the correlation length

$$\xi_\perp = \Big(\frac{\kappa_W}{|\mu|}\Big)^{1/\sigma},$$

so $\nu_\perp = 1/\sigma$. In $\mathbb{Q}_p^N$ the norm takes only the values p^k, so this scale is a level $k^* = \log_p \xi_\perp$ of the hierarchy, and $\xi_\perp$ is defined up to a factor p; this is the length ℓ_W of the Ginzburg criterion (Sec. VI), whose discreteness produces log-periodic corrections without changing the exponent. In the Euclidean theory $A_w(q) \simeq Dq^2$, and the same argument gives $\xi_\perp = (D/|\mu|)^{1/2}$, i.e. $\nu_\perp = \frac12$; this is B-C's diffusion length [Eqs. (52)–(53) of Ref. [13]].

7. $z = \sigma$

At $\mu = 0$ the only rate left is $A_w(\xi) \simeq \kappa_W\|\xi\|_p^\sigma$. A fluctuation of size L, supported on a ball of radius L ($\|\xi\|_p \approx 1/L$), relaxes in a time $T \approx L^\sigma/\kappa_W$. Time scales as (length)$^\sigma$, so $z = \sigma$; equivalently, $z = \nu_\parallel/\nu_\perp = \sigma$. This is the dynamic exponent of the power counting in Sec. V. In the Euclidean theory $T \approx L^2/D$ (diffusion), and $z = 2$.

8. Ginzburg estimate

We derive Eq. (67). By the correspondence between Eqs. (B2) and (B4) (Appendix B 3), the action (59) is the response-field action of the Langevin equation

$$\partial_t\psi = -\mu\psi - \boldsymbol{W}\psi - \frac{g}{2}\psi^2 + \eta,$$
$$\langle \eta(x,t)\eta(x',t')\rangle = g\,\psi(x,t)\,\delta(x-x')\delta(t-t') :$$

the term $-\frac{g}{2}\tilde\psi^2\psi$ of Eq. (59) is the noise, with variance proportional to the activity (multiplicative noise), and the term $\frac{g}{2}\tilde\psi\psi^2$ is the saturation. Without noise this is Eq. (C1).

Take $\mu < 0$, so that mean-field theory predicts the homogeneous density $\rho^* = 2|\mu|/g$, and ask how large the fluctuations about ρ^* are. On scales smaller than $\xi_\perp$ the activity is correlated, so we average over a ball B of radius $\xi_\perp$, of volume $V = \xi_\perp^N$: $\rho_B(t) = V^{-1}\int_B \psi(x,t)\,d^Nx$. Modes supported on balls of radius at least $\xi_\perp$ have $A_w(\xi) \lesssim |\mu|$, so, up to factors of order one, $\boldsymbol{W}$ does not change the relaxation of ρ_B, and the deviation $\delta\rho = \rho_B - \rho^*$ obeys, by the linearization of Appendix C 5,

$$\frac{d\,\delta\rho}{dt} = -|\mu|\,\delta\rho + \eta_B, \qquad \eta_B = \frac{1}{V}\int_B \eta\, d^Nx.$$

Since η is uncorrelated in space, $\langle \eta_B(t)\eta_B(t')\rangle = (g\rho^*/V)\,\delta(t-t')$. This is an Ornstein–Uhlenbeck process: noise of intensity $g\rho^*/V$ is accumulated during the relaxation time $\xi_\parallel = 1/|\mu|$, and the stationary variance is

$$\langle \delta\rho^2\rangle = \frac{g\rho^*/V}{2|\mu|} = \frac{g\rho^*\,\xi_\parallel}{2V}.$$

Dividing by ρ^{*2} and using $\rho^* = 2|\mu|/g$, $\xi_\parallel = 1/|\mu|$ and $V = \xi_\perp^N = (\kappa_W/|\mu|)^{N/\sigma}$ gives

$$\frac{\langle\delta\rho^2\rangle}{\rho^{*2}} = \frac{g\,\xi_\parallel}{2\rho^*\,\xi_\perp^N} = \frac{g^2}{4\,\kappa_W^{N/\sigma}}\,|\mu|^{N/\sigma-2},$$

which is Eq. (67) with its prefactor. Mean-field theory is consistent when this ratio is small. As $\mu \to 0$ it tends to zero if $N > 2\sigma$ and diverges if $N < 2\sigma$; at $N = 2\sigma$ it is a constant of order g^2/κ_W^2, the marginal case. The estimate treats the ball B as well mixed and drops factors of order one; it locates the upper critical dimension but not the exponents below it. The same condition follows from the one-loop integral of Sec. VI and from power counting.

9. Comparison and validity

The exponents of the total activity (β, β', $\nu_\parallel$) are those of B-C, because they come from the equation $d\rho/dt = -\mu\rho - \frac{g}{2}\rho^2$, which does not see the geometry. The spatial exponents ($\nu_\perp$, z) come from the kinetic operator, Dq^2 in B-C and $\kappa_W\|\xi\|_p^\sigma$ here; the two sets coincide for $\sigma = 2$. Mean-field exponents are exact only when the cubic coupling is irrelevant, i.e. for $N > N_c = 2\sigma$ (with logarithmic corrections at $N = 2\sigma$); for $N < 2\sigma$ loop corrections change all five exponents. The mean-field avalanche exponents (sizes distributed as $S^{-3/2}$, durations as T^{-2}) come from the same branching process as β' and do not depend on σ; the simulations of Sec. VIII A test when they hold.

Appendix D: Basic facts on p-adic analysis

In this appendix we fix the notation and collect the results of p-adic analysis used in the paper. For detailed expositions see Refs. [33, 35–37, 87].

1. The field of p-adic numbers

Throughout, p denotes a fixed prime number. The field of p-adic numbers $\mathbb{Q}_p$ is the completion of the field of rational numbers $\mathbb{Q}$ with respect to the p-adic norm

$$|x|_p = \begin{cases} 0 & \text{if } x = 0, \\ p^{-\gamma} & \text{if } x = p^\gamma \, \dfrac{a}{b}, \end{cases} \tag{D1}$$

where a and b are integers coprime with p. The integer $\gamma = \mathrm{ord}(x)$, with $\mathrm{ord}(0) = +\infty$, is called the p-adic order of x.

Any p-adic number $x \neq 0$ has a unique expansion

$$x = p^{\mathrm{ord}(x)} \sum_{j=0}^{\infty} x_j p^j, \qquad x_j \in \{0, 1, \dots, p-1\}, \quad x_0 \neq 0, \tag{D2}$$

and $|x|_p = p^{-\mathrm{ord}(x)}$. The *fractional part* of $x \in \mathbb{Q}_p$ is the rational number

$$\{x\}_p = \begin{cases} 0 & \text{if } x = 0 \text{ or } \mathrm{ord}(x) \geq 0, \\ p^{\mathrm{ord}(x)} \displaystyle\sum_{j=0}^{-\mathrm{ord}(x)-1} x_j p^j & \text{if } \mathrm{ord}(x) < 0. \end{cases} \tag{D3}$$

The p-adic integers are the numbers with $|x|_p \leq 1$, i.e. those whose expansion has no negative powers of p; they form the ring $\mathbb{Z}_p$. The successive digits of the expansion (D2) select a path in a rooted tree in which every vertex has p children. This is the origin of the correspondence between $\mathbb{Z}_p$ and the leaves of an infinite rooted tree used in Sec. III A. This description holds for $\mathbb{Z}_p$ and, more generally, for any ball. It does not hold for $\mathbb{Q}_p$: the expansion of a general p-adic number starts at an arbitrary negative power of p, the balls $p^{-l}\mathbb{Z}_p$ grow without bound, and there is no root. The points of $\mathbb{Q}_p$ are organized in a tree-like structure, which is not a rooted tree.

The norm is extended to $\mathbb{Q}_p^N$ by

$$\|x\|_p = \max_{1 \leq i \leq N} |x_i|_p, \qquad x = (x_1, \dots, x_N) \in \mathbb{Q}_p^N. \tag{D4}$$

With $\mathrm{ord}(x) = \min_i \mathrm{ord}(x_i)$ one has $\|x\|_p = p^{-\mathrm{ord}(x)}$, so the norm takes only the values p^k, $k \in \mathbb{Z}$, and 0. It satisfies the *ultrametric inequality*

$$\|x + y\|_p \leq \max\{\|x\|_p, \|y\|_p\}, \tag{D5}$$

which is stronger than the triangle inequality, and $(\mathbb{Q}_p^N, \|\cdot\|_p)$ is a complete ultrametric space.

2. Balls, spheres and topology

For $r \in \mathbb{Z}$ and $a \in \mathbb{Q}_p^N$, the ball and the sphere of radius p^r with center a are

$$B_r^N(a) = \{x \in \mathbb{Q}_p^N : \|x - a\|_p \leq p^r\} = a + p^{-r}\mathbb{Z}_p^N, \tag{D6}$$

$$\begin{aligned} S_r^N(a) &= \{x \in \mathbb{Q}_p^N : \|x - a\|_p = p^r\} \\ &= B_r^N(a) \smallsetminus B_{r-1}^N(a), \end{aligned} \tag{D7}$$

and we write $B_r^N = B_r^N(0)$, $S_r^N = S_r^N(0)$. In particular $B_0^N = \mathbb{Z}_p^N$ is the unit ball. The ultrametric inequality has the following consequences, which have no Euclidean counterpart. Every point of a ball is a center of it. Two balls are either disjoint or one is contained in the other. Each ball of radius p^r is the disjoint union of p^N balls of radius p^{r-1}. The balls therefore form a tree under inclusion, and the distance $\|x - y\|_p$ is the radius of the smallest ball that contains x and y.

Balls and spheres are both open and closed, and they are compact. The space $\mathbb{Q}_p^N$ is locally compact and totally disconnected: its only connected subsets are the empty set and the points. We denote by $\Omega(p^{-r}\|x - a\|_p)$ the characteristic function of the ball $B_r^N(a)$; here Ω is the characteristic function of the interval $[0, 1]$.

3. The Haar measure

Since $(\mathbb{Q}_p^N, +)$ is a locally compact topological group, it has a Haar measure $d^N x$, which is invariant under translations, $d^N(x + a) = d^N x$. It is unique once normalized by $\int_{\mathbb{Z}_p^N} d^N x = 1$. With this normalization,

$$\mathrm{vol}\big(B_r^N(a)\big) = p^{rN}, \quad \mathrm{vol}\big(S_r^N(a)\big) = (1 - p^{-N})\, p^{rN}, \tag{D8}$$

and $d^N(cx) = |c|_p^N \, d^N x$ for $c \in \mathbb{Q}_p \smallsetminus \{0\}$. These formulas give Eq. (1) for the integral of a radial function.

4. Test functions and L^ρ spaces

A function $\varphi : \mathbb{Q}_p^N \to \mathbb{C}$ is *locally constant* if for every x there is an integer $l(x)$ such that

$$\varphi(x + x') = \varphi(x) \quad \text{for all } x' \in B_{l(x)}^N. \tag{D9}$$

A *test function* (or Bruhat–Schwartz function) is a locally constant function with compact support. Every test function is a finite linear combination of characteristic functions of balls. The space of test functions is denoted by $\mathcal{D}(\mathbb{Q}_p^N)$. For a test function, the integer $l(x)$ in Eq. (D9) can be chosen independently of x; the largest such integer is its parameter of constancy.

For integers $m \leq l$, we denote by $\mathcal{D}_m^l(\mathbb{Q}_p^N)$ the space of test functions supported in the ball $B_{-m}^N = p^m\mathbb{Z}_p^N$

and with parameter of constancy $\geq -l$. A function in $\mathcal{D}^l_m(\mathbb{Q}_p^N)$ is constant on the cosets of $p^l\mathbb{Z}_p^N$ in $p^m\mathbb{Z}_p^N$, i.e. on the $p^{N(l-m)}$ balls of radius p^{-l} contained in B^N_{-m}, and it is determined by its values on them. Therefore $\mathcal{D}^l_m(\mathbb{Q}_p^N)$ has finite dimension $p^{N(l-m)}$. The space $\mathcal{D}^l$ of Sec. III A, which describes the finite network G_l^N, is $\mathcal{D}^l_0(\mathbb{Q}_p^N)$ in this notation: functions on $\mathbb{Z}_p^N$ that are constant on the balls of radius p^{-l}; its dimension is p^{Nl}.

For $1 \leq \rho < \infty$, $L^\rho(\mathbb{Q}_p^N)$ is the space of functions g with

$$\|g\|_\rho = \Big(\int_{\mathbb{Q}_p^N} |g(x)|^\rho \, d^N x \Big)^{1/\rho} < \infty. \tag{D10}$$

We use $\rho = 1$ and $\rho = 2$. The test functions are dense in $L^\rho(\mathbb{Q}_p^N)$.

5. The Fourier transform

Set $\chi_p(y) = \exp(2\pi i \{y\}_p)$ for $y \in \mathbb{Q}_p$. The map χ_p is an additive character of $\mathbb{Q}_p$: it is continuous, $|\chi_p(y)| = 1$, and $\chi_p(y + y') = \chi_p(y)\chi_p(y')$. It is trivial on $\mathbb{Z}_p$. For $\xi, x \in \mathbb{Q}_p^N$ set $\xi \cdot x = \sum_{j=1}^N \xi_j x_j$. The Fourier transform of $\varphi \in \mathcal{D}(\mathbb{Q}_p^N)$ is

$$\hat\varphi(\xi) = (\mathcal{F}\varphi)(\xi) = \int_{\mathbb{Q}_p^N} \chi_p(\xi \cdot x)\, \varphi(x)\, d^N x. \tag{D11}$$

It is a linear isomorphism of $\mathcal{D}(\mathbb{Q}_p^N)$ onto itself, and

$$\big(\mathcal{F}(\mathcal{F}\varphi)\big)(\xi) = \varphi(-\xi), \tag{D12}$$

which is the inversion formula of Sec. III D. The Fourier transform maps $\mathcal{D}^l_m$ onto $\mathcal{D}^{-m}_{-l}$: a function that is constant on balls of radius p^l has a transform supported in $\|\xi\|_p \leq p^{-l}$. For the finite network G_l^N, whose fields are constant on balls of radius p^{-l}, this gives the momentum cutoff $\|\xi\|_p \leq p^l$. The basic example is Eq. (2), the transform of the characteristic function of a ball.

The Fourier transform extends to a unitary operator on $L^2(\mathbb{Q}_p^N)$, $\|\mathcal{F}g\|_2 = \|g\|_2$, and Eq. (D12) holds in L^2. For $g \in L^1(\mathbb{Q}_p^N)$ the integral defining $\hat g$ converges, and $\hat g$ is bounded and continuous. For $w, g \in L^1(\mathbb{Q}_p^N)$ the convolution

$$(w \star g)(x) = \int_{\mathbb{Q}_p^N} w(x - y)\, g(y)\, d^N y \tag{D13}$$

is in $L^1(\mathbb{Q}_p^N)$, with $\|w \star g\|_1 \leq \|w\|_1 \|g\|_1$ and $\widehat{w \star g} = \hat w\, \hat g$.

6. Distributions

A distribution is a linear functional on $\mathcal{D}(\mathbb{Q}_p^N)$; the space of distributions is denoted by $\mathcal{D}'(\mathbb{Q}_p^N)$. In contrast with the Euclidean case, no continuity condition is needed: every linear functional on $\mathcal{D}(\mathbb{Q}_p^N)$ is continuous. We write (T, φ) for the value of T on φ. A sequence T_j converges to T if $(T_j, \varphi) \to (T, \varphi)$ for every test function φ. Every locally integrable function g defines a distribution by

$$(g, \varphi) = \int_{\mathbb{Q}_p^N} g(x)\, \varphi(x)\, d^N x. \tag{D14}$$

The Dirac distribution is $(\delta, \varphi) = \varphi(0)$. It is the limit of the functions $\delta_k(x) = p^{Nk}\, \Omega(p^k \|x\|_p)$ as $k \to \infty$, which are the normalized characteristic functions of the balls B^N_{-k}. The function δ_l is the regularized delta function of the finite network G_l^N (Sec. III D).

The Fourier transform of a distribution T is defined by

$$(\mathcal{F}[T], \varphi) = (T, \mathcal{F}[\varphi]) \quad \text{for all } \varphi \in \mathcal{D}(\mathbb{Q}_p^N). \tag{D15}$$

It is a linear isomorphism of $\mathcal{D}'(\mathbb{Q}_p^N)$ onto itself, and it agrees with the previous definitions on L^1 and L^2. In particular $\mathcal{F}[\delta] = 1$ and $\mathcal{F}[1] = \delta$. A bounded function such as $e^{tf'(0)\hat w(\xi)}$, which does not decay at infinity, therefore has an inverse Fourier transform in $\mathcal{D}'(\mathbb{Q}_p^N)$ but not as a function. This is the sense of the propagator formula (31).

The convolution of a distribution T with a test function ψ is the locally constant function $(T \star \psi)(y) = (T(x), \psi(y - x))$; in particular $\delta \star \psi = \psi$. Whenever the convolution of two distributions F and G exists, for instance if one of them has compact support or if both are in L^1, one has

$$\mathcal{F}[F \star G] = \mathcal{F}[F] \cdot \mathcal{F}[G], \tag{D16}$$

see Ref. [33], Sec. 7.

7. Explicit constants in the heat-kernel estimate

Here we derive explicit admissible constants in Eq. (94). Reference [68] proves Eq. (94) with constants that depend on the doubling constant of the spectral distribution function (their Theorem 2.14 and Lemmas 2.19–2.21), from which explicit values can also be extracted; Ref. [69] proves the upper bound. Here we give a direct derivation from the series (D18) below.

The calculation matters for the paper for two reasons. First, Eq. (94) is the basis of several statements of Sec. X: the spectral dimension $d_s = 2N/\sigma$, the power-law tail of the propagator in hierarchical distance, and the spreading of activity in the active phase. With explicit constants these statements become inequalities valid for all $t > 0$ and all x, with constants that depend only on p, N and σ and not on κ_W, so that the coupling enters only through the combination $\kappa_W t$. Second, the argument works directly with the positive series (D18) and does not use the abstract theory of isotropic semigroups. It is therefore the natural starting point for the open problem of Remark 3, the heat kernel of the W-operator of

the B-C theory, whose symbol is bounded and to which Theorem 5.17 of Ref. [68] does not apply.

Set $s = \kappa_W t > 0$. The heat kernel of $\kappa_W \boldsymbol{D}^\sigma$ is

$$Z_t^{(\sigma)}(x) = \int_{\mathbb{Q}_p^N} e^{-s\|\xi\|_p^\sigma} \chi_p(\xi\cdot x)\, d^N\xi, \qquad x \in \mathbb{Q}_p^N. \quad \text{(D17)}$$

Its integrand is radial, so $Z_t^{(\sigma)}$ is given by the well-known formula for the Fourier transform of a radial function [33], Lemma 6 of Appendix F. With $a_k = e^{-sp^{k\sigma}}$ and $\|x\|_p = p^m$,

$$Z_t^{(\sigma)}(x) = (1-p^{-N}) \sum_{k\le -m} p^{kN} a_k - p^{-mN} a_{1-m}.$$

It is convenient to regroup the terms. Equivalently, decompose $\mathbb{Q}_p^N \smallsetminus \{0\}$ into the spheres $S_k^N = \{\|\xi\|_p = p^k\}$; by Eq. (2), $\int_{S_k^N} \chi_p(\xi\cdot x)\, d^N\xi = b_k - b_{k-1}$ with $b_k = p^{kN}\Omega(p^k\|x\|_p)$, so that $Z_t^{(\sigma)}(x) = \sum_k a_k(b_k - b_{k-1})$. A *summation by parts*, the discrete analogue of integration by parts, moves the difference from b to a: $\sum_k a_k(b_k - b_{k-1}) = \sum_k b_k(a_k - a_{k+1})$, since the boundary terms vanish ($b_k \to 0$ as $k \to -\infty$ and $a_k \to 0$ as $k \to +\infty$). This gives the positive series

$$Z_t^{(\sigma)}(x) = \sum_{k\in\mathbb{Z}} p^{kN}\, \Omega(p^k\|x\|_p)\, \big(e^{-sp^{k\sigma}} - e^{-sp^{(k+1)\sigma}}\big), \quad \text{(D18)}$$

valid for all $x \in \mathbb{Q}_p^N$, including $x = 0$. For $\|x\|_p = p^m$ the sum runs over $k \le -m$. Each term equals $T_k = p^{kN} e^{-y_k}\big(1 - e^{-(p^\sigma-1)y_k}\big)$, with $y_k = sp^{k\sigma}$. Let $F(s,x) = s\big(s^{1/\sigma} + \|x\|_p\big)^{-N-\sigma}$, so that $F_t(x) = F(\kappa_W t, x)$. Since $\max\{s^{1/\sigma}, \|x\|_p\} \le s^{1/\sigma} + \|x\|_p \le 2\max\{s^{1/\sigma}, \|x\|_p\}$,

$$2^{-N-\sigma} M(s,x) \le F(s,x) \le M(s,x), \quad \text{(D19)}$$

where $M(s,x) = \min\big\{s\|x\|_p^{-N-\sigma},\, s^{-N/\sigma}\big\}$.

Upper bound. Since $1 - e^{-z} \le z$, $T_k \le (p^\sigma - 1)\, s\, p^{k(N+\sigma)} e^{-y_k}$. Dropping e^{-y_k} and summing the geometric series over $k \le -m$ gives $Z_t^{(\sigma)}(x) \le C_1\, s\|x\|_p^{-N-\sigma}$, with $C_1 = (p^\sigma - 1)/(1 - p^{-N-\sigma})$. Alternatively, write $p^{kN} = (y_k/s)^{N/\sigma}$, so that $T_k \le (p^\sigma - 1) s^{-N/\sigma} g(\ln y_k)$, with $g(u) = e^{\beta u} e^{-e^u}$ and $\beta = 1 + N/\sigma$. The points $\ln y_k$ are equally spaced with step $\sigma \ln p$, and g is unimodal on $\mathbb{R}$, with maximum $(\beta/e)^\beta$ and $\int_{\mathbb{R}} g(u)\, du = \Gamma(\beta)$. Hence $\sum_k g(\ln y_k) \le (\beta/e)^\beta + \Gamma(\beta)/(\sigma\ln p)$, and $Z_t^{(\sigma)}(x) \le C_2\, s^{-N/\sigma}$, with $C_2 = (p^\sigma - 1)\big[(\beta/e)^\beta + \Gamma(\beta)/(\sigma\ln p)\big]$. Together with Eq. (D19), $Z_t^{(\sigma)}(x) \le \max\{C_1, C_2\} M(s,x) \le 2^{N+\sigma}\max\{C_1, C_2\} F(s,x)$.

Lower bound. Each T_k is nonnegative, so $Z_t^{(\sigma)}(x)$ is bounded below by any single term. We use $1 - e^{-az} \ge (1 - e^{-a})z$ for $z \in [0,1]$ and $a > 0$, which follows from concavity. If $s \le \|x\|_p^\sigma$, take $k = -m$, so $y_{-m} = s\|x\|_p^{-\sigma} \le 1$ and $Z_t^{(\sigma)}(x) \ge T_{-m} \ge e^{-1}\big(1 - e^{-(p^\sigma-1)}\big) s\|x\|_p^{-N-\sigma}$. If $s > \|x\|_p^\sigma$ (in particular if $x = 0$), take $k = -\lceil \sigma^{-1}\log_p s\rceil$. Then $p^{-\sigma} < y_k \le 1$ and $p^k \le s^{-1/\sigma} < \|x\|_p^{-1}$, so the term is present in Eq. (D18), and $p^{kN} = (y_k/s)^{N/\sigma} > p^{-N} s^{-N/\sigma}$. Hence $Z_t^{(\sigma)}(x) \ge T_k \ge e^{-1}p^{-N}\big(1 - e^{-(1-p^{-\sigma})}\big) s^{-N/\sigma}$. The second constant is the smaller one, and by Eq. (D19), $M \ge F$.

Then, we have established the following result.

Proposition 3. *The estimate*

$$c\, F_t(x) \le Z_t^{(\sigma)}(x) \le C\, F_t(x)$$

(Eq. (94)) holds for all $t > 0$ and $x \in \mathbb{Q}_p^N$ with

$$\begin{aligned} c &= e^{-1}p^{-N}\big(1 - e^{-(1-p^{-\sigma})}\big), \\ C &= 2^{N+\sigma}(p^\sigma - 1) \\ &\quad \times \max\Big\{\frac{1}{1-p^{-N-\sigma}}, \Big(\frac{\beta}{e}\Big)^\beta + \frac{\Gamma(\beta)}{\sigma\ln p}\Big\}, \end{aligned} \quad \text{(D20)}$$

where $\beta = 1 + N/\sigma$.

The constants depend only on p, N, and σ, and not on κ_W. They are admissible, not sharp. The sharp constants are the infimum and the supremum of $Z_t^{(\sigma)}(x)/F(s,x)$. This ratio is invariant under the dilation $(s,x) \mapsto (p^\sigma s, p^{-1}x)$, because both $Z^{(\sigma)}$ and F acquire the factor p^{-N}. Hence it suffices to compute the ratio for $\|x\|_p \in \{0,1\}$, as a function of $s > 0$. For example, for $p = 2$, $N = 1$, $\sigma = 1/2$, a numerical evaluation of Eq. (D18) gives a ratio between about 0.53 and 1.49, whereas Eq. (D20) gives $c \approx 0.047$ and $C \approx 8.3$.

Appendix E: Regional networks

In this appendix we prove Lemmas 1 and 2, and extend Sec. IV C to the regional networks of Sec. IV A, Eqs. (49)–(51). In Ref. [43], $N = 1$ and the regions are called basins. The proof of Lemma 1 occupies Appendices E 1–E 3: Appendix E 1 collects two facts used below, part 2 is proved in Appendix E 2, and parts 1 and 3 in Appendix E 3. The proof of Lemma 2 is given in Appendix E 5.

1. Two ultrametric facts

As in Sec. IV A, $B_a = B_{l_a}^N(a) = a + p^{-l_a}\mathbb{Z}_p^N$ denotes the ball of radius p^{l_a} with center a (region a), not a ball centered at the origin. (i) If $a \ne b$, then $\|x - y\|_p = \|a - b\|_p =: d_{ab}$ for all $x \in B_a$, $y \in B_b$, by the ultrametric inequality, since the balls are disjoint. (ii) For $x \in B_a$, $B_a - x = B_{l_a}^N$, so $\int_{B_a} w_a(\|x - y\|_p)\, d^N y = \omega_a$ does not depend on $x \in B_a$. By (i), a translation-invariant radial kernel w restricted to K defines a regional network with $w_a = w$ and $c_{ab} = w(d_{ab})$. With $\alpha_a = \alpha$ and $f_a = f$, Eq. (50) is then Eq. (41) for a network whose neurons occupy K; it is the restriction of Eq. (41) to K when $K + \operatorname{supp} w \subset K$.

2. Reduction to a system

In this section, we prove part 2 of Lemma 1. Write $u_a = u\,1_{B_a}$. Eq. (50) becomes the system

$$\begin{aligned}\partial_t u_a(x,t) = &-\alpha_a\,u_a(x,t)\\ &+ f_a'(0)\int_{B_a} w_a(\|x-y\|_p)\,u_a(y,t)\,d^N y\\ &+ f_a'(0)\sum_{b\neq a} c_{ab}\,U_b(t)\end{aligned}\tag{E1}$$

for $x \in B_a$, where $U_b(t) = \int_{B_b} u(y,t)\,d^N y$ is the total activity of region b (Lemma 1). The integral describes branching inside region a; the sum describes branching from region b into region a, which depends on region b only through its total activity U_b. Integrating over B_a and using (ii), with $\int_{B_a} w_a(\|x-y\|_p)\,d^N x = \omega_a$, gives

$$\begin{aligned}\frac{dU_a}{dt} &= \big(-\alpha_a + f_a'(0)\,\omega_a\big)U_a + f_a'(0)\,\mathrm{vol}(B_a)\sum_{b\neq a} c_{ab}\,U_b\\ &= \sum_b M_{ab}U_b,\end{aligned}$$

where M is the matrix of Eq. (51): $M_{aa} = -\alpha_a + f_a'(0)\,\omega_a$ and $M_{ab} = f_a'(0)\,\mathrm{vol}(B_a)\,c_{ab}$ for $a \neq b$. This is part 2 of Lemma 1. Activity is transferred from region b to region a at rate $f_a'(0)\,\mathrm{vol}(B_a)\,c_{ab}$. At the level of regions the dynamics is a finite continuous-time Markov chain (with creation and killing of activity), and each state of the chain is replaced by a ball carrying the radial kernel w_a. This is the structure of the transition networks of Ref. [43].

3. Spectrum and criticality

In this section, we prove parts 1 and 3 of Lemma 1. Recall that the neuron space is $K = \bigsqcup_{a\in A} B_a$, the disjoint union of the finitely many balls $B_a = B^N_{l_a}(a)$ [Eq. (49)]. The space $L^2(K)$ is spanned by the indicator functions 1_{B_a} and by the wavelets $\psi_{\gamma nj}$ [Eq. (4)] supported in some B_a, i.e. with $\gamma \le l_a$. A wavelet supported in B_a has mean zero, so the sum in Eq. (E1) annihilates it, and so do the equations for the other regions. By Eq. (5), applied to the kernel $w_a 1_{B^N_{l_a}}$, it is an eigenfunction with eigenvalue

$$-\alpha_a + f_a'(0)\,\hat{w}_a\big(p^{1-\gamma}\big) \le -\alpha_a + f_a'(0)\,\omega_a = M_{aa}.$$

This is part 1. On the span of the 1_{B_a}, the right-hand side of Eq. (50) acts by the matrix $\tilde{M} = D^{-1}MD$, with $D = \mathrm{diag}(\mathrm{vol}(B_a))$. The off-diagonal entries of M are nonnegative (a *Metzler* matrix), so for $c > 0$ large enough $M + cI$ is a nonnegative matrix, to which the Perron–Frobenius theorem applies [88, 89]. Hence $s(M) = s(\tilde{M})$ is a real eigenvalue with a nonnegative eigenvector (the Perron eigenvector), and $s(M) \ge M_{aa}$ for every a, since the Perron root of a nonnegative matrix is not smaller than its diagonal entries. Hence the leading growth rate of Eq. (50) is $s(M)$, which proves part 3. This completes the proof of Lemma 1.

4. Markov property

For $x \in B_a$, define the total input weight received by a neuron of region a,

$$\hat{w}_{(a)} := \int_K w(x,y)\,d^N y = \omega_a + \sum_{b\neq a}\mathrm{vol}(B_b)\,c_{ab}.$$

By fact (ii) of Appendix E 1, it does not depend on $x \in B_a$. It is the value at $\xi = 0$ of the Fourier transform, in y, of $w(x,\cdot)1_K$; for a single ball and a translation-invariant kernel it reduces to $\hat{w}(0)$. In general $\hat{w}_{(a)}$ differs from $\hat{w}_a(0) = \omega_a$, which counts only the input from inside region a. Since $\omega_a \ge 0$ and $c_{ab} \ge 0$, the condition $\hat{w}_{(a)} = 0$ holds if and only if $\omega_a = 0$ and $c_{ab} = 0$ for every $b \neq a$, i.e. $w_a \equiv 0$ on $B^N_{l_a}$ (w_a is continuous and nonnegative) and region a receives no input at all: its neurons are not excited by one another or by the other regions. By Eq. (50) their activity then simply decays at rate α_a, and, since $I = 0$, they do not fire spontaneously. Such a region is trivially subcritical and can be removed from the network. We therefore assume $\hat{w}_{(a)} > 0$ for every a; this is also needed to define the rates $j(y\,|\,x)$ below. Set $\mu_a = \alpha_a - f_a'(0)\,\hat{w}_{(a)}$. Then Eq. (50) is the master equation (56) on K with

$$\begin{aligned} j(x\,|\,y) &= f_a'(0)\,w(x,y),\\ j(y\,|\,x) &= \frac{\alpha_a}{\hat{w}_{(a)}}\,w(x,y) \qquad (x \in B_a).\end{aligned}$$

Using the covering (49), Hypothesis 1 of Ref. [43], $j(x\,|\,y) \le j(y\,|\,x)$ for all $x, y \in K$, reduces to the list of conditions

$$\mu_a \ge 0 \qquad \text{for every } a \in A.\tag{E2}$$

Under these conditions, Theorem 3.1 of Ref. [43] gives a Feller semigroup on $C(K,\mathbb{R})$ and a Markov process with state space K. We rewrite Eq. (50) as $\partial_t u(x,t) = Ju(x,t)$, where J is the linear operator on its right-hand side. Then $J1_K = -\mu_a$ on B_a: the killing rate is constant on each region, and the process is conservative if and only if $\mu_a = 0$ for every a. Conditions (E2) are also necessary, since a Feller semigroup requires $J1_K \le 0$.

The row sums of $\tilde{M}$ are $-\mu_a$, so by the row-sum bounds for the Perron root of a nonnegative matrix (applied to $\tilde{M} + cI$; see Ref. [88], Chap. 8),

$$-\max_a \mu_a \le s(M) \le -\min_a \mu_a.$$

Consequently: under conditions (E2) the dynamics is not supercritical; if $\mu_a = 0$ for every a, then $s(M) = 0$,

i.e. *a conservative dynamics is critical*. The converse holds when all μ_a are equal, in particular for a single ball containing the support of a translation-invariant kernel, where $\hat{w}_{(a)} = \hat{w}(0)$ and $\mu_a = \mu$ (Sec. IV C). If the μ_a are not all equal and M is irreducible, then at $s(M) = 0$ one has $\min_a \mu_a < 0 < \max_a \mu_a$, because for an irreducible matrix the row-sum bounds are strict unless all row sums are equal [88, 89]; the critical dynamics is then a Markov process only after the change of normalization given by the Perron eigenvector of $\tilde{M}$. When $c_{ab} = c_{ba}$, $\alpha_a = \alpha$ and $f_a = f$ (for instance, for a translation-invariant kernel), the column sums of M are also $-\mu_b$, so $\frac{d}{dt}\sum_a U_a = -\sum_b \mu_b U_b$, and a conservative process conserves the total activity.

The Markov process given by Theorem 3.1 of Ref. [43] describes the linear (mean) dynamics. By the standard many-to-one formula for branching processes, the mean activity equals the transition density of a single walker, the ancestral lineage of the branching process, weighted by its survival. The process adds no elementary events to Eq. (A1): its jumps are branching events, and its killing is the excess of decay over branching.

Finally, if every w_a is supported in $\{\|x\|_p \le p^r\}$, every region has radius at least p^r, and the regional network comes from a translation-invariant kernel, then $c_{ab} = w(d_{ab}) = 0$ for $a \neq b$, M is diagonal, and the regions are independent networks, each of the kind discussed before Lemma 1.

5. Proof of Lemma 2

Proof. For $\xi = 0$ both sides vanish, so let $\|\xi\|_p = p^m$ with $m \le -r$. Since $w \in L^1$,

$$\hat{w}(0) - \hat{w}(\xi) = \int_{\mathbb{Q}_p^N} w(\|x\|_p)\,\big[1 - \chi_p(\xi\cdot x)\big]\,d^N x.$$

Step 1: only the tail contributes. If $\|x\|_p \le \|\xi\|_p^{-1} = p^{-m}$, then $|\xi\cdot x|_p \le \|\xi\|_p\,\|x\|_p \le 1$, so $\xi\cdot x \in \mathbb{Z}_p$, $\{\xi\cdot x\}_p = 0$ and $\chi_p(\xi\cdot x) = 1$: the bracket vanishes on the ball B_{-m}^N. Since $p^{-m} \ge p^r$, this ball contains the region $\|x\|_p \le p^r$, where w is arbitrary. On the complement, $\|x\|_p > p^{-m} \ge p^r$, so $w(\|x\|_p) = C_w \|x\|_p^{-N-\sigma}$, and

$$\hat{w}(0) - \hat{w}(\xi) = C_w \sum_{k \ge 1-m} p^{-k(N+\sigma)} \int_{S_k^N} \big[1 - \chi_p(\xi\cdot x)\big]\,d^N x,$$

where we decomposed $\{\|x\|_p > p^{-m}\}$ into the spheres S_k^N, $k \ge 1-m$, on which $\|x\|_p = p^k$ is constant.

Step 2: the sphere integrals. By Eq. (2), $\int_{S_k^N} \chi_p(\xi\cdot x)\,d^N x = p^{kN}\Omega(p^{k+m}) - p^{(k-1)N}\Omega(p^{k-1+m})$, and $\mathrm{vol}(S_k^N) = (1-p^{-N})p^{kN}$. For $k = 1-m$ the character integral equals $-p^{-mN}$, and for $k \ge 2-m$ it vanishes. Hence

$$\int_{S_{1-m}^N} \big[1 - \chi_p(\xi\cdot x)\big]\,d^N x = (1-p^{-N})p^{(1-m)N} + p^{-mN} = p^{(1-m)N},$$

$$\int_{S_k^N} \big[1 - \chi_p(\xi\cdot x)\big]\,d^N x = (1-p^{-N})p^{kN},$$

the second for $k \ge 2-m$.

Step 3: summation. Therefore

$$\begin{aligned}\frac{\hat{w}(0) - \hat{w}(\xi)}{C_w} &= p^{-(1-m)\sigma} + (1-p^{-N}) \sum_{k \ge 2-m} p^{-k\sigma} \\ &= p^{(m-1)\sigma}\Big[1 + \frac{(1-p^{-N})\,p^{-\sigma}}{1-p^{-\sigma}}\Big] \\ &= p^{(m-1)\sigma}\,\frac{1-p^{-N-\sigma}}{1-p^{-\sigma}} \\ &= \frac{1-p^{-N-\sigma}}{p^\sigma - 1}\,p^{m\sigma},\end{aligned}$$

where the geometric series converges because $\sigma > 0$. Since $p^{m\sigma} = \|\xi\|_p^\sigma$, this is Eq. (52). Finally, $A_w(\xi) = f'(0)\big(\hat{w}(0) - \hat{w}(\xi)\big)$ by Eq. (46), which gives $A_w(\xi) = \kappa_W \|\xi\|_p^\sigma$ with $\kappa_W = f'(0)c_\sigma$; this is the symbol of $\kappa_W \boldsymbol{D}^\sigma$, Eq. (3). □

Appendix F: Evaluation of momentum integrals

In this appendix we state precisely which momentum integrals over $\mathbb{Q}_p^N$ can be reduced to one-dimensional sums, and prove it. The general claim that "all momentum integrals are evaluated by Eq. (1)" is *not* made. Eq. (1) applies to radial integrands. A one-loop integral with a nonzero external momentum is not radial in the loop momentum, but it still reduces to single sums (Lemma 7). Integrals with several independent external momenta, or with several loops, require repeated use of these lemmas. We do not claim a general closed form for them.

Throughout, $S_k^N = \{\xi \in \mathbb{Q}_p^N : \|\xi\|_p = p^k\}$ and $B_k^N = \{\|\xi\|_p \le p^k\}$, so $\mathrm{vol}(B_k^N) = p^{kN}$ and $\mathrm{vol}(S_k^N) = (1-p^{-N})p^{kN}$.

Lemma 5 (Radial integrals)**.** *Let $F : p^{\mathbb{Z}} \to \mathbb{C}$ with $\sum_k p^{kN}|F(p^k)| < \infty$. Then $\int_{\mathbb{Q}_p^N} F(\|\xi\|_p)\,d^N\xi = (1-p^{-N})\sum_{k\in\mathbb{Z}} p^{kN}F(p^k)$.*

Proof. $\mathbb{Q}_p^N \setminus \{0\}$ is the disjoint union of the spheres S_k^N, on which $F(\|\xi\|_p) = F(p^k)$ is constant. Since $S_k^N = B_k^N \setminus B_{k-1}^N$, $\mathrm{vol}(S_k^N) = p^{kN} - p^{(k-1)N}$. Absolute convergence justifies summing term by term. □

Lemma 6 (Fourier transform of radial functions)**.** *If $g \in L^1(\mathbb{Q}_p^N)$ is radial, then $\hat{g}$ is radial. Explicitly, for $\|\xi\|_p = p^m$,*

$$\hat{g}(\xi) = (1-p^{-N}) \sum_{k \le -m} p^{kN} g(p^k) - p^{-mN}\,g(p^{1-m}). \quad \text{(F1)}$$

Proof. By Eq. (2),

$$\int_{S_k^N} \chi_p(\xi\cdot x)\,d^N x = p^{kN}\Omega(p^{k+m}) - p^{(k-1)N}\Omega(p^{k-1+m}).$$

This equals $(1-p^{-N})p^{kN}$ if $k \le -m$, equals $-p^{-mN}$ if $k = 1-m$, and vanishes if $k \ge 2-m$. Sum over k; the result depends on ξ only through m. □

Equation (F1) is the formula used in Proposition 2.

Lemma 7 (One loop with an external momentum)**.** *Let $F, G : p^{\mathbb{Z}} \to \mathbb{C}$ be such that the integrals below converge absolutely, and let $\eta \in \mathbb{Q}_p^N$ with $\|\eta\|_p = p^m$. Then*

$$\begin{aligned}
&\int_{\mathbb{Q}_p^N} F(\|\xi\|_p)\,G(\|\eta-\xi\|_p)\,d^N\xi \\
&= G(p^m)\,(1-p^{-N})\sum_{k<m} p^{kN}F(p^k) \\
&+ (1-p^{-N})\sum_{k>m} p^{kN}F(p^k)\,G(p^k) \\
&+ F(p^m)\Big[\int_{B_m^N} G(\|u\|_p)\,d^N u - p^{(m-1)N}G(p^m)\Big], \quad \text{(F2)}
\end{aligned}$$

where $\int_{B_m^N} G(\|u\|_p)\,d^N u = (1-p^{-N})\sum_{k\le m} p^{kN}G(p^k)$ by Lemma 5. In particular the left-hand side depends on η only through $\|\eta\|_p$.

Proof. Split $\mathbb{Q}_p^N$ into $\{\|\xi\|_p < p^m\}$, $\{\|\xi\|_p > p^m\}$ and S_m^N. By the strong triangle inequality, $\|\eta-\xi\|_p = \max(\|\eta\|_p, \|\xi\|_p)$ whenever $\|\xi\|_p \ne \|\eta\|_p$. On the first region $\|\eta-\xi\|_p = p^m$, and on the second $\|\eta-\xi\|_p = \|\xi\|_p$. Lemma 5 gives the first two lines. On S_m^N, $F(\|\xi\|_p) = F(p^m)$, and we substitute $u = \eta - \xi$. The condition $\|\xi\|_p = p^m$ becomes $\|\eta - u\|_p = p^m$, i.e. $u \in B_m^N \smallsetminus (\eta + B_{m-1}^N)$. On the removed ball $\eta + B_{m-1}^N$ we have $\|u\|_p = \|\eta\|_p = p^m$, and its volume is $p^{(m-1)N}$. This gives the third line. □

Lemma 7 extends to integrands that are arbitrary functions of the two radii. The proof is a slight variation of that of Lemma 7; the result is used in the calculation of loop integrals with a radial integrand, in particular the two-loop integrals below.

Lemma 8 (One loop with a general radial integrand)**.** *Let $H : p^{\mathbb{Z}} \times p^{\mathbb{Z}} \to \mathbb{C}$ be such that the integral below converges absolutely, and let $\xi \in \mathbb{Q}_p^N$ with $\|\xi\|_p = p^m$. Then*

$$\begin{aligned}
&\int_{\mathbb{Q}_p^N} H\big(\|\zeta\|_p, \|\xi-\zeta\|_p\big)\,d^N\zeta \\
&= (1-p^{-N})\sum_{k<m} p^{kN}H(p^k, p^m) \\
&+ (1-p^{-N})\sum_{k>m} p^{kN}H(p^k, p^k) \\
&+ (1-p^{-N})\sum_{j\le m} p^{jN}H(p^m, p^j) \\
&- p^{(m-1)N}H(p^m, p^m). \quad \text{(F3)}
\end{aligned}$$

In particular, the left-hand side is a radial function of ξ.

Proof. As in the proof of Lemma 7, split $\mathbb{Q}_p^N$ into $\{\|\zeta\|_p < p^m\}$, $\{\|\zeta\|_p > p^m\}$ and S_m^N. On the first region $\|\xi-\zeta\|_p = p^m$, and on the second $\|\xi-\zeta\|_p = \|\zeta\|_p$. On S_m^N put $u = \xi - \zeta$, which runs over $B_m^N \smallsetminus (\xi + B_{m-1}^N)$; on the removed ball $\|u\|_p = p^m$, and its volume is $p^{(m-1)N}$. □

Lemma 7 is the case $H(a,b) = F(a)G(b)$.

Lemmas 5, 6, and 7 are well known; see Refs. [33, 37]. We include them here to provide a complete, self-contained presentation.

Lemma 9 (Iterated convolutions)**.** *Let $w \ge 0$ be radial and integrable, with $w_0 = \int_{\mathbb{Q}_p^N} w(\|x\|_p)\,d^N x$, and radially nonincreasing: $w(r) \ge w(r')$ for $r < r'$. Then, for all $n \ge 1$ and $x \ne 0$,*

$$\begin{aligned}
&\textit{(i)} \quad w^{\star n}(x) \le (2n-1)\,w_0^{n-1}\,w(\|x\|_p), \\
&\textit{(ii)} \quad \lim_{\|x\|_p\to\infty} \frac{w^{\star n}(x)}{w(\|x\|_p)} = n\,w_0^{n-1}, \quad \text{(F4)}
\end{aligned}$$

the limit holding whenever $w(r) > 0$ for all large r.

Proof. Induction on n; the case $n = 1$ is trivial. Let $G = w^{\star(n-1)} \ge 0$, which is radial by Lemma 7 and has $\int_{\mathbb{Q}_p^N} G(u)\,d^N u = w_0^{n-1}$. Fix $\|x\|_p = p^m$ and split $w^{\star n}(x) = \int_{\mathbb{Q}_p^N} w(\|y\|_p)\,G(x-y)\,d^N y$ as in the proof of Lemma 7. (a) On $\|y\|_p < p^m$, $G(x-y) = G(x)$, and this region contributes $G(x)\int_{\|y\|_p<p^m} w(\|y\|_p)\,d^N y \le G(x)\,w_0$. (b) On $\|y\|_p > p^m$, $\|x-y\|_p = \|y\|_p$ and $w(\|y\|_p) \le w(p^m)$, so this region contributes at most $w(p^m)\int_{\|u\|_p>p^m} G(u)\,d^N u$. (c) On $\|y\|_p = p^m$, $w(\|y\|_p) = w(p^m)$, and this region contributes $w(p^m)\big[\int_{B_m^N} G(u)\,d^N u - p^{(m-1)N}G(x)\big] \le w(p^m)\int_{B_m^N} G(u)\,d^N u$. Adding the three and using the induction hypothesis $G(x) \le (2n-3)w_0^{n-2}w(p^m)$ gives $w^{\star n}(x) \le \big[(2n-3)+2\big]w_0^{n-1}w(p^m)$. For the limit, divide by $w(p^m)$ and let $m \to \infty$. By induction, term (a) tends to $(n-1)w_0^{n-2}\cdot w_0$. Term (b) is at most $\int_{\|u\|_p>p^m} G(u)\,d^N u \to 0$. Term (c) tends to $\int_{\mathbb{Q}_p^N} G(u)\,d^N u = w_0^{n-1}$. Indeed $\int_{B_m^N} G(u)\,d^N u \to w_0^{n-1}$, and the subtracted term satisfies

$$\begin{aligned}
p^{(m-1)N}G(x) &\le (2n-3)w_0^{n-2}\,p^{(m-1)N}w(p^m) \\
&= (2n-3)w_0^{n-2}\,\frac{W_m}{p^N-1} \to 0,
\end{aligned}$$

with W_m as in Eq. (87). Hence the limit is $n\,w_0^{n-1}$. □

Proposition 4 (The zero-loop propagator)**.** *Let $w \ge 0$ be radial and integrable, and set $c = f'(0) \ge 0$.*

1. *In $\mathcal{D}'(\mathbb{Q}_p^N)$, $\mathcal{F}^{-1}_{\xi\to x}\big(e^{-\lambda_0(\xi)t}\big) = e^{-\alpha t}\big[\delta + K_t\big]$ with $K_t = \sum_{n\ge1}(ct)^n w^{\star n}/n! \in L^1(\mathbb{Q}_p^N)$.*

2. *If in addition $\hat w \in L^1(\mathbb{Q}_p^N)$, then K_t is continuous and, for $\|x\|_p = p^k$,*

$$K_t(x) = (1-p^{-N}) \sum_{j\le -k} p^{jN} g_t(p^j) - p^{-kN} g_t(p^{1-k}), \quad \text{(F5)}$$

where $g_t(\|\xi\|_p) = e^{ct\hat w(\xi)} - 1$. If $\hat w$ is supported in $\|\xi\|_p \le p^L$, the sum is finite ($-L \le j \le -k$).

3. *If w is radially nonincreasing and positive at large distances, then $K_t(x) \simeq ct\, e^{ctw_0}\, w(\|x\|_p)$ as $\|x\|_p \to \infty$, and hence $\Delta(x,t) \simeq \theta(t)\, ct\, e^{-\lambda_0(0)t}\, w(\|x\|_p)$, with $\lambda_0(0) = \alpha - cw_0$.*

Proof. (1) $e^{-\lambda_0(\xi)t} = e^{-\alpha t} e^{ct\hat w(\xi)}$, and $e^{ct\hat w} = 1 + \sum_{n\ge1} (ct)^n \hat w^{\,n}/n!$ with the series converging uniformly, since $|\hat w| \le w_0$. The inverse Fourier transform of 1 is δ. Since $\hat w^{\,n} = \widehat{w^{\star n}}$ and $\|w^{\star n}\|_1 \le w_0^n$, the series $\sum_{n\ge1} (ct)^n w^{\star n}/n!$ converges in L^1, and its Fourier transform is $e^{ct\hat w} - 1$. (2) $|g_t| \le ct\, |\hat w|\, e^{ctw_0}$, so $g_t \in L^1$ and $K_t = \mathcal{F}^{-1} g_t$ is continuous. Since g_t is radial and real, $\mathcal{F}^{-1} g_t = \mathcal{F} g_t$, and Eq. (F5) is Eq. (F1) of Lemma 6 applied to g_t. (3) By Lemma 9, $(ct)^n w^{\star n}(x)/(n!\, w(\|x\|_p))$ is bounded by $(ct)^n (2n-1) w_0^{n-1}/n!$, which is summable, and tends to $(ct)^n n\, w_0^{n-1}/n!$. Dominated convergence gives $K_t(x)/w(\|x\|_p) \to \sum_{n\ge1} (ct)^n w_0^{n-1}/(n-1)! = ct\, e^{ctw_0}$. □

Remark 4. Let $w \in L^1(\mathbb{Q}_p^N)$ be radial and constant on B_r^N. Then $w(x+x') = w(x)$ for all $x \in \mathbb{Q}_p^N$ and all $x' \in B_r^N$: for $\|x\|_p \le p^r$ both points lie in B_r^N, and for $\|x\|_p > p^r$ the ultrametric inequality gives $\|x+x'\|_p = \|x\|_p$. By the translation invariance of the Haar measure, $\hat w(\xi) = \chi_p(\xi\cdot x')\, \hat w(\xi)$, i.e. $\hat w(\xi)\big(1 - \chi_p(\xi\cdot x')\big) = 0$ for all $x' \in B_r^N$. If $\hat w(\xi) \ne 0$, then $\chi_p(\xi\cdot x') = 1$ for all $x' \in B_r^N = p^{-r}\mathbb{Z}_p^N$, i.e. $\xi \cdot x' \in \mathbb{Z}_p$ for all such x', which holds if and only if $\|\xi\|_p \le p^{-r}$. Hence $\hat w$ is supported in the ball $\|\xi\|_p \le p^{-r}$. If $w \ne 0$, then $\hat w \not\equiv 0$, since the Fourier transform is injective on $L^1(\mathbb{Q}_p^N)$ and w, being locally constant, is continuous; thus the support of $\hat w$ is a nonempty subset of this ball. In particular $\hat w$ has compact support, and, being bounded, it is in $L^1(\mathbb{Q}_p^N)$, so part 2 of Proposition 4 applies. See Ref. [37], Theorem 1.161.

Proof of Lemma 4. (1) Since w is radial and real, $A_w(\xi) = f'(0) \int_{\mathbb{Q}_p^N} w(\|x\|_p)\, \big[1 - \cos\big(2\pi\{\xi\cdot x\}_p\big)\big]\, d^N x$, and the bracket is non-negative. Suppose $A_w(\xi) = 0$ for some $\xi \ne 0$. Then $\xi\cdot x \in \mathbb{Z}_p$ for almost every x in the support of w. This support is a union of spheres. If it met a sphere of radius $p^j > \|\xi\|_p^{-1}$, some x on that sphere would satisfy $|\xi\cdot x|_p > 1$. Hence w is supported in the ball of radius $\|\xi\|_p^{-1}$, and then $\hat w(0) - \hat w(\xi') = 0$ for every $\|\xi'\|_p \le \|\xi\|_p$, which contradicts Eq. (60).

(2) and (3). Split the sum (73) at $k = k_0$, where $k_0 \le -r$ is chosen so that $|\varepsilon(\xi)| \le \frac12$ for $\|\xi\|_p \le p^{k_0}$.

Terms with $k_0 < k \le \Lambda$. A_w is constant on spheres, positive by (1), and tends to $f'(0)\hat w(0) > 0$ as $\|\xi\|_p \to \infty$. Hence $\inf_{\|\xi\|_p > p^{k_0}} A_w > 0$, and these finitely many terms are bounded uniformly in $\mu \ge 0$, by a constant times $p^{\Lambda N}$. Relative to $\mu^{N/\sigma-2}$, they give the error $O(\mu^{2-N/\sigma})$.

Terms with $k \le k_0$. Here $A_w(p^k) \ge \frac12 \kappa_W p^{k\sigma}$, and

$$\Big|\frac{1}{(\mu + A_w(p^k))^2} - \frac{1}{(\mu + \kappa_W p^{k\sigma})^2}\Big| \le C'\, \frac{p^{k\delta}\, \kappa_W p^{k\sigma}}{\big(\mu + \frac12 \kappa_W p^{k\sigma}\big)^3}.$$

The sum over $k \le k_0$ of p^{kN} times the right-hand side is $O\big(\mu^{(N+\delta)/\sigma-2}\big)$ if $N + \delta < 2\sigma$, $O(|\ln \mu|)$ if $N + \delta = 2\sigma$, and $O(1)$ otherwise. This is the first error term.

The pure power law. At $\mu = 0$, the sum $\sum_{k\le k_0} p^{kN}/(\kappa_W p^{k\sigma})^2 = \kappa_W^{-2} \sum_{k\le k_0} p^{k(N-2\sigma)}$ converges if and only if $N > 2\sigma$, which, with the bounds above, proves (2). For $N < 2\sigma$, the sum of $p^{kN}/(\mu + \kappa_W p^{k\sigma})^2$ over $k > k_0$ is bounded by $\kappa_W^{-2} \sum_{k>k_0} p^{k(N-2\sigma)} < \infty$, so we may sum over all $k \in \mathbb{Z}$ at the cost of another $O(1)$. Finally, with $\mu = \kappa_W p^{\sigma s}$,

$$\sum_{k\in\mathbb{Z}} \frac{p^{kN}}{(\mu + \kappa_W p^{k\sigma})^2} = \kappa_W^{-N/\sigma}\, \mu^{N/\sigma-2}\, \Phi(s),$$

which gives Eq. (74). If only $\varepsilon \to 0$ is known, dominated convergence applied to the rescaled sum gives the error $o(1)$. □

Fourier series of Φ. Write $\Phi(s) = \sum_k F(k-s)$, with $F(u) = p^{uN}/(1+p^{\sigma u})^2$. By the Poisson summation formula, $\Phi(s) = \sum_n \hat F(n) e^{-2\pi i n s}$, with $\hat F(n) = \int_{\mathbb{R}} F(u) e^{-2\pi i n u}\, du$. The substitution $t = p^{\sigma u}$ turns this into a Beta integral, $\hat F(n) = (\sigma \ln p)^{-1} \int_0^\infty t^{a - i\omega_n - 1} (1+t)^{-2}\, dt = (\sigma \ln p)^{-1}\, \Gamma(a - i\omega_n)\, \Gamma(2 - a + i\omega_n)$, which is Eq. (78).

Two-loop integrals. We now write the two-loop self-energy of the action (59). The propagator is $\theta(t) e^{-\lambda(\xi)t}$, with $\lambda(\xi) = \mu + A_w(\xi)$. All lines run forward in time, and integrating over the times between successive vertices gives, for each interval in which the lines $\xi_1, \dots, \xi_r$ are present, a factor $\big(-i\omega + \lambda(\xi_1) + \cdots + \lambda(\xi_r)\big)^{-1}$, where η and ω are the external momentum and frequency. A two-loop self-energy has two splitting and two merging vertices, and there are two topologies. In the nested one, a one-loop self-energy is inserted on an internal line: $\eta \to \xi + (\eta - \xi)$, $\xi \to \zeta + (\xi - \zeta)$, $\zeta + (\xi - \zeta) \to \xi$, $\xi + (\eta - \xi) \to \eta$. In the overlapping one, a vertex is corrected: $\eta \to \xi + (\eta - \xi)$, $\xi \to \zeta + (\xi - \zeta)$, $(\xi - \zeta) + (\eta - \xi) \to \eta - \zeta$, $\zeta + (\eta - \zeta) \to \eta$. With

$$\begin{aligned} D_1 &= -i\omega + \lambda(\xi) + \lambda(\eta - \xi), \\ D_2 &= -i\omega + \lambda(\zeta) + \lambda(\xi - \zeta) + \lambda(\eta - \xi), \\ D_3 &= -i\omega + \lambda(\zeta) + \lambda(\eta - \zeta), \end{aligned}$$

the two contributions are

$$\begin{aligned} \Sigma_A(\eta,\omega) &= c_A\, g^4 \iint \frac{d^N\xi\, d^N\zeta}{D_1^2\, D_2}, \\ \Sigma_B(\eta,\omega) &= c_B\, g^4 \iint \frac{d^N\xi\, d^N\zeta}{D_1\, D_2\, D_3}, \end{aligned} \quad \text{(F6)}$$

where both integrals are over $\{\|\xi\|_p \le p^\Lambda,\ \|\zeta\|_p \le p^\Lambda\}$, and c_A, c_B are combinatorial factors.

Proposition 5 (Two loops at zero external momentum)**.** *Write* $\lambda_k = \lambda(p^k)$*. Then*

$$\Sigma_A(0,0) = \frac{c_A g^4}{4}\,(1-p^{-N})\sum_{m\le\Lambda}\frac{p^{mN}K_m}{\lambda_m^2},$$
$$\Sigma_B(0,0) = \frac{c_B g^4}{4}\,(1-p^{-N})\sum_{m\le\Lambda}\frac{p^{mN}L_m}{\lambda_m}, \qquad \text{(F7)}$$

where

$$K_m = 2(1-p^{-N})\sum_{k<m}\frac{p^{kN}}{\lambda_k+2\lambda_m} + \frac{(1-2p^{-N})\,p^{mN}}{3\lambda_m} + (1-p^{-N})\sum_{m<k\le\Lambda}\frac{p^{kN}}{2\lambda_k+\lambda_m},$$
$$L_m = (1-p^{-N})\sum_{k<m}\frac{p^{kN}}{\lambda_k(\lambda_k+2\lambda_m)} + (1-p^{-N})\sum_{m<k\le\Lambda}\frac{p^{kN}}{\lambda_k(2\lambda_k+\lambda_m)} + \frac{1-p^{-N}}{\lambda_m}\sum_{k<m}\frac{p^{kN}}{\lambda_k+2\lambda_m} + \frac{(1-2p^{-N})\,p^{mN}}{3\lambda_m^2}.$$

Proof. At $\eta = 0$, $\omega = 0$, since λ is radial, $D_1 = 2\lambda(\xi)$, $D_3 = 2\lambda(\zeta)$ and $D_2 = \lambda(\zeta)+\lambda(\xi-\zeta)+\lambda(\xi)$. The inner integrals over ζ are of the form (F3), with $H(a,b) = \big(\lambda(a)+\lambda(b)+\lambda_m\big)^{-1}$ for Σ_A and $H(a,b) = \lambda(a)^{-1}\big(\lambda(a)+\lambda(b)+\lambda_m\big)^{-1}$ for Σ_B. Lemma 8 gives K_m and L_m; in K_m the terms with $k<m$ from the first and third sums of Eq. (F3) coincide, and the terms with $k=m$ combine into $\big[(1-p^{-N})p^{mN} - p^{(m-1)N}\big]/(3\lambda_m) = (1-2p^{-N})p^{mN}/(3\lambda_m)$. The remaining integral over ξ is radial, and Lemma 5 applies. □

For $p=2$ and $N=1$ the factor $1-2p^{-N}$ vanishes: then no ζ on the sphere of radius p^m satisfies $\|\xi-\zeta\|_p = p^m$. With $\lambda_k = \mu + \kappa_W p^{k\sigma}$, both sums in Eq. (F7) scale as $g^4\mu^{2N/\sigma-3}$ times a function periodic in s. Their singular part is finite at $\mu=0$ if $N > 3\sigma/2$; like $J(0)$ in Sec. VI, it only shifts the critical point. The μ-derivative behaves as $g^4\mu^{2N/\sigma-4} = \big(g^2\mu^{N/\sigma-2}\big)^2$, the square of the one-loop Ginzburg ratio, consistently with $N_c = 2\sigma$.

Scope. Lemma 5 covers the one-loop integrals at zero external momentum used in the Ginzburg criterion (Sec. VI) and the spectral sums of Sec. X. Lemma 7 covers one-loop self-energy and vertex integrals with a single external momentum, when the propagators are radial functions of the momentum, for example $\big(\mu + A_w(\xi)\big)^{-1}$ after the frequency integral. Lemma 8 and Proposition 5 reduce the two-loop self-energy at zero external momentum to explicit double sums. One-loop integrals with two independent external momenta η_1, η_2, and two-loop integrals at nonzero external momentum, require one more step: the regions of integration are fixed by comparing $\|\xi\|_p$ with the sides of the ultrametric (isosceles) triangle with vertices 0, η_1, $-\eta_2$.

[1] E. Bullmore and O. Sporns, Nat. Rev. Neurosci. **10**, 186 (2009).
[2] C. C. Hilgetag and A. Goulas, Phil. Trans. R. Soc. B **375**, 20190319 (2020).
[3] J. M. Beggs and D. Plenz, J. Neurosci. **23**, 11167 (2003).
[4] S. Bornholdt and T. Röhl, Phys. Rev. E **67**, 066118 (2003).
[5] S. S. Schoenholz, J. Gilmer, S. Ganguli, and J. Sohl-Dickstein, in *International Conference on Learning Representations* (2017).
[6] S. Vock and C. Meisel, *Critical dynamics governs deep learning*, arXiv:2507.08527.
[7] O. Sporns, BioSystems **85**, 55 (2006).
[8] L. da F. Costa and O. Sporns, Eur. Phys. J. B **48**, 555 (2005).
[9] O. Sporns, G. Tononi, and R. Kötter, PLoS Comput. Biol. **1**, e42 (2005).
[10] D. J. Watts and S. H. Strogatz, Nature (London) **393**, 440 (1998).
[11] D. S. Bassett and E. Bullmore, Neuroscientist **12**, 512 (2006).
[12] W. A. Zúñiga-Galindo and B. A. Zambrano-Luna, Entropy **25**, 949 (2023).
[13] M. A. Buice and J. D. Cowan, Phys. Rev. E **75**, 051919 (2007).
[14] M. Doi, J. Phys. A **9**, 1465 (1976); **9**, 1479 (1976).
[15] L. Peliti, J. Phys. (France) **46**, 1469 (1985).
[16] H. K. Janssen, Z. Phys. B **42**, 151 (1981).
[17] P. Grassberger, Z. Phys. B **47**, 365 (1982).
[18] J. L. Cardy and R. L. Sugar, J. Phys. A **13**, L423 (1980).
[19] H. Hinrichsen, Adv. Phys. **49**, 815 (2000).
[20] M. Helias and D. Dahmen, *Statistical Field Theory for Neural Networks*, Lecture Notes in Physics Vol. 970 (Springer, Cham, 2020).
[21] J. Halverson, A. Maiti, and K. Stoner, Mach. Learn.: Sci. Technol. **2**, 035002 (2021).
[22] K. T. Grosvenor and R. Jefferson, SciPost Phys. **12**, 081 (2022).
[23] W. A. Zúñiga-Galindo, Physica A **612**, 128492 (2023).
[24] W. A. Zúñiga-Galindo, J. Phys. A **58**, 125201 (2025); arXiv:2410.02206.
[25] M. A. Buice and J. D. Cowan, Prog. Biophys. Mol. Biol. **99**, 53 (2009).
[26] D. A. Roberts and S. Yaida, *The Principles of Deep Learning Theory* (Cambridge University Press, Cambridge, 2022).
[27] K. Segadlo, B. Epping, A. van Meegen, D. Dahmen, M. Krämer, and M. Helias, J. Stat. Mech. (2022) 103401.
[28] W. A. Zúñiga-Galindo, *Critical Organization of Deep Neural Networks, and p-Adic Statistical Field Theories*, arXiv:2601.19070.
[29] R. Rammal, G. Toulouse, and M. A. Virasoro, Rev. Mod. Phys. **58**, 765 (1986).

[30] A. Khrennikov, S. Kozyrev, and W. A. Zúñiga-Galindo, *Ultrametric Equations and its Applications*, Encyclopedia of Mathematics and its Applications Vol. 168 (Cambridge University Press, Cambridge, 2018).

[31] M. Mézard, G. Parisi, N. Sourlas, G. Toulouse, and M. Virasoro, J. Phys. (Paris) **45**, 843 (1984).

[32] H. Frauenfelder, S. G. Sligar, and P. G. Wolynes, Science **254**, 1598 (1991).

[33] V. S. Vladimirov, I. V. Volovich, and E. I. Zelenov, *p-Adic Analysis and Mathematical Physics* (World Scientific, Singapore, 1994).

[34] A. N. Kochubei, *Pseudo-Differential Equations and Stochastics over Non-Archimedean Fields* (Marcel Dekker, New York, 2001).

[35] S. Albeverio, A. Yu. Khrennikov, and V. M. Shelkovich, *Theory of p-adic Distributions: Linear and Nonlinear Models* (Cambridge University Press, 2010).

[36] W. A. Zúñiga-Galindo, *Pseudodifferential Equations over Non-Archimedean Spaces*, Lecture Notes in Mathematics Vol. 2174 (Springer, Cham, 2016).

[37] W. A. Zúñiga-Galindo, *p-Adic Analysis: Stochastic Processes and Pseudo-Differential Equations* (De Gruyter, Berlin, 2025).

[38] G. Parisi and N. Sourlas, Eur. Phys. J. B **14**, 535 (2000).

[39] V. A. Avetisov, A. H. Bikulov, S. V. Kozyrev, and V. A. Osipov, J. Phys. A **35**, 177 (2002).

[40] W. A. Zúñiga-Galindo, C. He, and B. A. Zambrano-Luna, Prog. Theor. Exp. Phys. **2023**, 063A01 (2023).

[41] W. A. Zúñiga-Galindo, Adv. Theor. Math. Phys. **28**, 679 (2024).

[42] B. A. Zambrano-Luna and W. A. Zúñiga-Galindo, p-adic cellular neural networks, J. Nonlinear Math. Phys. **30**, 34–70 (2023), doi:10.1007/s44198-022-00071-8.

[43] W. A. Zúñiga-Galindo, Physica A **597**, 127221 (2022); arXiv:2201.06452.

[44] H. K. Janssen, K. Oerding, F. van Wijland, and H. J. Hilhorst, Eur. Phys. J. B **7**, 137 (1999).

[45] W. A. Zúñiga-Galindo, Hierarchical connectivity sets the upper critical dimension of neural avalanches. In preparation; available at arXiv.org.

[46] S. V. Kozyrev, Izv. Math. **66**, 367 (2002).

[47] W. A. Zúñiga-Galindo, Physica A **602**, 127648 (2022).

[48] W. A. Zúñiga-Galindo, Rev. Math. Phys. **34**, 2250022 (2022).

[49] H. Hinrichsen and M. Howard, Eur. Phys. J. B **7**, 635 (1999).

[50] E. M. Stein, *Singular Integrals and Differentiability Properties of Functions* (Princeton University Press, Princeton, 1970).

[51] M. Kwaśnicki, Fract. Calc. Appl. Anal. **20**, 7 (2017).

[52] U. C. Täuber, *Critical Dynamics: A Field Theory Approach to Equilibrium and Non-Equilibrium Scaling Behavior* (Cambridge University Press, Cambridge, 2014).

[53] F. J. Dyson, Commun. Math. Phys. **12**, 91 (1969).

[54] P. M. Bleher and Ya. G. Sinai, Commun. Math. Phys. **33**, 23 (1973).

[55] É. Yu. Lerner and M. D. Missarov, Theor. Math. Phys. **78**, 177 (1989).

[56] A. Abdesselam, A. Chandra, and G. Guadagni, Rigorous quantum field theory functional integrals over the p-adics I: anomalous dimensions, arXiv:1302.5971.

[57] S. S. Gubser, C. Jepsen, S. Parikh, and B. Trundy, J. High Energy Phys. **11** (2017) 107.

[58] H. Erbin, V. Lahoche, and D. O. Samary, arXiv:2108.01403.

[59] M. A. Buice, J. D. Cowan, and C. C. Chow, Neural Comput. **22**, 377 (2010).

[60] M. A. Muñoz, R. Dickman, A. Vespignani, and S. Zapperi, Phys. Rev. E **59**, 6175 (1999).

[61] D. T. Gillespie, J. Comput. Phys. **22**, 403 (1976).

[62] P. A. W. Lewis and G. S. Shedler, Nav. Res. Logist. Q. **26**, 403 (1979).

[63] P. Grassberger and A. de la Torre, Ann. Phys. (N.Y.) **122**, 373 (1979).

[64] A. Clauset, C. R. Shalizi, and M. E. J. Newman, SIAM Rev. **51**, 661 (2009).

[65] V. Koval, R. Meester, and P. Trapman, Electron. J. Probab. **17**, 57 (2012).

[66] I. Benjamini and N. Berger, Random Struct. Algorithms **19**, 102 (2001).

[67] M. Biskup, Ann. Probab. **32**, 2938 (2004).

[68] A. Bendikov, A. Grigor'yan, Ch. Pittet, and W. Woess, Russ. Math. Surveys **69**, 589 (2014) [arXiv:1304.6271].

[69] J. J. Rodríguez-Vega and W. A. Zúñiga-Galindo, Pacific J. Math. **237**, 327 (2008).

[70] X. Cabré and J.-M. Roquejoffre, Commun. Math. Phys. **320**, 679 (2013).

[71] L. S. Ornstein and F. Zernike, Accidental deviations of density and opalescence at the critical point of a single substance, Proc. Akad. Sci. Amsterdam **17**, 793 (1914).

[72] N. G. van Kampen, *Stochastic Processes in Physics and Chemistry*, 3rd ed. (North-Holland, Amsterdam, 2007).

[73] M. Penttonen and G. Buzsáki, Thalamus Relat. Syst. **2**, 145 (2003).

[74] L. F. Abbott and S. B. Nelson, Synaptic plasticity: taming the beast, Nat. Neurosci. **3**, 1178 (2000).

[75] A. Levina, J. M. Herrmann, and T. Geisel, Nat. Phys. **3**, 857 (2007).

[76] P. Moretti and M. A. Muñoz, Nat. Commun. **4**, 2521 (2013).

[77] J. Cardy, *Renormalisation group approach to reaction-diffusion problems*, arXiv:cond-mat/9607163 (1996).

[78] U. C. Täuber, M. Howard, and B. P. Vollmayr-Lee, J. Phys. A **38**, R79 (2005).

[79] T. E. Harris, *The Theory of Branching Processes* (Springer, Berlin, 1963).

[80] B. Poole, S. Lahiri, M. Raghu, J. Sohl-Dickstein, and S. Ganguli, in *Advances in Neural Information Processing Systems 29* (2016).

[81] P. C. Martin, E. D. Siggia, and H. A. Rose, Phys. Rev. A **8**, 423 (1973).

[82] H. K. Janssen, Z. Phys. B **23**, 377 (1976).

[83] C. De Dominicis, J. Phys. (Paris) Colloq. **37**, C1-247 (1976).

[84] P. Grassberger and K. Sundermeyer, Phys. Lett. B **77**, 220 (1978).

[85] U. C. Täuber, Lect. Notes Phys. **716**, 295 (2007) [arXiv:cond-mat/0511743].

[86] K. A. Takeuchi, M. Kuroda, H. Chaté, and M. Sano, Phys. Rev. Lett. **99**, 234503 (2007).

[87] M. H. Taibleson, *Fourier Analysis on Local Fields* (Princeton University Press, Princeton, 1975).

[88] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed. (Cambridge University Press, Cambridge, 2013).

[89] E. Seneta, *Non-negative Matrices and Markov Chains*, revised printing (Springer, New York, 2006).